Fixed-Point Antrag Signatures and Their Application to Compact Identity-Based Signatures

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Abstract. We give concrete security bounds and a complete fixed-point implementation of Antrag [11], a compact NTRU signature scheme with annular trapdoors. The implementation covers key generation, key preparation, and randomized ordinary and message-recovery signing at degrees 512, 768, and 1024. We prove unforgeability in the classical random-oracle model under a multi-target short-preimage assumption for the accepted-key distribution. Certified transfers to the finite implementations account for preparation errors, finite sampling, observable retries, and failures at 2^{64} signing queries.

We provide two signers sharing the key structure, parameters, and signature formats. The Cholesky signer adapts the exact-center construction of concurrent work by Zhang et al. [32], extending it to degree 768 with certified interval preparation; the Crown signer realizes Prest's Hybrid-Sampler with continuous perturbations. On Apple M5 and Intel Core i7-13700, Cholesky achieves 1.54–2.00 times the ordinary prepared-signing throughput of the measured $Q_{64.64}$ Falcon baseline at degrees 512 and 1024, although key generation and core preparation remain slower. Crown's core preparation is 3.0–3.3 times as fast as Cholesky's, favoring keys that sign few messages.

On an STM32F407 Cortex-M4, with shared SHAKE256 and lazy acceptance, Cholesky takes about 2–3% more signing cycles than FN-DSA fx32.64 at degrees 512 and 1024. Both Antrag signers use 6–9% less SRAM. Cholesky also uses less code flash but needs a prepared key 1.6–1.7 times the size of FN-DSA's; Crown's key is comparable in size to FN-DSA's, with slower signing.

Finally, message recovery in the private-key generator's certificate yields a certificate-based identity-based signature scheme with complete signatures of 1,941, 2,960, and 3,884 bytes at the three degrees, respectively.

Keywords: identity-based signatures · lattice signatures · message recovery · fixed-point arithmetic

1 Introduction

Hash-and-sign NTRU signatures [9] publish a ring element h and use a secret short basis B to sample a short preimage of a salted message hash. An ordinary

signature contains the salt and one short polynomial; verification reconstructs the missing coordinate using h and checks the joint norm. Falcon [13], which underlies FN-DSA, exemplifies this compact design. Compactness is particularly valuable when a public key travels with each signature. In certificate-based identity-based signatures (IBS) [3], for example, each signature carries a private-key generator (PKG) certificate, a user public key, and an application signature. Message recovery further reduces the transmitted user-key material by recovering part of it from the certificate.

Falcon samples preimages by traversing a secret LDL tree built through Fast Fourier Orthogonalization. At its leaves, a one-dimensional discrete Gaussian sampler draws integers with key-dependent widths and centers, computed in IEEE-754 binary64 arithmetic [13]. Prior work addresses this sampler from several angles: isochronous sampling targets output-independent running time [17], Rényi-divergence bounds control the effect of approximate Gaussians [12,18], and a sensitivity analysis quantifies the effect of floating-point errors [22]. The arithmetic around the sampler also needs protection. Zhou et al. exploit signed-exponent leakage when the floor of a sampling center is converted back to binary64 [33]. Fixed-point arithmetic avoids this conversion. De Almeida Braga et al. give a fixed-point Falcon signer [6]: additional key-generation rejection checks on Falcon's Gaussian keys, including a loose bound on Antrag's hybrid-quality parameter, enable a priori bounds on all fixed-point intermediates, while the precision analysis remains partly empirical. Halmans et al. give a constant-time 32.64 Falcon signer for the Cortex-M4 and show that the precision of fixed-point Falcon depends on its operation chain [16].

Falcon's recursive sampler also limits parallelism and complicates extension beyond power-of-two cyclotomic rings. Mitaka [10] uses Prest's Hybrid-Sampler [29], based on Peikert's perturbation approach [24], to support parallelism and masking; its variant MitakaZ replaces continuous perturbations with integral ones [8]. Antrag [11] improves the trapdoor quality through annular key generation and keeps Mitaka's signing and verification [11, full version, Section 5.2]. As its authors do, we call the resulting signature scheme Antrag; [32] calls it Mitaka. SOLMAE [20] likewise combines the Hybrid-Sampler with annular key generation, with different parameters. Antrag's public discrete width allows proposal tables to be shared across keys, and each row's coordinates can be sampled in parallel. The two-row construction also supports 3-smooth cyclotomic rings, including $X^{768} - X^{384} + 1$. As in Falcon, however, the centers of the first nearest-plane step are linear in the secret key with publicly computable coefficients; the perturbation adds only small noise. A binary64 implementation that leaks the sign and exponent of these centers would therefore face the attack of [33] at comparable cost; this motivates a fixed-point signer.

The original Antrag work analyzes annular trapdoors and key-recovery attacks and provides a floating-point implementation. For signature forgery, Mitaka gives an asymptotic GPV reduction to Module-SIS and notes that, at $q = 12289$, the Euclidean SIS bound makes this assumption vacuous [10, Section 5]. We extend the Antrag work with concrete signature-security bounds, message

recovery, and a fixed-point realization. The analysis accounts for the key distribution induced by completion and rejection, numerical errors, and observable retries. Our contributions are threefold.

1. **Security of Antrag.** Following the Falcon+ analysis of [12], we reduce ordinary and message-recovery forgery against the ideal signer to the multi-target Antrag ring inhomogeneous short integer solution (A-R-ISIS) problem for the specified accepted-key law in the classical random-oracle model. Uniform smoothing and acceptance bounds allow the reductions to account for collisions, retries, and the modeled count transcript, including in degree 768.
2. **Fixed-point realization and evaluation.** We implement Antrag key generation, key preparation, and signing using integer and fixed-point arithmetic at degrees 512, 768, and 1024. Two signers share each profile's parameters and formats: a Cholesky signer based on [32] and a Crown signer realizing Prest's Hybrid-Sampler [4, Appendix B]. Certified transfers cover preparation errors, finite sampling, observable retries, and failures at 2^{64} signing queries, with losses below 1.82 bits for the Cholesky signer and 2.47 bits for the Crown signer at the reference-event bounds of Table 3. The signers offer different trade-offs between preparation cost and signing throughput (Section 5.3).
3. **Compact identity-based signatures.** We instantiate the certificate construction of [3] with Antrag, combining message recovery in the PKG certificate with an ordinary user signature to reduce transmitted user-key material.

Concurrent work. Independently and concurrently, Zhang et al. [32] implemented Mitaka signing with Antrag trapdoors in fixed point at degrees 512 and 1024, using an improved MitakaZ sampler. Their sampler forms the perturbation from a rounded Cholesky factor applied to integer Gaussian samples, so that every sampling center can be computed exactly using number-theoretic transforms (NTTs) and Chinese remainder theorem (CRT) reconstruction. The two works differ in scope. Their fixed-point implementation covers signing with a prepared key, while key generation and key preparation use floating-point arithmetic; this work covers the complete scheme, including key generation and certified interval key preparation, at degrees 512, 768, and 1024. Their analysis bounds the sampler construction under perfect integer sampling and exact Cholesky rounding; the transfer here includes the implemented preparation, finite sampling, and observable retries. This work also gives unforgeability reductions for Antrag, message recovery, and the identity-based application. Parameter choices also differ: we use one inner width r_{in} across all degrees, so degrees 512 and 1024 share an online sampler and evaluator certificate. The Cholesky signer, a fixed-point realization of their construction, signs faster than the Crown signer but requires more key preparation, so the implementation offers both signers. Section 5.3 compares the signers and a reimplementation of their method under matched signing requirements.

Section 2 specifies the algorithms, including both signers; Section 3 gives their security bounds and certificate composition. Sections 4 and 5 develop the

fixed-point realization, evaluate its size and performance, and describe the implementation safeguards. Appendix A contains the security proofs, B the lattice-cost calibration, and C the random tapes, reference key generation, and numerical analysis of both signers. Appendices D and E analyze lazy Bernoulli timing and the precision choices of both signers, respectively. Appendix F gives the benchmark methods and additional results, and Appendix G analyzes an ideal deterministic certificate mode.

2 The Antrag Signature

2.1 Notation

Let $N \in \{512, 768, 1024\}$ be the ring degree and $q = 12289$ the modulus. For the indeterminate X, write $\varphi = X^N + 1$ when N is a power of two and $\varphi = X^{768} - X^{384} + 1$ when $N = 768$. Set $R = \mathbb{Z}[X]/(\varphi)$, $R_q = R/qR$, and $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$; $R_q^\times$ is the group of invertible elements. Polynomial multiplication is reduced modulo φ, and a_i denotes coefficient i of a, indexed from 0 to $N-1$. The ring adjoint is $a^*(X) = a(X^{-1}) \bmod \varphi$, extended to real coefficients; on matrices, $*$ also transposes the entries.

We write $U(S)$ for the uniform law on a finite set or interval. In the algorithms, an arrow denotes either assignment or sampling from the displayed law; sampling from a set is uniform. The symbol $\perp$ denotes failure, and $\|$ between byte strings denotes concatenation. We use ln and log for natural logarithms and $\log_2$ for base-two logarithms.

An NTRU public key is an element $h \in R_q$. Write $s = (s_0, s_1) \in R^2$ for a response pair. Vectors are written as coordinate tuples and treated as columns in matrix products. The associated lattice and the signing coset of a syndrome $c \in R_q$ are

$$\Lambda_h = \{(s_0, s_1) \in R^2 : s_0 + s_1 h = 0 \bmod q\}, \qquad \mathcal{L}_h(c) = \Lambda_h + (c, 0).$$

The coefficient lattice Λ_h has rank $2N$ and index q^N in R^2. Its covolume in the normalized canonical metric is q^N in the power-of-two rings and $q^N(3/4)^{N/2}$ in degree 768 (Appendix B.1).

A trapdoor is a short basis of Λ_h. For a short pair $(f, g) \in R^2$ with f invertible in R_q, set $h = gf^{-1} \bmod q$; then $(g, -f) \in \Lambda_h$. NTRUSolve [28] uses field-norm descent to solve in a smaller ring and lift a second pair (F, G) satisfying $fG - gF = q$. In power-of-two rings, the descent repeatedly applies $f(X) \mapsto f(X)f(-X)$ down to degree 1; in degree 768, it starts with a degree-3 norm to degree 256. The basis vectors $b_1 = (g, -f)$ and $b_2 = (G, -F)$ are the columns of $B = \left(\begin{smallmatrix} g & G \\ -f & -F \end{smallmatrix}\right)$, and $fG - gF = q$ gives $B^{-1} = q^{-1}\left(\begin{smallmatrix} -F & -G \\ f & g \end{smallmatrix}\right)$.

The canonical embedding Φ evaluates a polynomial at the complex roots of φ. For $a \in R \otimes_{\mathbb{Z}} \mathbb{R}$, define the normalized canonical norm by summing over all N embeddings ϕ:

$$\|a\|_{\mathrm{can}}^2 = \frac{1}{N} \sum_{\phi} |\phi(a)|^2.$$

The corresponding real inner product is $\langle a, b\rangle_{\mathrm{can}} = N^{-1}\sum_{\phi}\mathrm{Re}(\phi(a)\overline{\phi(b)})$, where the bar denotes complex conjugation. Coefficient norms $\|\cdot\|_2$ and $\|\cdot\|_\infty$ denote Euclidean length and maximum absolute coefficient.

The canonical norm on R^2 is

$$\|(s_0, s_1)\|_{\mathrm{can}} = \left(\|s_0\|_{\mathrm{can}}^2 + \|s_1\|_{\mathrm{can}}^2\right)^{1/2}.$$

Let G_R be the Gram matrix of the coefficient basis. This basis is orthonormal for $X^N + 1$. For $X^{768} - X^{384} + 1$, G_R has blocks $\left(\begin{smallmatrix} 1 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right)$ on positions $(j, j+384)$, with inverse blocks $\left(\begin{smallmatrix} 4/3 & -2/3 \\ -2/3 & 4/3 \end{smallmatrix}\right)$. In this ring, the canonical norm of a response pair is the square root of

$$\sum_i (s_{0,i}^2 + s_{1,i}^2) + \sum_{i<N/2} (s_{0,i}s_{0,i+N/2} + s_{1,i}s_{1,i+N/2}).$$

Write Φ_j for one representative of conjugate pair j, with $0 \le j < N/2$. For the packed evaluation matrix $E_{j,i} = \Phi_j(X^i)$, the inverse on real polynomials is $\Phi^{-1}(z) = (2/N)G_R^{-1}\,\mathrm{Re}(E^* z)$, componentwise on tuples. Here E^* is the conjugate transpose.

Let $\mathbb{K} = \mathbb{Q}[X]/(\varphi)$ be the field of fractions of R. The Gram–Schmidt coefficient

$$u = \frac{f^*F + g^*G}{ff^* + gg^*} \in \mathbb{K}$$

makes $\tilde{b}_2 = b_2 - ub_1$ orthogonal to b_1 at every embedding.

For a lattice or lattice coset Λ, a center d in its ambient real space, and a width $\sigma > 0$, $D_{\Lambda,\sigma,d}$ assigns to $x \in \Lambda$ mass proportional to $\exp(-\|x - d\|_{\mathrm{can}}^2/(2\sigma^2))$. We omit a zero center and sometimes add the superscript can to emphasize the metric; on $\mathbb{Z}$ the norm is absolute value. Signing targets such a Gaussian on $\mathcal{L}_h(c)$.

A profile π specifies the ring, key-generation policy, Gaussian widths, and encoding parameters; the three profiles, labeled 1f, 3f, and 5f, have $N = 512$, 768, and 1024 (Table 1). In particular, $\mathcal{C}_\pi \subset \mathbb{Z}^N$ is the set of centered coefficient vectors accepted by its fixed-length response codec, and ℓ_{salt} is its salt length in bits. For an encoded polynomial a, $\mathsf{Can}(a) = 1$ means that its encoding is canonical. Write $[x]_q$ for the centered representative of $x \in R_q$, with coefficients in $\{-\lfloor (q-1)/2\rfloor, \ldots, \lfloor q/2\rfloor\}$.

The codec is Falcon's compression [13, Section 3.11.2] with a coefficient bound. It allocates v bytes per polynomial, using a sign bit, b low magnitude bits, and a terminated unary quotient per coefficient. Let L be the coefficient-magnitude bit limit and $\mathcal{B}_\pi = \{x : |x_i| < 2^L \text{ for all } i\}$. Membership in $\mathcal{C}_\pi$ requires $x \in \mathcal{B}_\pi$ and $N(b+2) + \sum_i \lfloor |x_i|/2^b\rfloor \le 8v$; decoding rejects negative zero and nonzero padding.

Let $\mathcal{K}_\pi$ be the reference generator's output law conditioned on success. The generator applies the profile's deterministic candidate and completion maps to independent uniform words and returns the first admitted key within its trial

cap. Appendix C.2 specifies these maps, including the completion policy. Section 3 analyzes the ideal signer, which draws each response exactly from the coset Gaussian of Section 2.1, on keys drawn from $\mathcal{K}_\pi$; Section 4 compares the finite signer with this ideal signer.

Each profile and suite has a canonical public-key digest $\mathrm{pkh}_{\pi,h}$. The ordinary oracle $H_{\pi,h}$ maps a salt $r \in \{0,1\}^{\ell_{\mathrm{salt}}}$ and message $M \in \{0,1\}^*$ to a syndrome in R_q. Its input binds the profile and public key. Message-recovery mode (MRM) uses independent domains $H_{0,\pi,h}$ and $H_{1,\pi,h}$. We suppress fixed subscripts; Appendix A.5 gives the concrete digest construction.

2.2 Hybrid quality

At each representative embedding, write $\nu_j = |\Phi_j(f)|^2 + |\Phi_j(g)|^2$. The Gram–Schmidt vectors b_1 and $\tilde{b}_2$ of Section 2.1 have embedding lengths $\sqrt{\nu_j}$ and $q/\sqrt{\nu_j}$ [11, Lemma 2]; they control the smoothing of Λ_h and, after rounding, the spectrum of $\hat{B}\hat{B}^*$ (Section 2.5). Their maximum defines the pair quality:

$$Q_{\mathrm{hyb}}(f,g) = \max_j \max\bigl\{\sqrt{\nu_j},\, q/\sqrt{\nu_j}\bigr\}.$$

We also write $Q_{\mathrm{hyb}}(B) = Q_{\mathrm{hyb}}(f,g)$. For $\alpha \geq 1$, the pair has quality α when $Q_{\mathrm{hyb}}(f,g) \leq \alpha\sqrt{q}$, equivalently

$$\frac{q}{\alpha^2} \leq \nu_j \leq \alpha^2 q \qquad \text{for every } j.$$

The upper bound controls b_1 at every embedding, while the lower bound controls the dual lengths $q/\sqrt{\nu_j}$. An admitted key also satisfies

$$\|(F,G)\|_{\mathrm{can}}^2 \leq q\alpha^2 k_N, \qquad k_N = \begin{cases} N/4+1, & N = 512, 1024, \\ N/6+1, & N = 768. \end{cases} \tag{1}$$

The $O(N)$ integer test in Eq. (1) bounds b_2. Cauchy–Schwarz then bounds its projection onto b_1 and the signing centers. Appendix C.2 derives k_N from the ring covering radius and relates the threshold to the completion algorithm.

The public inner discrete width is $r_{\mathrm{in}} = 1.315 = 263/200$, and the response width and verification radius are

$$\sigma_{\mathrm{sig}} = 1.32\,\alpha\sqrt{q}, \qquad \beta_{\mathrm{sig}}^2 = \lfloor 1.042^2\,\sigma_{\mathrm{sig}}^2 \cdot (2N) \rfloor.$$

The tail factor 1.042, as in Mitaka [10, Section 3.5], balances expected signing trials against forgery cost. In a rounded-Gaussian Monte Carlo model of the MRM filter, the fixed-length encodings of s_0 and s_1 reject about 4% of samples regardless of the factor, and at 1.042 the norm test adds at most two percentage points. Falcon's factor 1.1 would cut expected trials by at most 2% but lower the random-SIS cost by two to four bits; 1.02 would gain at most two bits for 8–16% more trials.

Annular sampling uses convex combinations of the normalized radii α^{-1} and α, with $\xi = 1/3$:

$$r_{\mathrm{lo}} = \frac{1}{2}\big(\alpha + \alpha^{-1} - \xi(\alpha - \alpha^{-1})\big), \qquad r_{\mathrm{hi}} = \frac{1}{2}\big(\alpha + \alpha^{-1} + \xi(\alpha - \alpha^{-1})\big).$$

Continuous squared radii are uniform on $[qr_{\mathrm{lo}}^2,\, qr_{\mathrm{hi}}^2]$.

2.3 Key generation

Let $A^+(a,b) = \{(x,y) \in \mathbb{R}^2_{\geq 0} : a^2 \leq x^2 + y^2 \leq b^2\}$, with $0 \leq a \leq b$, be a first-quadrant annulus. At each conjugate pair of embeddings, [11, Algorithm 4] samples magnitudes in $A^+(r_{\mathrm{lo}}\sqrt{q}, r_{\mathrm{hi}}\sqrt{q})$ and independent uniform phases (Figure 1 uses $\mathrm{i}^2 = -1$). After inverse embedding, Round rounds coefficients to the nearest integer, with ties to even. For $N = 512, 1024$, OddSum changes an even coefficient sum to odd by adjusting the coefficient with the largest absolute rounding residual by one in the residual's direction. Ties use the least index; a zero residual moves negatively. Odd sums are unchanged. Degree 768 uses plain rounding. Figure 1 describes this geometry; the finite-word map, decision cutoff, and trial cap in Appendix C.2 define $\mathcal{K}_\pi$.

Key generation adds basis completion and the norm test in Eq. (1) to the annular test of [11, Algorithm 5]. The norm test supplies the bound on b_2 needed for fixed-point signing.

2.4 Signing and verification

Antrag is a GPV signature on Λ_h [14]. Each trial draws a fresh salt r and an independent private sampler tape, sets $c = H_{\pi,h}(r, M)$, and represents its coefficients in $\{0, \ldots, q-1\}$. Write $\bar{c} = (c, 0)$. Both signing modes use $prep = \mathsf{PreKey}(f, g, F, G)$, retrieved from the cache or computed by the selected sampler's preparation routine (Section 2.5). The call $\mathsf{PreSmp}(prep, \bar{c})$ returns coefficients $(x_1, x_2) \in R^2$ from which the signer forms

$$s_0 = c - x_1 g - x_2 G, \qquad s_1 = x_1 f + x_2 F.$$

Ordinary signing publishes (r, s_1). Verification recovers the centered $s_0 = [c - s_1 h]_q$ and tests the joint norm; MRM publishes both components. Replacing s_1 by $-s_1$ gives the sign convention of [11, Algorithm 1] without changing the canonical length.

The ordinary signing filter is $\mathcal{A}_\pi^{\mathrm{ord}} = \{s : \|s\|_{\mathrm{can}} \leq \beta_{\mathrm{sig}},\ s_0 \in \mathcal{B}_\pi,\ s_1 \in \mathcal{C}_\pi\}$; its bound on s_0 ensures exact centered reconstruction, including in degree 768. The MRM filter is $\mathcal{A}_\pi = \{s : \|s\|_{\mathrm{can}} \leq \beta_{\mathrm{sig}},\ s_0, s_1 \in \mathcal{C}_\pi\}$. Ordinary verification accepts exactly the pairs in $\mathcal{A}_\pi^{\mathrm{ord}}$, which lie in the relaxed set $\mathcal{V}_\pi = \{s : \|s\|_{\mathrm{can}} \leq \beta_{\mathrm{sig}},\ s_1 \in \mathcal{C}_\pi\}$ used as the solution predicate, so $\mathcal{A}_\pi \subseteq \mathcal{A}_\pi^{\mathrm{ord}} \subseteq \mathcal{V}_\pi$.

Message recovery follows the hash-and-sign approach of [25,15]. Fix positive integer dimensions $\lambda_{\mathrm{MR}}, \gamma_{\mathrm{MR}}$ for the hash prefix and nonce, with $\lambda_{\mathrm{MR}} + \gamma_{\mathrm{MR}} \leq N$, and set $\mu = N - \lambda_{\mathrm{MR}} - \gamma_{\mathrm{MR}}$. The independent oracles

$$H_{0,\pi,h} : \mathbb{Z}_q^{\gamma_{\mathrm{MR}}} \times \mathbb{Z}_q^{\mu} \times \{0,1\}^* \to \mathbb{Z}_q^{\lambda_{\mathrm{MR}}}, \qquad H_{1,\pi,h} : \mathbb{Z}_q^{\lambda_{\mathrm{MR}}} \to \mathbb{Z}_q^{N - \lambda_{\mathrm{MR}}}$$

Fig. 1. Geometric description of KeyGen. Appendix C.2 defines the finite reference law $\mathcal{K}_\pi$.

1. **repeat**
2. **for** $j = 0, \dots, N/2 - 1$ **do**
3. $u_j \leftarrow U([qr_{\mathrm{lo}}^2,\ qr_{\mathrm{hi}}^2])$
4. $\theta_j \leftarrow U([0, \pi/2])$
5. $\vartheta_{f,j}, \vartheta_{g,j} \leftarrow U([0, 2\pi])$
6. $z_j \leftarrow \sqrt{u_j}\cos\theta_j\,(\cos\vartheta_{f,j} + \mathrm{i}\sin\vartheta_{f,j})$
7. $w_j \leftarrow \sqrt{u_j}\sin\theta_j\,(\cos\vartheta_{g,j} + \mathrm{i}\sin\vartheta_{g,j})$
8. $(\tilde{f}, \tilde{g}) \leftarrow \Phi^{-1}(z, w)$
9. $(f, g) \leftarrow \big(\mathsf{Round}(\tilde{f}), \mathsf{Round}(\tilde{g})\big)$
10. **if** $N \in \{512, 1024\}$ **then** $(f, g) \leftarrow \big(\mathsf{OddSum}(f), \mathsf{OddSum}(g)\big)$
11. **for** $j = 0, \dots, N/2 - 1$ **do**
12. $\nu_j \leftarrow |\Phi_j(f)|^2 + |\Phi_j(g)|^2$
13. **if** some $\nu_j \notin [q/\alpha^2,\ \alpha^2 q]$ **then continue**
14. **if** $f \notin R_q^\times$ **then continue**
15. $h \leftarrow g f^{-1} \bmod q$
16. $(F, G) \leftarrow \mathsf{NTRUSolve}(f, g)$
17. **if** $(F, G) = \perp$ **then continue**
18. **if** $\|(F, G)\|_{\mathrm{can}}^2 > q\alpha^2 k_N$ **then continue**
19. **return** $sk = (f, g, F, G),\ pk = h$

Fig. 2. $\mathsf{Sign}((f, g, F, G), M)$

1. Retrieve or compute $prep = \mathsf{PreKey}(f, g, F, G)$
2. **repeat**
3. $r \leftarrow \{0,1\}^{\ell_{\mathrm{salt}}}$
4. $c \leftarrow H_{\pi,h}(r, M), \quad \bar{c} \leftarrow (c, 0)$
5. $(x_1, x_2) \leftarrow \mathsf{PreSmp}(prep, \bar{c})$
6. $s_0 \leftarrow c - x_1 g - x_2 G$
7. $s_1 \leftarrow x_1 f + x_2 F$
8. **until** $\|(s_0, s_1)\|_{\mathrm{can}} \le \beta_{\mathrm{sig}}$
9. **and** $s_0 \in \mathcal{B}_\pi,\ s_1 \in \mathcal{C}_\pi$
10. **return** (r, s_1)

Fig. 3. $\mathsf{Verify}(pk, M, (r, s_1))$

1. **if** $r \notin \{0,1\}^{\ell_{\mathrm{salt}}}$ **then return** 0
2. **if** $\mathsf{Can}(s_1) = 0$ or $s_1 \notin \mathcal{C}_\pi$ **then return** 0
3. $c \leftarrow H_{\pi,h}(r, M)$
4. $s_0 \leftarrow [c - s_1 h]_q$
5. **return** 1 iff $\|(s_0, s_1)\|_{\mathrm{can}} \le \beta_{\mathrm{sig}}$ and $s_0 \in \mathcal{B}_\pi$

are independent of $H_{\pi,h}$. Let $M_0 \in \mathbb{Z}_q^\mu$ be the recoverable message field and M_1 the explicit byte string. Split $c = (c_0, c_1)$ into blocks of lengths λ_{MR} and $N - \lambda_{\mathrm{MR}}$. Each trial draws a fresh nonce $\rho \in \mathbb{Z}_q^{\gamma_{\mathrm{MR}}}$ and masks the concatenation

(ρ, M_0) with $H_{1,\pi,h}(c_0)$. All MRM reductions modulo q use representatives in $\{0, \ldots, q-1\}$.

An r_{rec}-byte payload can be encoded injectively as M_0, by a map we call RecEnc, when $2^{8r_{\mathrm{rec}}} \le q^{\mu}$; decoding rejects values outside that image. We write $\mathrm{Enc}_{\pi}(M_0, M_1)$ for the injective encoding of the profile, public-key digest, fixed-width recoverable field, and length-prefixed explicit message. The attempt stream binds the signing type (Appendix C.1).

Fig. 4. $\mathsf{SignMRM}((f, g, F, G), M_0, M_1)$

1. Retrieve or compute $prep = \mathsf{PreKey}(f, g, F, G)$
2. **repeat**
3. $\quad \rho \leftarrow U(\mathbb{Z}_q^{\gamma_{\mathrm{MR}}})$
4. $\quad c_0 \leftarrow H_{0,\pi,h}(\rho, M_0, M_1)$
5. $\quad c_1 \leftarrow \big(H_{1,\pi,h}(c_0) + (\rho, M_0)\big) \bmod q$
6. $\quad c \leftarrow (c_0, c_1)$
7. $\quad \bar{c} \leftarrow (c, 0)$
8. $\quad (x_1, x_2) \leftarrow \mathsf{PreSmp}(prep, \bar{c})$
9. $\quad s_0 \leftarrow c - x_1 g - x_2 G$
10. $\quad s_1 \leftarrow x_1 f + x_2 F$
11. **until** $\|(s_0, s_1)\|_{\mathrm{can}} \le \beta_{\mathrm{sig}}$
12. $\quad$ **and** $s_0, s_1 \in \mathcal{C}_{\pi}$
13. **return** (s_0, s_1)

Fig. 5. $\mathsf{VerifyMRM}(pk, M_1, (s_0, s_1))$

1. **if** $\mathsf{Can}(s_0) = 0$ or $\mathsf{Can}(s_1) = 0$ **then return** 0
2. **if** $s_0 \notin \mathcal{C}_{\pi}$ or $s_1 \notin \mathcal{C}_{\pi}$ **then return** 0
3. **if** $\|(s_0, s_1)\|_{\mathrm{can}} > \beta_{\mathrm{sig}}$ **then return** 0
4. $c \leftarrow (s_0 + s_1 h) \bmod q$
5. $(c_0, c_1) \leftarrow c$
6. $(\rho, M_0) \leftarrow \big(c_1 - H_{1,\pi,h}(c_0)\big) \bmod q$
7. **if** $H_{0,\pi,h}(\rho, M_0, M_1) \neq c_0$ **then return** 0
8. **return** $(1, M_0)$

2.5 Two perturbation samplers

Both signers realize the sampling step by Peikert's convolution approach [24] in the two-row form of Prest's Hybrid-Sampler [29]: a perturbation is combined with two online discrete-Gaussian draws at centers determined by the prepared basis, so that the response is close to $D_{\mathcal{L}_h(c),\sigma_{\mathrm{sig}}}$. The signers differ in how the perturbation is drawn and in what preparation caches. The Cholesky signer uses the construction of Zhang et al. [32, Algorithm 3]: a rounded Cholesky factor applied to integer Gaussian samples, which makes every center an exact rational. The Crown signer draws continuous perturbations with key-dependent widths and computes centers in fixed-point Fourier arithmetic. Both signers share the key structure, the parameters of Table 1, the online proposal tables, the random-stream generation framework and online proposal-record layouts, and the signature formats. An encoded key binds its signing suite, whose identifier also separates the key-generation streams (Appendix C.2). We write $\mathsf{PreKey}^{\mathrm{ch}}$ and $\mathsf{PreSmp}^{\mathrm{ch}}$ for the Cholesky signer and $\mathsf{PreKey}^{\mathrm{cr}}$ and $\mathsf{PreSmp}^{\mathrm{cr}}$ for the Crown signer; PreKey and PreSmp denote the selected signer's pair.

Cholesky signer. The Gram–Schmidt coefficient u of Section 2.1 is rounded to $\hat{u} = \lfloor p_u u \rceil / p_u$ with $p_u = 2^{16}$, coefficientwise. The vectors $\hat{b}_1 = b_1$ and $\hat{b}_2 = b_2 - \hat{u} b_1$ span the same $\mathbb{K}$-module as B, but need not span the same R-lattice because $\hat{u}$ need not belong to R; every lattice point satisfies $x_1 b_1 + x_2 b_2 = (x_1 + \hat{u} x_2)\hat{b}_1 + x_2 \hat{b}_2$. Let $\hat{B}$ be the 2×2 matrix over $\mathbb{K}$ with columns $\hat{b}_1, \hat{b}_2$, and put

$$\tilde{s} = \frac{\sigma_{\mathrm{sig}}}{r_{\mathrm{in}}} = \frac{264}{263}\,\alpha\sqrt{q}, \qquad \Sigma_p = \tilde{s}^2 I_2 - \hat{B}\hat{B}^*.$$

For every admitted key, Σ_p is positive definite at every embedding (Section 4.1). $\mathsf{PreKey}^{\mathrm{ch}}$ computes its lower Cholesky factor embeddingwise and rounds the coefficients of the resulting matrix A, whose entries lie in $R \otimes_{\mathbb{Z}} \mathbb{R}$, to $\hat{A} = \lfloor p_A A \rceil / p_A$, with $p_A = 2^{43}$ for 1f and 2^{44} for 3f and 5f. These rounding formulas and Figure 6 are idealized; Section 4.1 specifies the implemented interval-midpoint rounding and the error bounds used in the transfer.

Each trial draws a fresh zero-centered integer pair $v = (v_0, v_1) \leftarrow D^{\mathrm{can}}_{R^2, L_{\mathrm{p}} r_{\mathrm{in}}}$ with $L_{\mathrm{p}} = 2$ and sets the perturbation $p = \hat{A} v / L_{\mathrm{p}}$. The perturbation and online Gaussian weights have parameter matrices $r_{\mathrm{in}}^2 \hat{A}\hat{A}^*$ and $r_{\mathrm{in}}^2 \hat{B}\hat{B}^*$, whose sum is close to $\sigma_{\mathrm{sig}}^2 I$; Lemma 2 bounds the resulting output-law error. Because $\hat{u}$ and $\hat{A}$ have power-of-two denominators and qB^{-1} is integral, the sampling centers are exact rationals with fixed denominators; the perturbation requires no further rounding. The input v has independent integer coefficients in the power-of-two rings and correlated coefficient pairs in degree 768, as do the two online ring Gaussians conditional on their centers; the perturbation p itself has correlated rational coefficients. Figures 6 and 7 give $\mathsf{PreKey}^{\mathrm{ch}}$ and $\mathsf{PreSmp}^{\mathrm{ch}}$; in the latter, d_2 and d_1 are the centers of the two online draws.

Fig. 6. $\mathsf{PreKey}^{\mathrm{ch}}(f, g, F, G)$, Cholesky signer, idealized rounding

1. $u \leftarrow (f^* F + g^* G)/(f f^* + g g^*)$
2. $\hat{u} \leftarrow \lfloor p_u u \rceil / p_u$
3. $\hat{b}_1 \leftarrow b_1, \quad \hat{b}_2 \leftarrow b_2 - \hat{u} b_1$
4. $\Sigma_p \leftarrow \tilde{s}^2 I_2 - \hat{B}\hat{B}^*$
5. **for** $j = 0, \ldots, N/2 - 1$ **do**
6. $\quad (\sigma_{ik})_{i,k \in \{0,1\}} \leftarrow \Phi_j(\Sigma_p)$
7. $\quad \ell_{00} \leftarrow \sqrt{\sigma_{00}}, \quad \ell_{10} \leftarrow \sigma_{10}/\ell_{00}$
8. $\quad L_j \leftarrow \left(\begin{smallmatrix} \ell_{00} & 0 \\ \ell_{10} & \sqrt{\sigma_{11} - |\ell_{10}|^2} \end{smallmatrix}\right)$
9. $A \leftarrow \Phi^{-1}\big((L_j)_{j < N/2}\big)$, entrywise
10. $\hat{A} \leftarrow \lfloor p_A A \rceil / p_A$
11. **return** $(\hat{u}, \hat{A}, \hat{B})$

Fig. 7. $\mathsf{PreSmp}^{\mathrm{ch}}(prep, \bar{c})$, Cholesky signer

1. Parse $(\hat{u}, \hat{A}, \hat{B})$ from $prep$
2. $v \leftarrow D^{\mathrm{can}}_{R^2, L_{\mathrm{p}} r_{\mathrm{in}}}$
3. $p \leftarrow \hat{A} v / L_{\mathrm{p}}$
4. $(\hat{c}_1, \hat{c}_2) \leftarrow \hat{B}^{-1}(\bar{c} - p)$
5. $d_2 \leftarrow \hat{c}_2$
6. $x_2 \leftarrow D^{\mathrm{can}}_{R, r_{\mathrm{in}}, d_2}$
7. $d_1 \leftarrow \hat{c}_1 - \hat{u} x_2$
8. $x_1 \leftarrow D^{\mathrm{can}}_{R, r_{\mathrm{in}}, d_1}$
9. **return** (x_1, x_2)

Crown signer. Each Hybrid-Sampler stage adds continuous Gaussian noise to its center before drawing a discrete ring Gaussian. The latter has independent coefficients in power-of-two rings and correlated pairs in degree 768. In $\mathsf{PreSmp}^{\mathrm{cr}}$, $y_i = (y_{i,j})_{j<N/2}$ denotes the perturbation, $z_i = (z_{i,j})_{j<N/2}$ the shifted center, and $d_i = \Phi^{-1}(z_i)$ its coefficient representation, for $i = 1, 2$. In $\mathsf{PreKey}^{\mathrm{cr}}$, $\gamma_j = -\Phi_j(u)$ is the negated embedding of the Gram–Schmidt coefficient and $\tau_{1,j}, \tau_{2,j}$ are the residual perturbation widths. The law $\mathcal{N}(0, 512)$ has mean zero and variance 512; multiplying a draw by $\omega = \sqrt{N/1024}$ gives variance $N/2$ per real embedding coordinate. All displayed normal draws are independent. Figures 8 and 9 give $\mathsf{PreKey}^{\mathrm{cr}}$ and $\mathsf{PreSmp}^{\mathrm{cr}}$; Section 4 gives its width condition, its comparison with the ideal signer (Lemma 3), and its fixed-point realization; Appendix C.5 bounds its finite stages.

Fig. 8. $\mathsf{PreKey}^{\mathrm{cr}}(f, g, F, G)$, Crown signer

for $j = 0, \ldots, N/2 - 1$ **do**
  $\nu_j \leftarrow |\Phi_j(f)|^2 + |\Phi_j(g)|^2$
  $\chi_j \leftarrow \Phi_j(f)/q$
  $\psi_j \leftarrow \overline{\Phi_j(g)}/\nu_j$
  $\gamma_j \leftarrow \frac{\Phi_j(G)\overline{\Phi_j(g)} + \Phi_j(F)\overline{\Phi_j(f)}}{-\nu_j}$
  $\tau_{1,j} \leftarrow \sqrt{\sigma_{\mathrm{sig}}^2/\nu_j - r_{\mathrm{in}}^2}$
  $\tau_{2,j} \leftarrow \sqrt{\sigma_{\mathrm{sig}}^2 \nu_j/q^2 - r_{\mathrm{in}}^2}$
$\omega \leftarrow \sqrt{N/1024}$
return $(\chi_j, \psi_j, \gamma_j, \tau_{1,j}, \tau_{2,j})_{0 \le j < N/2}$ and ω

Fig. 9. $\mathsf{PreSmp}^{\mathrm{cr}}(prep, \bar{c})$, Crown signer

Parse $\omega, (\chi_j, \psi_j, \gamma_j, \tau_{1,j}, \tau_{2,j})_{j<N/2}$
for $j = 0, \ldots, N/2 - 1$ **do**
  $y_{2,j} \leftarrow \omega\tau_{2,j}\big(\mathcal{N}(0, 512) + \mathrm{i}\mathcal{N}(0, 512)\big)$
  $y_{1,j} \leftarrow \omega\tau_{1,j}\big(\mathcal{N}(0, 512) + \mathrm{i}\mathcal{N}(0, 512)\big)$
  $z_{2,j} \leftarrow \Phi_j(c)\,\chi_j - y_{2,j}$
$d_2 \leftarrow \Phi^{-1}(z_2)$
$x_2 \leftarrow D^{\mathrm{can}}_{R, r_{\mathrm{in}}, d_2}$
for $j = 0, \ldots, N/2 - 1$ **do**
  $z_{1,j} \leftarrow \Phi_j(c)\,\psi_j + \Phi_j(x_2)\,\gamma_j - y_{1,j}$
$d_1 \leftarrow \Phi^{-1}(z_1)$
$x_1 \leftarrow D^{\mathrm{can}}_{R, r_{\mathrm{in}}, d_1}$
return (x_1, x_2)

3 Security of Antrag Signatures

The bounds in this section apply to the ideal signer, whose trials draw the response exactly from $D^{\mathrm{can}}_{\mathcal{L}_h(c), \sigma_{\mathrm{sig}}}$ in place of PreSmp. Section 4 bounds the divergence of the specified sampler, including its finite arithmetic, from this ideal signer. We use UF-CMA for existential unforgeability under chosen-message attack. Write $\mathrm{Adv}^{\mathrm{UF\text{-}CMA}}_{\mathsf{S}}(t, Q_H, Q_S)$ for the maximum probability of forging on a fresh complete message in time t, with at most Q_H hash queries and Q_S signing queries. Subscripts A and A-MRM select the two signing modes. For a fixed profile, reduction times count expected local computation, oracle calls are counted separately, and $\widetilde{O}$ suppresses polylogarithmic factors. Oracle queries are classical, as in [12]. A proof in the quantum random-oracle model is known for Falcon+ [21]; the analysis here remains in the classical model. Proofs appear in Appendix A.

Signing redraws rejected sampler proposals and retries trials that fail the signing filter, so its runtime can reveal their counts. Both reference experiments expose the signing-filter decisions, auxiliary proposal counts, failure flags, and refill positions; rejected targets and samples remain private. The auxiliary counts follow fixed public laws independent of the key, target, and samples. The reductions follow [12]; because the simulator generates every trial, exposing this view adds no reduction loss. The bounds also apply when only signatures are revealed. Section 4 bounds the divergence between the implementation's counts and the reference laws.

The bounds concern randomized single-mode signing of messages or collision-resistant prehashes; deterministic, mixed-mode and external-representative signing oracles are outside their scope (Appendix A.5). Appendix G sketches a deterministic certificate mode with a weaker ideal-model bound.

3.1 Ideal sampling and uniform comparisons

For a lattice Λ, let $\Lambda^\vee$ be its dual under the canonical inner product: its elements have integer inner product with every vector of Λ. For $0 < \varepsilon < 1$, the smoothing parameter in standard-deviation units is

$$\eta_\varepsilon(\Lambda) = \inf\left\{\sigma > 0 : \sum_{y \in \Lambda^\vee \setminus \{0\}} e^{-2\pi^2\sigma^2\|y\|_{\mathrm{can}}^2} \le \varepsilon\right\}.$$

We use the same notation $\eta_\varepsilon(R)$ for the ring, in Prest's normalized standard-deviation convention [29]. Define

$$\rho_h(N,\varepsilon) = \frac{1}{\pi}\sqrt{\frac{\ln(2N(1+1/\varepsilon))}{2}}, \qquad \rho_u(N,\varepsilon) = \frac{1}{\pi}\sqrt{\frac{\ln(4N(1+1/\varepsilon))}{2}}.$$

Lemma 1 (canonical sampling). For every admitted basis B of Λ_h and $\varepsilon_p, \varepsilon_u \in (0,1)$, the smoothing parameters in standard-deviation units satisfy

$$\eta_{\varepsilon_p}(R) \le \rho_h(N,\varepsilon_p), \qquad \eta_{\varepsilon_u}(\Lambda_h) \le \rho_u(N,\varepsilon_u) Q_{\mathrm{hyb}}(B). \tag{2}$$

A canonical ring Gaussian factors into scalar laws in the power-of-two rings and correlated pair laws in degree 768, each admitting exact rejection sampling.

For all admitted keys, require

$$r_{\mathrm{in}} \ge \rho_h(N,\varepsilon_p), \qquad \sigma_{\mathrm{sig}} \ge \rho_u(N,\varepsilon_u) Q_{\mathrm{hyb}}(B). \tag{3}$$

The first inequality is used by the sampler comparison in Section 4.3; the second smooths the syndrome. Let Q_{syn} be the law of $s_0 + s_1 h \bmod q$ for $s \leftarrow D^{\mathrm{can}}_{R^2,\sigma_{\mathrm{sig}}}$, and put $c_u = \frac{1+\varepsilon_u}{1-\varepsilon_u}$. Under the ideal signer, the response conditioned on its syndrome c is exactly $D_{\mathcal{L}_h(c),\sigma_{\mathrm{sig}}}$. Quotient smoothing [12] bounds the ratio of $U(R_q)$ to Q_{syn} by $c_u^{\pm 1}$.

For probability laws P, Q and order $a > 1$, use multiplicative Rényi divergence

$$R_a(P\|Q) = \left(\sum_x P(x)^a Q(x)^{1-a} \right)^{1/(a-1)},$$

replacing the sum by an integral for continuous laws. It is infinite unless P is absolutely continuous with respect to Q. Let $r_u(a)$ bound $R_a(U(R_q)\|Q_{\mathrm{syn}})$ uniformly over admitted keys. With $\delta_u = c_u - 1 < 1$, [30, Lemma 3] gives

$$r_u(a) \le \mathcal{R}_a(\delta_u), \qquad \mathcal{R}_a(\delta) = \left(1 + \frac{a(a-1)\delta^2}{2(1-\delta)^{a+1}} \right)^{1/(a-1)}. \tag{4}$$

For every event E, Hölder gives $P(E) \le (R_a(P\|Q)Q(E))^{(a-1)/a}$. Uniform conditional divergences multiply along adaptive transcripts [2].

Let p_π be the infimum, over admitted keys, of the probability that one ideal signing trial with a uniform syndrome returns $s \in \mathcal{A}_\pi$. It also lower-bounds ordinary trial acceptance. For $Q_S \ge 1$ signing queries and an integer global trial cap $C_S \ge Q_S$, define

$$\Delta_{\mathrm{rep}}(C_S) = \sum_{i=0}^{Q_S-1} \binom{C_S}{i} p_\pi^i (1-p_\pi)^{C_S-i}. \tag{5}$$

Before an oracle collision, fresh challenges and independent sampler randomness give conditional acceptance at least p_π. Continuing with independent uniform challenges after a collision yields an auxiliary process whose exhaustion probability bounds that of the original process before collision. Appendix A.3 proves $p_\pi > 0.58$, giving $\Delta_{\mathrm{rep}}(2Q_S) \le \exp(-4Q_S(0.08)^2)$.

3.2 Ordinary unforgeability

Assumption 1 (accepted-key A-R-ISIS hardness). For $m \ge 1$, the m-target A-R-ISIS problem samples $(B, h) \leftarrow \mathcal{K}_\pi$ and independent $c_1, \ldots, c_m \leftarrow U(R_q)$; given $(h, c_1, \ldots, c_m)$, a solver must return $j \in \{1, \ldots, m\}$ and $s \in \mathcal{L}_h(c_j) \cap \mathcal{V}_\pi$. Its key law and norm predicate distinguish it from Falcon's problem [12, Definition 10]. Write $\mathrm{Adv}_\pi^{m\text{-A-R-ISIS}}$ for its maximal success probability at the stated resources. For each profile and resource tuple used in a concrete security claim, we assume the stipulated upper bound on this advantage.

Assumption 1 is stated for the accepted-key law defined by the candidate map, completion policy, and all admission predicates. Section 4 handles operational key-generation failures; Corollary 1 charges cryptographic expansion of the candidate and signing tapes through $\varepsilon_{\mathrm{tape}}$. For the MRM reduction, let $\mathrm{A}^\star$ be an auxiliary ordinary scheme with the same key generation, hash, output format, and verifier as A, but signing filter $\mathcal{A}_\pi$.

Theorem 1 (ordinary Antrag). Assume $p_\pi > 0$ and Eq. (3). Consider an adversary running in time t, making at most Q_H queries to $H_{\pi,h}$ and $Q_S \ge 1$

signing queries to the ideal signer. For any integer $C_S \geq Q_S$ and order $a_u > 1$, put $\theta_u = (a_u - 1)/a_u$. Evaluating the A-R-ISIS advantage at expected local time $t + \widetilde{O}(C_S + Q_H)$, with oracle calls counted separately, gives

$$\begin{aligned} \mathrm{Adv}_{\mathrm{A}}^{\text{UF-CMA}} \leq & \left\{ r_u(a_u)^{C_S} \, \mathrm{Adv}_{\pi}^{(Q_H+1)\text{-A-R-ISIS}} \right\}^{\theta_u} \\ & + \Delta_{\mathrm{rep}}(C_S) + \left(\binom{C_S}{2} + C_S(Q_H + 1) \right) 2^{-\ell_{\mathrm{salt}}}. \end{aligned} \tag{6}$$

The same bound holds with A replaced by $\mathrm{A}^\star$.

3.3 Message-recovery unforgeability

The MRM reduction couples $\mathrm{A}^\star$ and MRM trials, using Enc_π from Section 2.4 to preserve message freshness.

Theorem 2 (Antrag-MRM). The auxiliary signer $\mathrm{A}^\star$ and MRM use the same ideal response sampler and filter, with independent private tapes. Consider an MRM adversary running in time t and making at most $Q_S \geq 1$ signing queries. Let Q_0, Q_1 bound its distinct queries to $H_{0,\pi,h}, H_{1,\pi,h}$. The adversary has MRM signing access only under the challenge key. For an integer $C_S \geq Q_S$, set

$$\begin{gathered} D_\rho = q^{\gamma_{\mathrm{MR}}}, \qquad D_s = 2^{\ell_{\mathrm{salt}}}, \qquad D_0 = q^{\lambda_{\mathrm{MR}}}, \\ A_C = Q_0 + C_S, \qquad K_C = Q_0 C_S + \binom{C_S}{2}, \\ t' = t + \widetilde{O}(Q_0 + Q_1 + C_S). \end{gathered}$$

If $p_\pi > 0$ and $A_C + 1 \leq \min\{D_\rho, D_s\}$, there is an adversary against $\mathrm{A}^\star$ satisfying

$$\begin{aligned} \mathrm{Adv}_{\text{A-MRM}}^{\text{UF-CMA}} \leq & \; \mathrm{Adv}_{\mathrm{A}^\star}^{\text{UF-CMA}}(t', Q_0 + Q_S, Q_S) + \Delta_{\mathrm{rep}}(C_S) \\ & + K_C \left((D_\rho - A_C)^{-1} + (D_s - A_C)^{-1} \right) \\ & + \frac{(Q_0 + 2C_S)(Q_1 + 1) + \binom{A_C}{2} + 1}{D_0}. \end{aligned} \tag{7}$$

3.4 Certificate composition

The certificate construction of [3] realizes IBS by composing two independent signature instances. The PKG publishes $\mathsf{pk}_{\mathrm{PKG}}$ as the master public key and retains $\mathsf{sk}_{\mathrm{PKG}}$. In the escrowed construction, it generates one independent user pair $(\mathsf{pk}_U, \mathsf{sk}_U)$ per identity, certifies the public key, and delivers the secret key over a protected channel.

For an identity id and byte string x, let $\mathrm{Bind}(\mathrm{id}, x)$ injectively encode the component profiles, length-prefixed identity, and x in a separate domain. An ordinary certificate signs $\mathrm{Bind}(\mathrm{id}, \mathsf{pk}_U)$.

For MRM, split the coefficients of the user key: $\mathsf{pk}_U^{(r)}$ holds the first $\min\{N_U, \mu\}$, with N_U the user's degree and μ of the PKG profile, and $\mathsf{pk}_U^{(e)}$

packs the rest. The PKG signs $\mathsf{pk}_U^{(r)}$, zero-padded to μ coefficients, as the recoverable field and $\mathrm{Bind}(\mathrm{id}, \mathsf{pk}_U^{(e)})$ as the explicit field. Write the certificate as σ_{PKG} and the user's signature on a message as σ_U. The complete IBS signature is

$$\sigma_{\mathrm{SC}} = \mathsf{pk}_U^{(e)} \| \sigma_{\mathrm{PKG}} \| \sigma_U,$$

where $\mathsf{pk}_U^{(r)}$ is empty for an ordinary certificate. The component profiles fix all field lengths. Given the identity and master public key, verification authenticates the certificate, reconstructs and validates the canonical user key, and verifies σ_U under that key. Recovery rejects nonzero padding and non-canonical explicit coefficients. The verifier thus needs no separately authenticated user-key directory.

The security experiment maintains at most one user-key record per identity, answers signing queries, and reveals secret keys on corruption. A forgery must authenticate a new complete application message for an identity with no corrupted record. Denote the composition, selected certificate scheme, and user-signature scheme by SC, Cert, and User, respectively.

Theorem 3 (composition). Let Q_C bound the number of identities whose user keys the PKG generates and certifies. In the escrowed experiment, with domain-separated component oracles,

$$\mathrm{Adv}_{\mathrm{SC}}^{\text{UF-CMA}} \le \mathrm{Adv}_{\mathrm{Cert}}^{\text{UF-CMA}} + Q_C \, \mathrm{Adv}_{\mathrm{User}}^{\text{UF-CMA}}. \tag{8}$$

The certificate term uses the issuance budget, and the user term a uniform per-record signing budget. Both include simulation of the other component and its independent oracle domains. When both component premises cover the observations defined at the start of Section 3, the composition bound includes them. At fixed resources, the factor Q_C reduces the user term's success-probability exponent by $\log_2 Q_C$.

3.5 Parameters and concrete interpretation

Table 1. Antrag parameters. All profiles have $q = 12289$ and $r_{\mathrm{in}} = 1.315$; v is in bytes. The labels target NIST security Levels 1, 3, and 5.

Profile	N	α	(b, L, v)	$(\lambda_{\mathrm{MR}}, \gamma_{\mathrm{MR}})$	Salt bits
1f	512	1.15	$(7, 10, 619)$	$(19, 15)$	192
3f	768	1.19	$(7, 10, 948)$	$(29, 19)$	256
5f	1024	1.23	$(7, 11, 1245)$	$(38, 24)$	320

Choose $(\varepsilon_p, \varepsilon_u) = (2^{-39}, 2^{-38})$ for 1f and $(2^{-38}, 2^{-37})$ for 3f and 5f. Both normalized smoothing bounds are below 1.311735 in the power-of-two profiles and 1.306168 in degree 768. Since $Q_{\mathrm{hyb}}(B) \le \alpha\sqrt{q}$, $r_{\mathrm{in}} = 1.315$ and $\sigma_{\mathrm{sig}} = 1.32\,\alpha\sqrt{q}$ satisfy Eq. (3) uniformly. At $Q_S = 2^{64}$ and $C_S = 2^{65}$, the proved acceptance bound gives $\Delta_{\mathrm{rep}} < 2^{-1024}$.

Table 2 reports the random short integer solution (SIS) model of [12] and, as the selected cost, the smaller of that estimate and the annular key-recovery estimate of [11]. The SIS model uses an unstructured q-ary lattice in dimension $2N$ and determinant q^N as a heuristic model for the structured multi-target problem. Normalizing the degree-768 canonical covolume to q^N gives the effective radius $B_{\mathrm{eff}} = (4/3)^{1/4}\beta_{\mathrm{sig}}$; the power-of-two profiles use $B_{\mathrm{eff}} = \beta_{\mathrm{sig}}$ (Appendix B.1).

Table 2. Heuristic lattice costs in classical/quantum bits, and the illustrative ideal-signer exponent b_{cal} of Eq. (6) at $C_S = 2^{65}$.

Profile	B_{eff}/q	Random SIS	Selected cost	b_{cal}
1f	0.457	122/111	122/111	121.172/110.210
3f	0.622	195/177	195/177	192.909/175.008
5f	0.691	277/252	264/240	261.566/237.679

The selected cost is the smaller of the random-SIS and direct key-recovery estimates [11, Tables 2 and 3(a)], assuming completion and rejection preserve the modeled hardness. These costs guide parameter selection; Assumption 1 separately stipulates the multi-target success bound used in the reduction.

For a success-probability exponent $\lambda > 0$, if $\mathrm{Adv}_{\pi}^{m\text{-A-R-ISIS}} \le 2^{-\lambda}$ at the stated resources, the leading term in Eq. (6) is bounded by $2^{-b_{\mathrm{cal}}}$, where $b_{\mathrm{cal}} = \theta_u(\lambda - C_S \log_2 r_u(a_u))$. Table 2 substitutes the selected cost exponents for λ and optimizes the integer order $2 \le a_u \le 512$. Appendix B.2 gives the conditional work-to-success interpretation. The full ideal-signer bound also includes collisions, exhaustion, and composition; Section 4.3 gives the additional loss of the specified sampler and its finite arithmetic.

Salt lengths follow $\ell_{\mathrm{salt}} = \lambda_{\mathrm{target}} + 64$ for targets 128, 192, and 256, following Falcon's salt rule $k = \lambda + \log_2 Q_S$ [13, Section 2.2.2], read in the work-to-success convention of [12, Section 5.4]. Falcon uses 320 bits at every level for API uniformity. Appendix B.2 derives the query-cost normalization and the allowance for the factor-two trial cap.

We select $q = 12289$ rather than $q = 3329$. At $q = 3329$ and $N = 512$, the repetition-rate model of [11, Section 4.1] gives about four annular candidates per key at $\alpha = 1.29$ (their Table 2 lists 1.23); this value of α lowers the random-SIS estimate to 117/107 classical/quantum bits. At $N = 768$ and 1024, the small-modulus models of [7,1] give at most 178 and 254 classical bits, below the Level 3 and 5 targets. Weak-smoothness analyses permit narrower sampling widths for Falcon-type signatures [31,18] and might restore these targets at $q = 3329$.

4 Fixed-Point Antrag

4.1 Arithmetic, key admission, preparation, and side channels

The fixed-point realization uses the certified arithmetic framework of [4, Appendix B] and the parameters of Table 1. For an integer fractional precision $t \geq 0$, an integer z in Q_t represents $z2^{-t}$; products use widened integer types and specified rounding points. The notation $Q_{m.f}$ specifies f fractional bits and absolute value below 2^m (Appendix E). For an integer bit length $\ell \geq 1$, U_ℓ denotes an independent uniform integer in $\{0, \ldots, 2^\ell - 1\}$. The distributional analysis uses independent records of these words; Appendix C.1 specifies their cryptographic expansion and consumption. Offline directed-interval certificates at 384-bit precision, cross-checked at 512 bits, bound table, rounding, evaluator, acceptance and count errors; tests check implementation conformance.

Key generation evaluates each candidate once in directed Q_{64} intervals and rejects any undetermined rounding, OddSum, or annular decision. This cutoff belongs to $\mathcal{K}_\pi$ (Appendix C.2); Appendix B.3 estimates, without certifying, its effect relative to exact-precision admission. Write K_π^{fx} and K_π^{ref} for the finite and reference generators and ω_{kg} for their common random tape. Under the same oracle environment, the transfer assumes

$$K_\pi^{\mathrm{fx}}(\omega_{\mathrm{kg}}) \neq \perp \implies K_\pi^{\mathrm{fx}}(\omega_{\mathrm{kg}}) = K_\pi^{\mathrm{ref}}(\omega_{\mathrm{kg}}). \tag{9}$$

For a set $\mathcal{E}$ of keys, let $\mu_\pi^{\mathrm{fx}}(\mathcal{E})$ and $\mu_\pi^{\mathrm{ref}}(\mathcal{E})$ be the unconditioned probabilities of returning a key in $\mathcal{E}$. Eq. (9) gives $\mu_\pi^{\mathrm{fx}}(\mathcal{E}) \leq \mu_\pi^{\mathrm{ref}}(\mathcal{E})$. Arithmetic-capacity and resource failures terminate the experiment, allowing this coupling to extend to adaptive key-generation calls. This coupling requires exact implementation of the reference integer graph up to terminal failure. The leakage transfer assumes that observable execution is determined by the declared trial decisions, proposal counts and public refill policy; arithmetic bounds and implementation tests support these premises (Appendix C.2).

Cholesky preparation. $\mathsf{PreKey}^{\mathrm{ch}}$ is evaluated in a single Q_{96} midpoint-radius pass, with signed 128-bit centers and 64-bit radii. It encloses u, rounds the coefficient centers to Q_{16}, and recomputes the embeddings of the resulting exact rational $\hat{u}$. It then encloses the lower Cholesky factor of Σ_p at every embedding, transforms it back to coefficients, and rounds each center to the nearest multiple of p_A^{-1}, with ties to even. The stored error is therefore at most half an output-grid unit plus the Q_{96} radius allowance; we do not assume that the mathematical Cholesky factor is rounded exactly. With radius limits 2^{-56} for u and 2^{-46}, 2^{-44}, 2^{-46} for A in 1f, 3f, 5f, the transfer uses the effective precisions

$$p'_u = \frac{2^{16}}{1 + 2^{-39}}, \qquad p'_A = \frac{p_A}{1 + 2^{a_A - 45}} \text{ (1f, 5f)}, \qquad p'_A = \frac{2^{44}}{3} \text{ (3f)},$$

where $p_A = 2^{a_A}$. Integer majorants derived from annularity and Eq. (1) bound every interval radius for all admitted keys, and runtime guards check them, so preparation adds no rejection predicate to $\mathcal{K}_\pi$. Let $\beta(p_u)$ bound $\|\hat{B}\|$ for every

annular key of quality α [32, Lemma 2]. Directed arithmetic gives $\tilde{s}^2 - \beta(p'_u)^2 >$ 123.59, 131.39, and 140.83 for 1f, 3f, and 5f, so Σ_p is positive definite at every embedding with a wide margin and every Cholesky radicand exceeds 100. The exact integer maps for the sampling centers are computed with fixed-schedule NTTs and CRT reconstruction.

Crown preparation. $\mathsf{PreKey}^{\mathrm{cr}}$ uses the same Q_{96} interval arithmetic. In 1f/5f, coefficients round to Q_{64} and residual widths to Q_{62}. In 3f, both round to Q_{72} and then narrow to Q_{64}. Appendix E bounds these errors for every admitted key, so preparation adds no rejection.

Key admission. Desktop secret-key admission checks the NTRU equation, completion norm, coefficient bounds and annular quality; preparation reuses this validation. Compact preparation assumes an already valid key, with optional relation and norm checks at loading. Security assumes keys drawn from $\mathcal{K}_\pi$.

Side channels. The execution model requires secret-dependent arithmetic to use schedules and memory-access patterns determined by public parameters. Key generation combines branch-free interval operations, masked table scans, and fixed-schedule NTTs for $h = g/f$; the degree-768 transform modulo q is an incomplete NTT. Preparation uses fixed-schedule Newton reciprocals, guarded square roots, and transforms. By default, signing uses full scans of the sampling tables with masked selection, complete-word acceptance tests, fixed-schedule center transforms (Section 4.2), and center conversion without secret-dependent branches or divisions. The signing filter evaluates the norm, both magnitude bounds, and the codec lengths required by the mode before branching on their combined result. Component taint tests cover x86-64/AArch64 samplers and Cortex-M4 key generators; the latter also pass basic-block trace checks. Whole-program conformance of the measured desktop builds remains an implementation obligation.

Sections 3 and 4.2 account for the variable numbers of signing trials and online proposals. In reference key generation with independent candidate tapes, discarded candidates and their count are independent of the returned key; the execution premise requires completion of the accepted candidate to follow the public schedules of Appendix C.2. Fixed-point and exact integer arithmetic eliminate the conversion leakage of [33]. Protection against power or electromagnetic leakage, including the Hamming weight of center integer parts, would require masking, which this implementation does not provide.

4.2 Sampling and observable counts

Sampling proceeds in two phases. Phase 1 draws the perturbation, which does not depend on the target: the Cholesky signer draws an integer vector v from fixed tables, and the Crown signer draws grid approximations to continuous Gaussian perturbations. The perturbation and online draws read separate streams (Appendix C.1), so Phase 1 can run first, even though Figure 9 interleaves the crown draws, and offline if its tape is not derived from the message. Phase 2 runs online, once the target is known. It forms a sampling center from the perturbation and the target, draws a ring Gaussian there by rejection sampling, and repeats

for a second center that also depends on the first draw. The Cholesky signer computes the centers with exact integer arithmetic and the Crown signer with fixed-point Fourier arithmetic; both use the same proposal tables and complete-word comparisons.

Phase 1, Cholesky signer: perturbation. In the power-of-two profiles, each coordinate of v approximates an independent draw from $D_{\mathbb{Z},L_{\mathrm{p}}r_{\mathrm{in}}}$: a magnitude from a 96-bit cumulative table with cutoff 28 (1f) or 29 (5f), and an independent sign. Positive-magnitude masses are rounded down and the remainder is assigned to zero; directed 384- and 512-bit arithmetic generates the tables. In degree 768, a canonical coefficient pair (a, b) has squared norm $a^2 + ab + b^2$. Writing $b = 2\mathsf{k} + e$ and $a = \mathsf{t} - \mathsf{k}$ with a parity bit e gives

$$a^2 + ab + b^2 = (\mathsf{t} + e/2)^2 + 3(\mathsf{k} + e/2)^2.$$

The sampler draws e with a 96-bit approximation of its theta-weighted probability, then t and k from fixed tables at widths $L_{\mathrm{p}}r_{\mathrm{in}}$ and $L_{\mathrm{p}}r_{\mathrm{in}}/\sqrt{3}$ and centers $-e/2$; table selection is masked. Every perturbation coordinate or pair consumes a fixed-length record and reveals no count.

Phase 1, Crown signer: perturbation. The Crown signer draws each real perturbation coordinate from a rejection sampler for $\mathcal{N}(0, 512)$ on the 2^{-53} grid in $(-320, 320)$ and scales it by $\omega\tau_{i,j}$. Its proposal is a step-function envelope of the Gaussian density, which we call the crown: a draw selects one of 80 public blocks of width 4 with a tabulated probability, then a uniform grid point within the block and a sign. The rejection law is independent of the key.

Phase 2, Cholesky signer: exact centers. The centers combine the perturbation with the target through exact integer maps; the terms in c carry the target and the term in x_2 the first draw. Following [32, Section 4.3], put $d_p = L_{\mathrm{p}}p_A$, $u_{\mathrm{int}} = p_u\hat{u}$, and $A_{\mathrm{int}} = p_A\hat{A}$ with entries A_{00}, A_{10}, A_{11}. Preparation computes the integer ring elements

$$\begin{aligned} m_{11} &= u_{\mathrm{int}}f - p_uF, & m_{12} &= u_{\mathrm{int}}g - p_uG, \\ C_{10} &= m_{11}A_{00} + m_{12}A_{10}, & C_{11} &= m_{12}A_{11}, \\ C_{20} &= fA_{00} + gA_{10}, & C_{21} &= gA_{11}, \end{aligned}$$

and splits each coefficient as $C = d_pC^{\mathrm{hi}} + C^{\mathrm{lo}}$ with $-d_p/2 \le C^{\mathrm{lo}} < d_p/2$. With $H_2 = fc - C_{20}^{\mathrm{hi}}v_0 - C_{21}^{\mathrm{hi}}v_1$, $\Lambda_2 = -C_{20}^{\mathrm{lo}}v_0 - C_{21}^{\mathrm{lo}}v_1$, $H_1 = m_{11}c - C_{10}^{\mathrm{hi}}v_0 - C_{11}^{\mathrm{hi}}v_1 - qu_{\mathrm{int}}x_2$, and $\Lambda_1 = -C_{10}^{\mathrm{lo}}v_0 - C_{11}^{\mathrm{lo}}v_1$, the two sampling centers of $\mathsf{PreSmp}^{\mathrm{ch}}$ are

$$\hat{c}_2 = \frac{d_pH_2 + \Lambda_2}{qd_p}, \qquad \hat{c}_1 - \hat{u}x_2 = \frac{d_pH_1 + \Lambda_1}{qd_pp_u}.$$

Products use NTTs modulo the 31-bit primes $p_0 = 2147346433$ and $p_1 = 2147309569$, both congruent to 1 modulo 18432 so that they serve all three rings, with centered CRT reconstruction; H_2 needs only p_0. The carrier bounds follow from annularity and Eq. (1), without a coefficientwise size-reduction premise. In 1f, 3f, and 5f they give $\|H_2\|_\infty < 2^{25.4}$, $2^{26.9}$, and $2^{26.0}$; $\|H_1\|_\infty < 2^{49.7}$, $2^{52.7}$, and $2^{51.6}$; and $\|\Lambda_j\|_\infty < 2^{57.9}$, $2^{60.7}$, and $2^{59.9}$, respectively. All are below the

centered CRT magnitude bound $p_0 p_1/2 \approx 2^{61}$. Only the fractional part of each center is rounded down to Q_{64}; zero fractions stay zero. Appendix C.6 bounds the reconstructed responses.

Phase 2, Crown signer: centers. The Crown signer computes its centers in fixed-point Fourier arithmetic from the prepared projections, with the formats of Appendix E; Appendix C.5 bounds the crown evaluator error, the replacement of grid outputs by continuous perturbations, and the center errors.

Phase 2, both signers: Gaussian sampling. The signers share two proposal tables, for widths r_{in} and $2r_{\mathrm{in}}/\sqrt{3}$; the power-of-two profiles use only the first. Set $A_0 = \lfloor 2^{64}/(2r_{\mathrm{in}}^2) \rfloor$ and $A_1 = \lfloor 3 \cdot 2^{64}/(8r_{\mathrm{in}}^2) \rfloor$. For $i = 0, 1$, define the normalizer $Z_i = \sum_{j \geq 0} e^{-A_i j^2/2^{64}}$ and the proposal law $P_i(j) = e^{-A_i j^2/2^{64}}/Z_i$ on nonnegative integers j. A proposal selector and a sign bit determine a scalar candidate, and the acceptance test cancels the proposal weight to leave the target Gaussian mass; in degree 768, one joint test cancels both weights and gives the canonical pair law without a separate center-dependent normalization. The finite tables use 96-bit masses with supports $0 \ldots 15$ and $0 \ldots 17$. For the Cholesky signer in 1f and 5f, the acceptance probability is evaluated with a two-floor Q_{64} exponent and the degree-12 Q_{63} exponential polynomial of fixed-point Falcon [6], without a common acceptance scale; 3f uses a degree-20 Q_{64} pair evaluator (Appendix C.3).

Acceptance tests. Every online scalar or pair proposal consumes all fields of its Table 8 record, including a 56-bit acceptance word. The 1f/5f scalar evaluator forms the 56-bit threshold directly; the 3f pair evaluator returns W on the 2^{96} probability scale and uses $W \gg 40$. Both comparisons follow a fixed integer schedule; Lemma C.1 bounds threshold rounding. Proposal acceptance probabilities exceed 0.767 for Cholesky scalars and 0.60 for pairs, and a cap of 1024 proposals per requested output abandons the trial on exhaustion. A crown proposal uses a 72-bit acceptance word with threshold $W \gg 24$, accepts with probability above 0.92, and is capped at 256 proposals per output; the Crown signer's scalar proposals carry a public acceptance scale (Appendix C.3) and accept with probability above 0.745.

Observable counts. Fix a scalar or pair center. Let T count proposals through the first acceptance and Z be the accepted output; let w_z be the probability that one proposal accepts and returns z, and put $p = \sum_z w_z$. Independent proposals give $\Pr[T = k, Z = z] = (1-p)^{k-1} w_z = \Pr[T = k] \Pr[Z = z]$, so output and count are independent conditional on the center, even when conditioning on success within a cap, while p may depend on the center. Write $G_p(k) = p(1-p)^{k-1}$ for the geometric law on $k \geq 1$. Each reference trial augments the ideal response sampler with $2N$ scalar counts, or N pair counts in degree 768, drawn independently from G_{p_*} for a public $p_* \in (0, 1)$ fixed by signer and profile (Appendix C.3); for the Crown signer it also carries $2N$ independent crown counts from their public uncapped implementation law. These auxiliary counts are independent of the key, target, and samples; a public buffer policy maps records to refill positions (Appendix C.1). Before the cap, an implementation count follows G_p, whose order-a divergence from the reference law

is

$$\log_2 R_a(G_p \| G_{p_*}) = \frac{1}{a-1} \log_2 \frac{p^a / p_*^{a-1}}{1-(1-p)^a/(1-p_*)^{a-1}}. \tag{10}$$

The denominator is positive on an acceptance interval if it is positive at its lower endpoint, and in the parameter $\log(1-p)$ the divergence decreases toward p_* and increases away from it, so its maximum is attained at an endpoint. Uniform conditional bounds compose adaptively, including refills and batch boundaries determined by the counts; evaluating Eq. (10) at the certified endpoints gives the count stage of the transfer (Appendix C.3).

4.3 Transfer to fixed-point signing

For each signer, the transfer compares three levels: the ideal coset-Gaussian signer, the exact perturbation sampler (Lemma 2 or Lemma 3), and the finite implementation. The reference experiment is the stopped signing experiment with ideal records.

Stage composition and stopped view. Order the stages from the ideal signer toward the implementation, so that stage 1 is the sampler construction (Lemma 2 or Lemma 3) and the last stage is the observable counts; each stage compares the law after it with the law before it. For stages $i = 1, \ldots, m$ with Rényi orders $a_i > 1$ and lifetime bounds D_i on $\log_2 R_{a_i}$, put $\theta_i = (a_i - 1)/a_i$ and initialize $\Theta = 1$, $\Gamma = 0$. Each stage updates

$$\Theta \leftarrow \theta_i \Theta, \qquad \Gamma \leftarrow \theta_i(\Gamma + D_i).$$

Successive applications of Hölder's inequality give this recurrence. Appendices C.4 and C.5 list the orders and lifetime bounds.

Fix a deterministic trial budget $C_S \geq Q_S$. In the stopped signing view, both experiments halt with a public flag before sampling trial $C_S + 1$ or at a freshness

Table 3. Transfer losses in bits for a reference-event bound $2^{-b_{\mathrm{ref}}}$, at $Q_S = 2^{64}$ and $C_S = 2^{65}$, rounded upward. *Construction* gives the sampler construction alone: Lemma 2 for the Cholesky signer and Lemma 3 for the Crown signer. *Complete transfer* adds the finite tables, center errors, evaluators, acceptance words, and observable counts. Here b_{ref} is an input parameter; the loss increases with b_{ref}, so larger values are conservative. The loss columns quantify the multiplicative factor in Eq. (11); additive failures are bounded separately by $\varepsilon_{\mathrm{joint}}$.

Signer	Profile	b_{ref}	Construction	Complete transfer
Cholesky	1f	128	0.621024	0.649243
Cholesky	3f	256	1.812961	1.813429
Cholesky	5f	256	1.720724	1.776978
Crown	1f	128	0.424309	0.898876
Crown	3f	256	1.199714	1.797804
Crown	5f	256	1.199714	2.464213

event charged in Theorems 1 and 2. For ordinary signing, these are repeated trial salts, new trial salts matching the salt field of an earlier adversarial hash query, and later discoveries of unpublished trial salts. MRM also includes the corresponding nonce events, collisions with earlier prefixes or occupied $H_{1,\pi,h}$ cells, and discoveries of erased rejected-trial prefixes. Surviving these checks does not make a syndrome uniform; the stopped view keeps the transcript conditioned on the earlier events. Let $\varepsilon_{\mathrm{joint}}$ bound the probability that the finite experiment reaches a local proposal cap, the per-signature limit of 272 trials, an arithmetic guard, or MRM nonce-decoder exhaustion (Appendix C.1) before the reference experiment stops. Then, for every event E in the stopped view,

$$\Pr[E_{\mathrm{fx}}] \le 2^{\Gamma} \Pr[E_{\mathrm{ref}}]^{\Theta} + \varepsilon_{\mathrm{joint}}. \tag{11}$$

Section 3 bounds the ideal signer. For a reference-event bound $2^{-b_{\mathrm{ref}}}$, with $b_{\mathrm{ref}} \ge 0$, Eq. (11) gives a multiplicative loss of $(1-\Theta)b_{\mathrm{ref}}+\Gamma$ bits. Table 3 evaluates it at $C_S = 2Q_S$ under independent ideal records, including the acceptance words and observable counts. For 5f, applying the transfer to Table 2's classical calibration requires evaluating the loss at $b_{\mathrm{ref}} = 261.566$. With the specified outer cap of 272 trials, Appendix C bounds $\varepsilon_{\mathrm{joint}} < 2^{-276}$ at these budgets; the cap term dominates. For forgery, apply the transfer to $F \cup B$, where F is an unflagged forgery and B a charged freshness or global-trial stop. The proofs of Theorems 1 and 2 bound this reference event, and Corollary 1 composes the transferred bounds.

Cholesky signer. The first stage compares the exact Cholesky sampler, with the prepared $\hat{u}$ and $\hat{A}$ but exact Gaussian draws and exact centers, with the ideal coset Gaussian. For $e \in (0,1)$ write $\kappa(e) = (1+e)/(1-e)$. Let $\Sigma_e = \Sigma_p - \hat{A}\hat{A}^*$ be the covariance error of the rounded factor, and let $\bar{\tau} \ge \|\Sigma_e\|/\tilde{s}^2$ be its certified bound: in 1f and 5f the parameter $\bar{\tau}(p'_u, p'_A)$ of [32, Section 3.3], and in 3f the bound derived in Appendix C.4. It is below $3.86 \cdot 10^{-13}$, $1.14 \cdot 10^{-12}$, and $4.95 \cdot 10^{-13}$ in 1f, 3f, and 5f.

Lemma 2 (sampler construction [32, Theorem 1]). Let ε_p, $\varepsilon_v = 2^{-64}$, and $\varepsilon_\Lambda \le 1/2$ be the smoothing errors of the online normalizers, the perturbation, and the lattice mass, and put $\ell = 2\log\kappa(\varepsilon_p) + \log\kappa(\varepsilon_v)$. Let $a > 1$ satisfy $(2a-1)\bar{\tau} < 1$. If $\tilde{s} > \beta(p'_u)$ and the three smoothing conditions of [32, Theorem 1] hold for $(\varepsilon_p, \varepsilon_v, \varepsilon_\Lambda)$, then for every admitted key and every syndrome c, the exact sampler's response law $P_{B,c}$ satisfies

$$\log R_a\big(P_{B,c} \,\|\, D_{\mathcal{L}_h(c),\sigma_{\mathrm{sig}}}\big) \le \frac{a\ell^2}{4} + \frac{a\|\Sigma_e\|_F^2}{2\tilde{s}^4(1+\bar{\tau})(1-(2a-1)\bar{\tau})} + \frac{5a\bar{\tau}^2}{(1-(2a-1)\bar{\tau})^2}.$$

In the power-of-two profiles, [32, Lemma 3] gives

$$\|\Sigma_e\|_F \le \frac{2\sqrt{3}\,N\sin(\pi/(2N))}{1+\sqrt{5}}\,\bar{\tau}\tilde{s}^2,$$

so the middle term is at most

$$\frac{3aN^2\sin^2(\pi/(2N))\,\bar{\tau}^2}{(3+\sqrt{5})(1+\bar{\tau})(1-(2a-1)\bar{\tau})},$$

the form of [32, Theorem 1]; degree 768 uses the Frobenius bound of Appendix C.4.

The bound is Theorem 1 of [32] with separate smoothing errors for the two online normalizers and the perturbation, and with the effective precisions in place of exact rounding; Appendix C.4 adapts its proof to all three rings and gives the explicit form of the three smoothing conditions. We use $\varepsilon_\Lambda = 2^{-38}$ in 1f and 2^{-37} in 3f and 5f, with ε_p as in Section 3.5.

The remaining stages compare the finite sampler with this exact sampler: finite perturbation tables, rounding of the center fractions, the finite online proposal tables and evaluator, truncation to the acceptance words, and the observable counts of Eq. (10).

For the Cholesky signer, the construction stage dominates the loss. In degree 512, the construction-stage bound is 0.621024 bits, compared with a complete-transfer bound of 0.649243 bits (Table 3). The finite stages add at most 0.057 bits at any degree.

Crown signer. The Crown signer's transfer uses the same stopped view, freshness events, and stage recurrence. In addition to Eq. (3), the Crown signer's Hybrid-Sampler requires

$$\sigma_{\mathrm{sig}} > r_{\mathrm{in}} Q_{\mathrm{hyb}}(B), \tag{12}$$

which makes the residual variances $\tau_{i,j}^2$ positive. Since $Q_{\mathrm{hyb}}(B) \le \alpha\sqrt{q}$, the parameters of Table 1 satisfy it for every admitted key, with normalized residual variances at least $1.32^2 - 1.315^2 = 527/40000$.

Lemma 3 (two-stage comparison). Under Eqs. (3) and (12), for every admitted B, every $c \in R_q$, and every $s \in \mathcal{L}_h(c)$, the exact Hybrid-Sampler's response law $P_{B,c}^{\mathrm{hyb}}$ satisfies

$$\left(\frac{1-\varepsilon_p}{1+\varepsilon_p}\right)^2 \le \frac{P_{B,c}^{\mathrm{hyb}}(s)}{D_{\mathcal{L}_h(c),\sigma_{\mathrm{sig}}}(s)} \le \left(\frac{1+\varepsilon_p}{1-\varepsilon_p}\right)^2.$$

Appendix C.5 proves Lemma 3. Its proof writes the exact Hybrid-Sampler law as the ideal coset Gaussian reweighted by a factor whose logarithm ranges over an interval of length $\ell_{\mathrm{h}} = 2\ln\frac{1+\varepsilon_p}{1-\varepsilon_p}$. The bounded-log-weight argument of Eq. (C.3) therefore gives, uniformly over admitted keys, syndromes and preceding states,

$$D_{\mathrm{hyb}} = \frac{C_S a_{\mathrm{h}} \ell_{\mathrm{h}}^2}{8\ln 2}. \tag{13}$$

At order a_{h}, this replaces Lemma 2. Its finite stages cover the crown evaluator and grid lift, center errors, the discrete evaluator, acceptance rounding, and observable counts (Appendix C.5). At $Q_S = 2^{64}$, the Crown signer's complete loss exceeds the Cholesky signer's by 0.25 and 0.69 bits at degrees 512 and 1024, and is 0.02 bits smaller at degree 768. The Crown construction bound is smaller at every degree; its larger numerical contribution in the power-of-two rings accounts for the difference. These are comparisons of certified upper bounds, which depend on the analysis, not of the samplers' actual distances.

The one-trial total-variation bounds of Appendices C.4 and C.5 give acceptance above 0.57 and global-stop probability at most $\exp(-4Q_S(0.07)^2)$ at $C_S = 2Q_S$. Their sharper certificates bound rejection by 0.4200000002. Thus the 272-attempt cap contributes at most $Q_S(0.4200000002)^{272} < 2^{-276.418544}$ for $Q_S \le 2^{64}$; including local-cap, guard and nonce-decoder terms gives $\varepsilon_{\text{joint}} < 2^{-276}$.

4.4 Transfer to the identity-based composition

Let C and U index the certificate and user components; a subscript $j \in \{C, U\}$ on a transfer quantity selects that component's parameters and budget. Let $\varepsilon_{\text{tape}}$ bound the advantage of replacing the key-generation and signing expansion of Appendix C.1 by independent ideal records in the same public-oracle environment. Write SC, fx for the composition with finite components.

Corollary 1 (fixed-point composition). Suppose both Antrag components satisfy Eq. (9), the arithmetic and record specifications, and the trial and primitive-call budgets. Assume the hypotheses of Theorems 1–3 with the stopped signing view of Section 4.3, and terminate on operational key-generation failure. For each component j, let $S_j \in [0, 1]$ bound the union of an unflagged reference forgery and a freshness or global-trial stop, including the digest-composition term $\varepsilon_{\text{dig},j}$ of Appendix A.5 and truncating at one. For $S \in [0, 1]$, define $\mathcal{F}_j(S) = \min\{1, 2^{\Gamma_j} S^{\Theta_j} + \varepsilon_{\text{joint},j}\}$. Then

$$\mathrm{Adv}^{\text{UF-CMA}}_{\mathrm{SC},\mathrm{fx}} \le \mathcal{F}_C(S_C) + Q_C \mathcal{F}_U(S_U) + \varepsilon_{\text{tape}}. \tag{14}$$

The certificate term uses its issuance budget and the user term a uniform per-key budget. With several user profiles, take their maximum transferred bound.

5 Performance Evaluation and Comparison

5.1 Falcon comparison

At degrees 512 and 1024, the Falcon baseline adapts the Type 1 implementation of [6]. Its scalar sampler follows [17, full version, Algorithms 1 and 2], with a 72-bit proposal and a lazy 64-bit Bernoulli comparison. The analysis of [6] assumes global center and width error bounds and an idealized scalar sampler; part of the precision assessment is empirical [6, Sections 5.4 and 7.5]. The baseline uses $Q_{32.32}$ for degree-512 completion [26], Q_{80} F-only reduction at degree 1024, and $Q_{64.64}$ for the LDL tree and signing. The wider completion can change the accepted Gaussian-key distribution; both paths retain the fixed-point key-admission checks of [6].

All implementations draw sampling randomness from SHAKE256, which in Falcon replaces the reference ChaCha20 generator. On Apple M5, both use ARMv8 SHA3 assembly, Antrag expanding its four sampler streams in pairs and Falcon one sequential stream. On x86-64, Antrag prefills four streams with an AVX2 permutation; Falcon uses XKCP's fully unrolled scalar C permutation

[19] for one sequential stream. The comparison includes random-stream generation.

The optimized Falcon-512/1024 baseline in Table 5[1] caches mirrored leaf widths and uses vector NTTs for key generation and exact response reconstruction. Preparation reuses the decoded integer basis; sampling skips unused final FFT merges. Its public interface uses NTT-form keys and public-key-bound message hashing, sharing verification kernels and decoders with Antrag. Hash-to-point expansion is batched, and the 72-bit proposal scan uses NEON on M5 and scalar code on x86-64. These arithmetic optimizations preserve fixed-point values and sampler-byte consumption for a fixed input stream; MRM follows [15]. The measured interface bounds both response components by 840 in magnitude and caps signing at 27 attempts.

At degrees 512 and 1024, Antrag key generation on desktop and Cortex-M4 adapts Pornin's F-only reduction [26, Section 2], approximating F/f in the Fourier domain. Antrag's annularity bounds the combined energy of f and g, so individual components of f can be too small for Falcon's $Q_{32.32}$ quotient arithmetic. We retain all integer limbs in the conversion, use Q_{80} transforms, and scale the reciprocal by 2^{-48} to accommodate small Fourier components. The reduction uses 20-bit steps. Antrag and the Falcon baseline both use integer AVX2 NTT kernels for key generation on i7 and NEON NTTs on M5. Degree 768 uses mixed-degree F-only completion on both platforms, followed by the final joint reduction of Appendix C.2. Appendix E lists the arithmetic formats of both Antrag signers.

Falcon-512/1024 and Antrag's Cholesky signer reconstruct the response products exactly by NTT multiplication modulo a single auxiliary prime followed by a centered lift: $p_{\mathrm{NTT}} = 2147473409$ for Falcon and $p_0 = 2147346433$ for the Cholesky signer. The Crown signer also uses p_0, with the bounds of Appendix C.6. Falcon's prepared-key product bounds of 192,537,600 and 575,490,048 at degrees 512 and 1024 follow a worst-case recursion over its sampling tree. The Cholesky signer's response coefficients stay below 102,979, 407,631, and 166,328 at degrees 512, 768, and 1024, because its perturbation and proposal supports are finite. All lie below half of the respective prime, ensuring unique recovery (Appendix C.6).

Falcon-768 is an experiment included for comparison, not a proposal of this paper. It uses Gaussian key candidates in the ring $\mathbb{Z}[X]/(X^{768} - X^{384} + 1)$, with $q = 12289$, coefficient standard deviation 3.309394, signing width $1.32 \cdot 1.17\sqrt{q} \approx 171.205639$, squared response bound 48883555 (tail factor 1.042, as for Antrag), and a 40-byte salt. Key coefficients follow an exact 16-bit cumulative table of the Gaussian law (cutoff 14). The NTRU completion is one code for the desktop and the Cortex-M4: exact norms (31-bit residues, number-theoretic transforms of the trinomial rings), Q_{80} Babai quotients with exact updates on the mixed tower $768 \to 256 \to \cdots \to 2 \to 1$, and ntrugen's two-prime top level; an accepted key satisfies the NTRU equation by a congruence-and-bound argument. It feeds a recursive sampler with Q_{96} preparation, Q_{63} internal coefficients, Q_{64} centers,

[1] Source code: `https://github.com/OlymSec/fxp-fndsa`.

Table 4. Ordinary and complete identity-based signature sizes in bytes. The PKG certificate embeds μ user-key coefficients, which saves r_{pk} key bytes.

Component	N	User key	Ordinary signature	Total with ordinary cert.	r_{pk}	Total with MRM cert.
Antrag	512	896	643	2182	836	1941
Falcon	512	897	666	2229	836	1978
Antrag	768	1344	980	3304	1260	2960
Antrag	1024	1792	1285	4362	1683	3884
Falcon	1024	1793	1280	4353	1683	3869

complete 56-bit acceptance words, and exact two-prime reconstruction. As the 512/1024 baseline derives its right subtree from the left, preparation computes one LDL branch and defines the other as its mirror: reversed children, negated coefficients, one product per top-level $(0, 2)$ coefficient, and widths K_t/w. Response decoding uses buffered reads and leading-zero counts. Its parameter security and numerical transfer remain uncertified.

Falcon-512/1024's lazy Bernoulli test [17, full version, Algorithm 2] compares uniform bytes with a 64-bit threshold until the first unequal byte or the eighth byte. Its comparison length has a threshold-independent marginal law, but can depend on the returned scalar after conditioning on acceptance. Under fixed positive costs per proposal and compared byte, Appendix D gives a counterexample to the perfect output isochrony claimed in [17, full version, Section 6]. The effect on complete Falcon signatures remains to be analyzed. Antrag compares a complete acceptance word per proposal on a fixed schedule (Section 4.2), making each proposal's work independent of its outcome; Eq. (11) accounts for its observable proposal counts.

5.2 Signature sizes

Let $|x|$ denote byte length. Table 4 uses the same profile for the PKG and user. A user key packs N coefficients modulo q in 14 bits each. To save space, the MRM certificate embeds the first μ of them directly as its recoverable field M_0 and sends the rest in e_{pk} bytes: $r_{\mathrm{pk}} = |\mathsf{pk}_U| - e_{\mathrm{pk}}$ and $|\sigma_{\mathrm{SC}}| = |\sigma_{\mathrm{PKG}}| + |\mathsf{pk}_U| - r_{\mathrm{pk}} + |\sigma_U|$, where $e_{\mathrm{pk}} = \lceil 14(N-\mu)/8 \rceil$, plus Falcon's key-header byte. Even an optimal byte encoding through RecEnc recovers 25, 38, and 50 fewer bytes, respectively. Ordinary certificates have $r_{\mathrm{pk}} = 0$. Sizes omit Antrag's two-byte component and key headers: ordinary and MRM components occupy $v + \ell_{\mathrm{salt}}/8$ and $2v$ bytes. User signatures are ordinary; identity and component formats come from context. Falcon rows use the lengths of [13,15] and this μ.

At degree 512, the salt length accounts for 32 of the 47-byte difference with an ordinary certificate and 16 of the 37-byte difference with an MRM certificate; shorter responses and omitted headers account for the rest. At degree 1024, both schemes use 40-byte salts, and Antrag's larger response allocation makes its totals 9 and 15 bytes longer.

5.3 Desktop performance

Table 5 compares throughput on Apple M5 with Apple Clang 21 and on Intel Core i7-13700 with GCC 15.2. Both builds use `-O3` and native arithmetic and SHAKE instructions; Antrag uses no external arithmetic library. Turbo Boost was enabled on i7. Both Antrag signers use Q_{64} centers. PreKey measures core geometry and cached transforms from validated keys in preallocated storage. Signing and verification cover the public interfaces, with prepared signing keys and no verifier cache; Appendix F.1 gives the protocol. Cholesky's ordinary signing throughput is 1.54 and 1.66 times Falcon's at degrees 512 and 1024 on M5, and 1.85 and 2.00 times on i7. Cholesky core preparation takes 0.4–0.9 ms on M5 and 0.8–1.6 ms on i7, once per key using Q_{96} intervals. Both schemes serialize NTT-form public keys, so uncached verification uses two transforms. MRM hashes bind the serialized key.

Table 5. Desktop throughput (operations/s). PreKey measures core preparation; signing uses prepared keys and Q_{64} centers. Antrag supports $Q_S = 2^{64}$ (Table 3). Falcon-768 is a comparison experiment with Q_{63} internal coefficients (Section 5.1).

Key generation and verification (Antrag pooled)

Profile	M5 Keygen	M5 Verify	M5 MRM verify	i7 Keygen	i7 Verify	i7 MRM verify
1f	320.5	149368	99125	229.8	114117	85604
3f	15.05	107253	69006	14.47	84809	62656
5f	96.01	75221	49429	71.68	56730	42833
Falcon-512	448	143534	104699	619.3	112233	89558
Falcon-768	111.7	102853	73781	89.67	82070	64600
Falcon-1024	104.9	72031	52410	119.4	56023	45230

Preparation and prepared signing

Profile	Signer	M5 PreKey	M5 Sign	M5 MRM sign	i7 PreKey	i7 Sign	i7 MRM sign
1f	Cholesky	2361	13059	12147	1276	9953	9331
3f	Cholesky	1123	4862	4673	655	4251	4080
5f	Cholesky	1127	6899	6575	608	5211	4964
1f	Crown	7118	10805	10263	4205	6650	6331
3f	Crown	3640	3939	3825	2020	2702	2644
5f	Crown	3444	5433	5253	2010	3297	3208
Falcon-512	Falcon	8576	8475	8201	6240	5386	5214
Falcon-768	Falcon	2646	2978	2915	1777	2156	2121
Falcon-1024	Falcon	4450	4165	4052	2897	2605	2537

For the experimental degree-768 extension, Cholesky's ordinary signing throughput is 1.63 times Falcon-768's on M5 and 1.97 times on i7. The recursive extension costs more per signature than Falcon-1024 despite its smaller degree.

For Antrag, comparisons of 1,000 randomized key generations per configuration show that F-only completion increases key-generation throughput over full-(F, G) completion by 15–16% at degree 512 and 56–60% at degree 1024 across the two processors. The mixed-degree F-only path reduces complete degree-768 key-generation time by 23% on M5 and 27% on i7, retaining the fixed round cutoff. The gains arise from reduced arithmetic and, at power-of-two degrees, fewer rejected annular candidates.

Comparison of the two signers. The Cholesky signer raises ordinary signing throughput over Crown by factors of 1.21, 1.23, and 1.27 at degrees 512, 768, and 1024 on M5, and 1.50, 1.57, and 1.58 on i7. Crown's core preparation is 3.0–3.3 times as fast. With these costs, Cholesky amortizes the difference after about 8–18 ordinary signatures per key.

Comparison with [32]. We independently implemented the method of [32] (Appendix F.2). Using their parameters, ChaCha20 tapes, lazy acceptance, norm-only filtering, and variable-length encoding, prepared signing on the i7-13700 takes 71.8 μs at degree 512 and 141.0 μs at degree 1024, close to the respective published timings of 74.3 and 150.3 μs on an i7-1360P.

In the comparison harness of Appendix F.2 (Table 16, first row), the Cholesky signer takes 97.9/187.6 μs at degrees 512/1024 on the i7-13700 (GCC 15.2), using our parameters and AVX2 SHAKE256. Table 5 measures the public v1 interface; Table 16 separates the effects of parameters, filtering and encoding.

5.4 Embedded performance

Table 6 compares key generation, preparation, and signing for the two Antrag signers, the Falcon-768 comparison experiment (Section 5.1), and Pornin's fixed-point FN-DSA, c-fn-dsa-fx32.64 [27] (Type 1 32.64 in the terminology of [6]), on an STM32F407 Cortex-M4 at 84 MHz. All signers use SHAKE256 on the same XKCP Keccak-p[1600] permutation and Falcon-512/1024's lazy Bernoulli test, which, unlike the complete-word acceptance of Section 4.2, is variable-time. All builds use GCC 14.2.1 at `-O3` with link-time optimization. Signing uses prepared keys in flash and workspaces in SRAM; FN-DSA uses its expanded key. Appendix F.1 defines the memory accounting; code flash is a signing-only footprint.

Cholesky signs 1.20–1.30 times as fast as Crown and uses less code flash. At degrees 512 and 1024, it takes about 3% and 2% more cycles than FN-DSA, whereas Crown takes 1.33 and 1.31 times as many cycles. Cholesky draws twice as many discrete Gaussians per candidate, but its fixed-width online sampler uses fewer proposals: about 1.3 per coordinate compared with 1.7 for FN-DSA in the degree-512 profile. This offsets part of the additional perturbation cost. Each Cholesky coordinate also consumes a 13-byte perturbation record (Table 8), increasing random-tape expansion. With ChaCha20 tapes, Cholesky needs about 41% fewer signing cycles at degrees 512 and 1024 (below).

On these samples, Antrag's key generation takes 3.1 and 2.0 times FN-DSA's cycles at degrees 512 and 1024, including rejected candidates. On the same seeds, the F-only completion takes 8%, 18%, and 22% fewer cycles than the full-(F, G) solver (191, 868, and 755 Mcc at degrees 512, 768, and 1024). Preparation of an

Table 6. Cortex-M4 costs: signing memory in bytes and mean times in millions of core cycles (Mcc), including retries. Sign: 100 signatures. Keygen: 100/30/30 Antrag keys at degrees 512/768/1024, 100/30 FN-DSA keys at 512/1024, and 30 Falcon-768 keys; shared by both Antrag signers, with the F-only completion of Appendix C.2. Prep: five Antrag and five Falcon-768 keys; five expansions of one FN-DSA key.

Signer	N	Key flash	Code flash	SRAM	Keygen	Prep	Sign
Cholesky	512	46160	25926	17840	175	137	6.10
Cholesky	768	69136	35338	26516	711	416	14.81
Cholesky	1024	92112	30130	34216	590	317	12.37
Crown	512	27600	32880	17576	175	56.5	7.91
Crown	768	42544	49624	25696	711	139	17.76
Crown	1024	55248	36912	33000	590	118	15.87
FN-DSA, fx32.64	512	26729	32344	18999	57.3	1.68	5.93
FN-DSA, fx32.64	1024	56425	44632	36407	298	3.57	12.11
Falcon-768	768	43888	48968	43216	267	31.1	21.42

already valid key, once per key, takes 56.5–139 Mcc for Crown and 137–416 Mcc for Cholesky, against 1.7–3.6 Mcc for FN-DSA's key expansion.

Crown's prepared key is about the size of FN-DSA's expanded key; Cholesky's is 1.6–1.7 times as large. Both Antrag signers use 6–9% less SRAM than FN-DSA, and Cholesky uses 20–32% less code flash. The 32.64 signer of [16] reports 9.17/19.49 Mcc at degrees 512/1024 on an STM32L4R5 at 20 MHz, including dynamic LDL-tree construction during signing; Table 6 measures signing with prepared keys. Degree 768 (pair sampling) remains slower than degree 1024, as on the desktop. Falcon-768 signs in 1.45 and 1.21 times Cholesky's and Crown's cycles at degree 768 with 1.6 times Cholesky's SRAM (its 37,976-byte workspace holds two Q_{64} centers and the recursion's temporaries); its exact Q_{96} preparation takes 31.1 Mcc, under a quarter of Crown's, and its key generation 267 Mcc, 0.38 times Antrag's, with about 31 Gaussian candidates per key and the NTRU completion shared with its desktop implementation. Its first port measured 1,175, 33.6 and 21.41 Mcc with 81,712 bytes of SRAM; the shared completion, the 16-bit key law and the in-place, 16-bit signing storage brought the key generation to 0.23 times that and the SRAM to 0.53 times, at the same signing cost.

With ChaCha20 sampling tapes (Appendix F.1), the Cholesky signer takes 3.62 and 7.35 Mcc at degrees 512 and 1024, 9% and 6% fewer cycles than the 3.98 and 7.82 Mcc that Zhang et al. [32, Table 8] report for their fixed-point Mitaka on an STM32F411 at the same clock.

6 Conclusion

We connect Antrag's annular trapdoors to concrete security bounds and fixed-point signing at degrees 512, 768, and 1024. The ordinary and message-recovery reductions use A-R-ISIS for the specified key law. Certified transfers from the ideal signer cover both perturbation samplers, their numerical errors, observable counts, and failures; at 2^{64} signing queries and every degree, the losses are

below 1.82 bits for the Cholesky signer and 2.47 bits for the Crown signer at the reference-event bounds of Table 3. On Apple M5 and Intel Core i7-13700, prepared signing with the Cholesky signer at degrees 512 and 1024 is 1.54 to 2.00 times as fast as the fixed-point Falcon baseline, at the cost of slower key generation and core preparation; Crown's core preparation is 3.0 to 3.3 times as fast as Cholesky's. Message recovery in PKG certificates yields compact identity-based signatures. On Cortex-M4, the Cholesky signer comes within about 3% of Pornin's fixed-point FN-DSA, with less SRAM and code flash.

Limitations. The Q_{64} cutoff comparison uses statistical confidence bounds (Appendix B.3). The selected tail factor has a certified acceptance bound (Appendix A.3), while comparisons of tail-factor performance use a Monte Carlo model (Section 2.2). The implementation has no masking against power or electromagnetic analysis.

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A Proofs

A.1 Correctness and centered reconstruction

Proposition A.1. Every successful ordinary or MRM signing call verifies, provided its message fields have valid encodings.

Proof. The NTRU equation gives $s_0 + s_1 h = c \bmod q$. Since $s_0 \in \mathcal{B}_\pi$ and $2^L \le 2048 < q/2$, ordinary verification reconstructs the same s_0 and tests the same norm, including its degree-768 cross terms. MRM reverses the mask to recover (ρ, M_0) and checks the original hash equation. The injective payload encoding recovers the original bytes. □

A.2 Canonical sampling (Lemma 1)

Proof. Let $\zeta_{2304} = e^{2\pi \mathrm{i}/2304}$ be a primitive 2304th root of unity. For $R = \mathbb{Z}[\zeta_{2304}]$ and $0 \le j, k < N$, the normalized trace of ζ_{2304}^{j-k} is one for $j = k$, $1/2$ for $j - k = \pm 384$, and zero otherwise. Pairing positions $(j, j + 384)$ therefore gives the Gram blocks of Section 2.1. Their Gram–Schmidt lengths are $1, \sqrt{3}/2$, while the power-of-two basis is orthonormal. Applying the Gram–Schmidt basis bound [14, Lemma 3.1], derived from [23], in dimension N, with widths divided by $\sqrt{2\pi}$, proves the ring bound.

For the NTRU bound, set $\Delta_f = ff^* + gg^*$. The Gram–Schmidt vectors of Section 2.1 are

$$\widetilde{b}_1 = (g, -f), \qquad \widetilde{b}_2 = (qf^*/\Delta_f, qg^*/\Delta_f).$$

Their embedding lengths are $\sqrt{\nu_j}$ and $q/\sqrt{\nu_j}$. Expand the basis vectors in the order $(X^j b_1)_{j<N}, (X^j b_2)_{j<N}$. The resulting lattice has index q^N in R^2, since $\det B = q$. The surjective syndrome map shows that this lattice is Λ_h. The spans $\mathbb{K}b_1$ and $\mathbb{K}\widetilde{b}_2$ are canonically orthogonal. Multiplication by either Gram–Schmidt vector has operator norm bounded by its largest embedding length. Projecting the second block away from the first therefore bounds the real Gram–Schmidt norm by $Q_{\mathrm{hyb}}(B)$. The dimension-$2N$ basis bound proves Eq. (2).

For width $r > 0$ and a real pair center (η_0, η_1), define the Gaussian mass function, for $t \in \mathbb{R}$, by

$$\Theta_r(t) = \sum_{k\in\mathbb{Z}} e^{-(k-t)^2/(2r^2)}.$$

Draw $m_1 \leftarrow D_{\mathbb{Z},2r/\sqrt{3},\eta_1}$ and set $\mu_0 = \eta_0 - (m_1 - \eta_1)/2$. Accept with probability $\Theta_r(\mu_0)/\Theta_r(0)$; on rejection, redraw m_1 and recompute μ_0. After acceptance, draw $m_0 \leftarrow D_{\mathbb{Z},r,\mu_0}$. Poisson summation gives

$$\Theta_r(t) = \sqrt{2\pi}\, r\Big(1 + 2\sum_{n\ge 1} e^{-2\pi^2 r^2 n^2}\cos(2\pi n t)\Big) \le \Theta_r(0).$$

The product of proposal, acceptance, and conditional masses cancels $\Theta_r(\mu_0)$ and is proportional to $\exp\big(-((m_0-\eta_0)^2 + (m_0-\eta_0)(m_1-\eta_1) + (m_1-\eta_1)^2)/(2r^2)\big)$. Independent pairs give the ring Gaussian. At $r = r_{\mathrm{in}}$, bounding the positive Poisson tail by $e^{-2\pi^2 r^2}/(1 - e^{-6\pi^2 r^2})$ shows correction acceptance greater than $1 - 5.999 \cdot 10^{-15}$. □

A.3 Uniform signing acceptance

Fix an admitted basis B and public key h. Let $J_B(c, s)$ be the joint law of a uniform syndrome c and the ideal signer's response $s \leftarrow D_{\mathcal{L}_h(c),\sigma_{\mathrm{sig}}}$. Let J_W instead draw $s \leftarrow D^{\mathrm{can}}_{R^2,\sigma_{\mathrm{sig}}}$ and set $c = s_0 + s_1 h \bmod q$. Conditioned on c, both laws give s the coset Gaussian $D_{\mathcal{L}_h(c),\sigma_{\mathrm{sig}}}$, so $J_B(c,s)/J_W(c,s) = U(R_q)(c)/Q_{\mathrm{syn}}(c)$ and quotient smoothing (Section 3.1) gives $J_B(c,s) \ge \frac{1-\varepsilon_u}{1+\varepsilon_u} J_W(c,s)$.

Under J_W, write p_{codec} for the probability that one response polynomial fails its codec and p_{norm} for the probability that the response pair exceeds β_{sig}. For one polynomial put $K = 8v - N(b+2)$ and $w_b(z) = \lfloor |z|/2^b \rfloor$ for $z \in \mathbb{Z}$. In a power-of-two ring, define

$$Z_{\mathrm{code}}(t) = \sum_{z\in\mathbb{Z}} e^{-z^2/(2\sigma_{\mathrm{sig}}^2) + t w_b(z)}, \qquad t \in \mathbb{R}.$$

For $t > 0$ and $z \leftarrow D_{\mathbb{Z},\sigma_{\mathrm{sig}}}$, Chernoff's inequality and a union bound for the magnitude cutoff give

$$p_{\mathrm{codec}} \le e^{-t(K+1)} \left(\frac{Z_{\mathrm{code}}(t)}{Z_{\mathrm{code}}(0)}\right)^N + N \Pr[|z| \ge 2^L].$$

We use $t = 111/1000$ for 1f and $76/1000$ for 5f.

For 3f, let (m_0, m_1) be one zero-centered canonical coefficient pair of width σ_{sig}. Write p_{mag} for the probability that both magnitudes are below 2^L, and m_w, v_w for the conditional mean and variance of $w_b(m_0) + w_b(m_1)$. Set $d = K+1-Nm_w/2 > 0$, $W_{\max} = 2\lfloor(2^L-1)/2^b\rfloor$, and $M_w = \max\{m_w, W_{\max}-m_w\}$. Bernstein's inequality gives

$$p_{\mathrm{codec}} \le 1 - p_{\mathrm{mag}}^{N/2}\left(1 - \exp\left[-\frac{d^2}{2((N/2)v_w + M_w d/3)}\right]\right).$$

For all profiles, put $\gamma_{\mathrm{tail}} = \beta_{\mathrm{sig}}^2/(2N\sigma_{\mathrm{sig}}^2)$ and $\kappa_0 = (1+2^{-128})/(1-2^{-128})$. Ambient smoothing and a theta-mass moment bound give

$$p_{\mathrm{norm}} \le \kappa_0 e^{-N(\gamma_{\mathrm{tail}}-1-\log\gamma_{\mathrm{tail}})}, \qquad \Pr_{J_W}[\mathcal{A}_\pi] \ge (1-p_{\mathrm{codec}})^2 - p_{\mathrm{norm}}.$$

Directed 384-bit summation with analytic omitted-tail bounds yields

profile	p_{norm}	p_{codec}	$\Pr_{J_W}[\mathcal{A}_\pi]$
1f	< 0.168332	< 0.126527	> 0.594623
3f	< 0.069064	< 0.189060	> 0.588559
5f	< 0.028336	< 0.112702	> 0.758961

Multiplying by the pointwise factors and taking the infimum over keys proves $p_\pi > 0.58$.

A.4 Ordinary unforgeability (Theorem 1)

Proof. Stop on trial exhaustion or repeated trial salts. Also stop if a new trial salt matches the salt field of an earlier adversarial hash query, or if a later external query discovers an unpublished trial salt; include final verification as one external query. Queries to a salt after its signature has been returned do not trigger this stop. There are at most $\binom{C_S}{2}$ trial/trial pairs and $C_S(Q_H+1)$ trial/external pairs. A new salt is uniform; a later test hits a hidden salt with probability $2^{-\ell_{\mathrm{salt}}}$ in a continuation that erases that salt's unpublished oracle cell. The original and erased continuations agree until the first hit. A union bound gives the salt term in Eq. (6), and the fresh-trial extension gives Eq. (5). At $C_S = 2Q_S$, Hoeffding's inequality gives $\Delta_{\mathrm{rep}}(2Q_S) \le \exp(-4Q_S(p_\pi - 1/2)^2) \le \exp(-4Q_S(0.08)^2)$, using $p_\pi > 0.58$.

In the stopped experiment, the ideal signer already draws each response from the exact coset Gaussian. Replace each uniform syndrome by the syndrome of an ambient Gaussian; a uniform sequential Rényi bound and one application of Hölder give the factor in Eq. (6). The final experiment is simulated without a trapdoor: sample an ambient Gaussian response, program its syndrome at the fresh signing cell, and apply the selected filter. Both $\mathcal{A}_\pi^{\mathrm{ord}}$ and $\mathcal{A}_\pi$ are contained in $\mathcal{V}_\pi$ and have acceptance at least p_π, so the same bound applies to A and $\mathrm{A}^\star$. Generate the independent reference counts and derive refills using the public policy. These common operations preserve the comparisons, including outer retries.

The remaining fresh oracle cells, including final verification, use at most $Q_H + 1$ challenge targets. A fresh-message forgery gives an A-R-ISIS solution for one of them. Absorbing flags remain in the transcript; the unflagged ideal forgery event is a subset of solver success. Restoring the original experiment adds collision and exhaustion probabilities without conditioning on survival. □

A.5 Message recovery (Theorem 2) and digest composition

Proof. The simulator maintains tables T_0, T_1 for $H_{0,\pi,h}, H_{1,\pi,h}$ and a partial injection $\iota : \mathbb{Z}_q^{\gamma_{\mathrm{MR}}} \rightharpoonup \{0,1\}^{\ell_{\mathrm{salt}}}$. Existing queries return their stored values. On a fresh $H_{0,\pi,h}(\rho, M_0, M_1)$ query, assign $\iota(\rho)$ uniformly outside its current image if undefined, and query

$$H_{\pi,h}(\iota(\rho), \mathrm{Enc}_\pi(M_0, M_1)) = (c_0, c_1).$$

Abort if $c_0 \in \mathrm{dom}\, T_1$; otherwise store $T_0(\rho, M_0, M_1) = c_0$ and $T_1(c_0) = c_1 - (\rho, M_0)$. A fresh $H_{1,\pi,h}$ query receives an independent uniform value.

On a valid signing query (M_0, M_1), submit $\mathrm{Enc}_\pi(M_0, M_1)$ to the A* signing oracle and obtain (r', s_1) and its complete reference view. Set $c = (c_0, c_1) = H_{\pi,h}(r', \mathrm{Enc}_\pi(M_0, M_1))$ and $s_0 = [c - s_1 h]_q$. Draw $\rho \leftarrow U(\mathbb{Z}_q^{\gamma_{\mathrm{MR}}})$ independently. Abort if $\rho \in \mathrm{dom}\, \iota$, $r' \in \mathrm{im}\, \iota$, or $c_0 \in \mathrm{dom}\, T_1$. Otherwise set $\iota(\rho) = r'$, store the corresponding entries in T_0, T_1, and return (s_0, s_1). Forward all trial count records and outer decisions, deriving refills using the public MRM buffer policy. Stop if forwarding a record would exceed C_S trials.

Couple A* and MRM trials through a common fresh uniform target, sampler tape, and independent public count record. Their response sampler and filter coincide, giving the same responses and acceptance decisions. Targets and tapes are independent of salt/nonce labels, so the accepted nonce may be drawn after receiving the source signature. Keep auxiliary records for rejected trials, but erase their oracle cells from the simulator. The experiments agree until an abort or a query encountering an erased cell.

Labels. Each nonce or salt domain has at most $Q_0 C_S$ query/trial pairs and $\binom{C_S}{2}$ trial/trial pairs. Charge each pair at its later creation or test. In the erased continuation, hidden labels remain exchangeable among values not excluded by earlier assignments and tests. At most A_C values are excluded, giving total probability $K_C((D_\rho - A_C)^{-1} + (D_s - A_C)^{-1})$. This also covers a later map assignment meeting a rejected source salt.

Prefixes. At most A_C prefixes are generated by adversarial $H_{0,\pi,h}$ queries and signing trials. Each is uniform at generation before a source-input collision. Testing against previous external $H_{1,\pi,h}$ queries and earlier prefixes contributes $[A_C(Q_1 + 1) + \binom{A_C}{2}]/D_0$, including final verification.

A later $H_{1,\pi,h}$ query may discover an erased rejected-trial prefix. Fix a trial and a continuation that erases its cells, retains its independent public count record, forces its outer decision to rejection, and draws subsequent randomness independently of that trial's target. For a later test value x, conditional on the preceding state and retained count record, x is independent of the target. On the

actual rejection branch the continuations agree until another charged collision. Therefore $\Pr[c_0 = x \text{ and the trial rejects}] \le \Pr[c_0 = x] = D_0^{-1}$. This joint-event bound remains valid even when rejection biases the hidden prefix. Summing over C_S trials and $Q_1 + 1$ tests gives the additional $C_S(Q_1 + 1)/D_0$ in Eq. (7).

Conversion. Let (s_0^*, s_1^*) be an accepted MRM forgery with explicit message M_1^*. Form $c^* = (c_0^*, c_1^*) = s_0^* + s_1^* h \bmod q$ and recover $(\rho^*, M_0^*) = c_1^* - T_1(c_0^*) \bmod q$. If its $H_{0,\pi,h}$ cell is fresh, the check succeeds with probability at most D_0^{-1}. Message freshness excludes a rejected signing cell on that same message pair. Otherwise the programmed equations give $H_{\pi,h}(\iota(\rho^*), \mathrm{Enc}_\pi(M_0^*, M_1^*)) = c^*$. Canonical decoding makes $(\iota(\rho^*), s_1^*)$ an accepting ordinary signature with the same centered s_0^*. Injectivity of Enc_π gives message freshness. The simulator makes at most $Q_0 + Q_S$ source hash queries and Q_S signing queries. Extending the fresh trial stream bounds exhaustion by Eq. (5).

Event accounting. Add the label, prefix, prefix-discovery and fresh final-query bounds above, and $\Delta_{\mathrm{rep}}(C_S)$ for global exhaustion. Published-cell re-queries incur no loss. Since $A_C + C_S = Q_0 + 2C_S$, their sum is Eq. (7). □

Digest composition for the randomized interfaces. The public interface factors its message oracle through a 512-bit representative. Set $\mathrm{pkh} = \mathrm{SHAKE256}(\texttt{'H'} \| \mathrm{Enc}(pk))[64]$, where $\mathrm{Enc}(pk)$ contains the two-byte v1 header and $\mathrm{Pack}_{14}(\hat{h})$ in the profile's fixed NTT order. For a fixed key and one randomized signing mode, let D encode the message mode, one-byte context length, context, optional prehash identifier, and message or digest. Put

$$I_{\mathrm{ord}}(D) = \mathrm{SHAKE256}(\texttt{'I'} \| \mathrm{pkh} \| D)[64],$$
$$I_{\mathrm{MR}}(M_0, D) = \mathrm{SHAKE256}(\texttt{'I'} \| \mathrm{pkh} \| \mathrm{Frame}(\mathrm{Pack}_{14}(M_0)) \| D)[64].$$

Here [64] denotes 64 output bytes, Pack_{14} is the canonical little-endian 14-bit coefficient encoding, and $\mathrm{Frame}(x) = \mathrm{LE16}(|x|) \| x$. Ordinary signing uses $\mathrm{HashToRing}(\texttt{'P'} \| r \| I_{\mathrm{ord}}(D))$. Randomized MRM uses

$$c_0 = \mathrm{HashToCoeffs}(\texttt{'0'} \| \texttt{04} \| I_{\mathrm{MR}}(M_0, D) \| \mathrm{Frame}(\mathrm{Pack}_{14}(\rho)), \lambda_{\mathrm{MR}}),$$
$$h_1 = \mathrm{HashToCoeffs}(\texttt{'1'} \| \texttt{04} \| \mathrm{pkh} \| \mathrm{Frame}(\mathrm{Pack}_{14}(c_0)), N - \lambda_{\mathrm{MR}}).$$

These coefficient maps use the residue-sampling rule of Appendix C.1. The canonical encoding is injective in the represented message; when D contains a prehash, freshness here concerns that digest, and security for the original message additionally requires collision resistance of the prehash.

Let Q_I count all distinct representative-oracle inputs evaluated in the experiment, including signing, final verification, and other queries to the same domain, and let Q_G count external queries to the outer target or prefix oracle. Under independent classical random-oracle domains, replacing the composed oracle by the direct-message oracle of Theorem 1 or 2 costs at most

$$\varepsilon_{\mathrm{dig}} = \left(\binom{Q_I}{2} + Q_I Q_G \right) 2^{-512}.$$

To prove this, assign representatives lazily and keep their reverse map. For an outer query whose representative is already assigned uniquely, use the corre-

sponding direct-message oracle cell; otherwise sample and remember an independent answer. Stop if two representative inputs collide or a newly assigned representative matches an earlier unmatched outer query. Each fresh representative hits at most the previously assigned representatives and unmatched query values, giving the displayed union bound. Until a stop, all target cells, signatures, trial decisions, counts, and verification results agree; injectivity preserves message freshness. For raw XOF access, a stream conditioned on a target coefficient is simulated by sampling rejected words and then one of its five accepted representatives, followed by an independent suffix. The simulator makes at most Q_G externally induced direct-message queries, in addition to the source signer's own queries, and $\widetilde{O}(Q_I + Q_G)$ table work; the component bounds use these induced budgets. This is a classical random-stream simulation, with the private-record replacement premise of Corollary 1. It does not establish deterministic or mixed-mode security.

A.6 Certificate composition (Theorem 3)

Proof. If the forged certificate authenticates an unissued identity–key binding, it forges the certificate component. Otherwise it identifies an honest uncorrupted record, and the fresh application message forges that user's component. Injective binding and canonical recovery make this case split unique. Guessing the targeted record loses at most Q_C; the reduction generates all other keys and forwards challenge signing views. The two cases give Eq. (8), as in [3, Theorem 3.5]. □

A.7 Fixed-point composition (Corollary 1)

Proof. Replace the cryptographic tapes by ideal records at cost $\varepsilon_{\mathrm{tape}}$, then apply the successful-output coupling at every key-generation call. Reference failures produce no forgery. Conditional on a returned reference key, its law is the one in the A-R-ISIS experiment. Apply the digest-composition argument of Appendix A.5 to bound the reference event, including its digest-collision stop in S_j. Apply Eq. (11) to each component's flagged event, uniformly over admitted keys. If its conditional reference probability is p_K, Jensen's inequality gives $\mathbb{E}[p_K^{\Theta_j}] \le (\mathbb{E}p_K)^{\Theta_j}$ since $0 < \Theta_j < 1$; hence the same transfer applies to the key mixture. Every forgery precedes a stop or belongs to a charged stop event. Theorem 3 composes the transferred bounds, retaining all terminal outcomes. □

B Lattice Calibration

B.1 Random-SIS model and Rényi orders

Each degree-768 Gram block has determinant 3/4, so the ring has covolume $(3/4)^{N/4}$ and Λ_h has covolume $q^N(3/4)^{N/2}$. Scaling lengths by $(4/3)^{1/4}$ gives the determinant normalization used in Table 2.

For an integer lattice-reduction block size $\beta \geq 2$, define the modeled root-Hermite factor $\delta_\beta = \left(\frac{(\pi\beta)^{1/\beta}\beta}{2\pi e}\right)^{1/(2(\beta-1))}$. The random-SIS model selects the least integer $40 \leq \beta \leq 2N$ satisfying $\delta_\beta^{2N-1}\sqrt{q} \leq B_{\text{eff}}$. The classical and quantum Core-SVP exponents are $\lfloor 0.292\beta \rfloor$ and $\lfloor \log_2(\sqrt{13/9})\beta \rfloor$. Optimizing the calibration in Section 3.5 gives the following Rényi orders:

profile	a_u classical	a_u quantum
1f	294	281
3f	186	177
5f	216	206

For all three profiles, $d = 2N$ is optimal within the determinant-q^N model over $N+1 \leq d \leq 2N$, with predicted length $\delta_\beta^{d-1} q^{N/d}$.

B.2 Probability bounds and work-to-success estimates

For attack work T and success probability ε, the work-to-success measure is T/ε [13,12].

For the ordinary salt term δ_s in Eq. (6), setting $C_S = 2Q_S$ and $W = Q_S + Q_H + 1$ gives

$$\delta_s = \frac{Q_S(2Q_S + 2Q_H + 1)}{2^{\ell_{\text{salt}}}}, \qquad \frac{\delta_s}{W} \leq \frac{2Q_S}{2^{\ell_{\text{salt}}}}.$$

An attack making these queries incurs at least W units of query effort. The factor-two trial cap therefore costs at most one bit in the salt contribution under the allocation of Section 3.5. Primitive costs and reduction overhead enter through the full work model.

For MRM, take $W_{\text{MR}} = Q_S + Q_0 + Q_1 + 1$ and normalize each collision term in Eq. (7) by W_{MR}. At $C_S = 2Q_S$, $K_C/W_{\text{MR}} \leq C_S$, so the combined nonce and salt contribution per unit of query effort is at most $C_S((D_\rho - A_C)^{-1} + (D_s - A_C)^{-1})$. For example, with $Q_S = 2^{64}$ and $Q_0 = Q_1 = 2^{\lambda_{\text{target}}}$ for $\lambda_{\text{target}} = 128, 192, 256$ in profiles 1f, 3f, 5f, the normalized salt terms are at most about $2^{1-\lambda_{\text{target}}}$ and the prefix terms approximately $2^{-130.53}$, $2^{-202.38}$, $2^{-260.65}$, respectively. The nonce terms are smaller than the salt terms because $D_\rho > D_s$ in every profile.

For the sampling calibration, put $K_u = \theta_u C_S \log_2 r_u(a_u)$, so $b_{\text{cal}} = \theta_u \lambda - K_u$. If the hardness model gives $\varepsilon_B \leq T_B 2^{-\lambda}$ and the reduction has $T_B \leq \kappa_{\text{red}} T$, with $T \geq 1$, then

$$\frac{2^{K_u}\varepsilon_B^{\theta_u}}{T} \leq \kappa_{\text{red}}^{\theta_u} 2^{-b_{\text{cal}}} T^{\theta_u - 1} \leq \kappa_{\text{red}}^{\theta_u} 2^{-b_{\text{cal}}}.$$

The overhead factor κ_{red} depends on the computational model. The collision, exhaustion, and composition terms are evaluated separately at the same budgets.

B.3 Multi-target hardness and conditioning

Assumption 1 must cover the specified target count and completion policy, including preprocessing and amortized attacks. For a base key law μ and an additional acceptance event A of probability $p_A > 0$, any fixed attack satisfies $\Pr_{\mu|A}[E] \le \Pr_{\mu}[E]/p_A$. A certified lower bound on p_A controls the conditioning loss. Changes to the candidate map require a separate comparison.

For the Q_{64} cutoff, let Adm denote full admission under the same completion policy without the cutoff, and let A be the event that the evaluator determines every required decision. Put $\varepsilon_{\mathrm{cut}} = \Pr[\neg A \mid \mathsf{Adm}]$, and let δ_{cut} be the per-candidate probability of an undetermined decision. The cutoff law $\mathcal{K}_\pi$ and exact-precision finite-word law $\mathcal{K}_\pi^*$ satisfy $\Pr_{\mathcal{K}_\pi}[E] \le \Pr_{\mathcal{K}_\pi^*}[E]/(1-\varepsilon_{\mathrm{cut}})$, with $\varepsilon_{\mathrm{cut}} \le \delta_{\mathrm{cut}}/\Pr[\mathsf{Adm}]$.

With the specified F-only completion policies, 20,000 independently seeded candidates per profile yielded 6,972, 5,287 and 4,688 admitted keys (1f, 3f, 5f), with no undetermined Q_{64} decision or operational failure. Under the independent candidate-tape sampling model, a zero-event binomial upper bound for δ_{cut} and a Hoeffding lower bound for admission, each with one-sided error $1/120$, give simultaneous 95% confidence bounds across all three profiles. Cutoff admission lower-bounds $\Pr[\mathsf{Adm}]$; hence $\varepsilon_{\mathrm{cut}}$ is below $7.09\cdot10^{-4}$, $9.45\cdot10^{-4}$ and $1.072\cdot10^{-3}$, respectively, with conditioning penalty below 0.00155 bits. These are statistical bounds under the sampling and arithmetic-conformance premises, not certified bounds on the key distribution; Assumption 1 applies directly to $\mathcal{K}_\pi$.

For independent identically distributed candidates and a fixed predicate, every positive trial cap gives the same conditional output law $\mu \mid A$; the cap changes the return probability. Operational failures remain part of the profile's reference generator.

C Completion and Numerical Bounds

C.1 Random tapes and sampling records

All multibyte fields are little-endian. Tables 7 and 8 specify the bytes consumed by each record; buffering may generate further bytes in advance.

Candidate streams. A candidate record consists of $2N$ unsigned 64-bit words $W_0, \ldots, W_{2N-1}$, grouped into blocks of $H_N = N/2$ words. Appendix C.2 specifies the seed expansion and candidate map.

Table 7. Candidate-stream layout for Antrag key generation. Each block contains $H_N = N/2$ words.

Word ($0 \le j < H_N$)	Width	Role at embedding j
W_j	64 bits	Squared radius
W_{j+H_N}	64 bits	Relative magnitudes of f and g
W_{j+N}	64 bits	Phase of f
W_{j+3H_N}	64 bits	Phase of g

The $16N$-byte record determines all candidate decisions. Rejection, including the precision cutoff of Appendix C.2, advances to a fresh seed, while operational failure terminates generation.

Signing streams. In randomized mode, each outer attempt draws fresh entropy $\eta \in \{0,1\}^{512}$. Let $\mathrm{kh} = \mathrm{SHAKE256}(\texttt{'B'} \| \mathrm{Enc}(sk))[64]$ and let $I = I_{\mathrm{ord}}(D)$ or $I_{\mathrm{MR}}(M_0, D)$ be the 64-byte representative of Appendix A.5. The attempt stream is

$$X = \mathrm{SHAKE256}(\texttt{'S'} \| \mathrm{octet}(\mathit{type}) \| \mathrm{kh} \| I \| \mathrm{LE16}(\mathit{attempt}) \| \eta),$$

with type `03` for ordinary signing and `04` for MRM. Ordinary signing reads its salt and then a 64-byte sampler root R. The optional seeded interface derives $\eta = \mathrm{SHAKE256}(\texttt{'S'} \| \texttt{FF} \| \mathit{seed} \| \mathrm{LE16}(\mathit{attempt}))[64]$ for reproducible measurements. Replacing these secret-key-bound streams by independent records is included in $\varepsilon_{\mathrm{tape}}$.

In MRM mode, the same derivation first supplies the recovery nonce. Successive 16-bit words w are accepted when $w < 61445 = 5q$ and mapped to $w \bmod q$ until γ_{MR} coefficients have been obtained. Rejected words remain consumed. The next 64 bytes supply the sampler root; failure to complete the nonce within 256 words rejects the outer attempt. Hash-to-ring expansion uses the same residue-sampling rule on a separate public SHAKE256 stream.

The four streams are $\mathcal{S}_j = \mathrm{SHAKE256}(\texttt{'E'} \| R \| \mathrm{octet}(j))$, $j = 0, 1, 2, 3$, where octet is one byte. For Cholesky, $\mathcal{S}_2, \mathcal{S}_3$ supply v_0, v_1, and $\mathcal{S}_0, \mathcal{S}_1$ supply x_2, x_1. For Crown, the stage order is $(y_2 : \mathcal{S}_1) \longrightarrow (x_2 : \mathcal{S}_3) \longrightarrow (y_1 : \mathcal{S}_0) \longrightarrow (x_1 : \mathcal{S}_2)$. Each outer trial starts fresh streams. Scalar and parallel providers preserve each stream's bytes.

Table 8. Random-tape fields consumed by one sampler draw or proposal, in stream order. The perturbation rows belong to the Cholesky signer and the crown row to the Crown signer.

Draw	Proposal selector(s)	Position/sign	Acceptance	Bytes
Antrag scalar	U_{96}	One sign byte	U_{56}	20
Antrag pair	U_{96}, U_{96}	One sign byte	U_{56}	32
Antrag crown	U_{16}	55 bits + sign	U_{72}	18
Perturbation, 1f/5f	U_{96}	One sign byte	—	13
Perturbation pair, 3f	U_{96} parity, U_{96}, U_{96}	Two sign bytes	—	38
Falcon scalar	U_{72}	One sign byte	Lazy U_{64}	11–18

The scalar selector uses P_0, with the sign in bit zero of the following byte. Pair selectors use P_1 then P_0, with signs in bits zero and one. Unused sign-byte bits are discarded. A crown record stores its 55-bit position in the low bits of seven bytes and its sign in the high bit. A perturbation record selects a magnitude by its table and a sign by bit zero of its sign byte; a 3f perturbation pair first selects the parity, then both magnitudes. Falcon's lazy test leaves unread comparison bytes for subsequent proposals.

Each Antrag perturbation coordinate or pair, and each online proposal, consumes a complete record. Rejection draws a fresh record, including both coordinates of a pair; acceptance advances to the next output without resetting the

stream. Unused prefetched suffixes may be rewound, but rejected records cannot be reused. Outer rejection increments the attempt counter and discards the remaining trial material.

Nonce-decoder exhaustion. Let $\varepsilon_{\text{nonce}}$ bound the probability of any nonce-decoder exhaustion over at most C_S outer trials. Since $\gamma_{\text{MR}} \le 24$, failure within 256 words requires at least 233 rejections; ordinary mode decodes no nonce, so $\varepsilon_{\text{nonce}} = 0$ there. For $C_S \le 2^{65}$,

$$\varepsilon_{\text{nonce}} \le C_S \sum_{k \le 23} \binom{256}{k} \left(\frac{4091}{65536}\right)^{233} < 2^{-758}.$$

The final bound uses $4091/65536 < 2^{-4}$ and $\sum_{k\le 23}\binom{256}{k} < 2^{109}$. Under independent ideal records, residue-sampling counts are independent of accepted values.

C.2 Reference key generation

OddSum ensures modulo-2 invertibility in power-of-two degrees. For degree 768, $\varphi \bmod 2 = (X^6 + X^3 + 1)^{128}$; invertibility depends on the residue modulo the irreducible factor $X^6 + X^3 + 1$, so candidate generation uses plain rounding. This departs from [4, Algorithm B.1], which retains the parity correction in its 3-smooth rings. The mixed-degree solver's Bézout computation handles either coefficient-sum parity.

Let *seed* be a 64-byte candidate seed. The one-byte suite identifier CPId is `01` for Crown or `03` for Cholesky; the one-byte profile identifier Prof is `01`, `03` or `05`. The candidate stream is

$$\text{SHAKE256}(\texttt{ANTRAG-FX-KG-v1}\|\text{CPId}\|\text{Prof}\|\textit{seed}).$$

Read its first $2N$ words W_j as in Appendix C.1. Using the radii of Section 2.2, define, for $0 \le j < H_N$,

$$\begin{aligned} u_j &= q r_{\text{lo}}^2 + q(r_{\text{hi}}^2 - r_{\text{lo}}^2) W_j 2^{-64}, & \theta_j &= \pi W_{j+H_N} 2^{-65}, \\ \vartheta_{f,j} &= 2\pi W_{j+N} 2^{-64}, & \vartheta_{g,j} &= 2\pi W_{j+3H_N} 2^{-64}, \\ z_j &= \sqrt{u_j}\cos\theta_j \, e^{\mathrm{i}\vartheta_{f,j}}, & w_j &= \sqrt{u_j}\sin\theta_j \, e^{\mathrm{i}\vartheta_{g,j}}. \end{aligned}$$

These values and their conjugates specify a real coefficient vector through the inverse canonical embedding. Let M_φ denote the conductor of φ: $M_\varphi = 2N$ in the power-of-two profiles and $M_\varphi = 2304$ in 3f. Write $\zeta_{M_\varphi} = e^{2\pi\mathrm{i}/M_\varphi}$. The permutation ord reverses the $\log_2 H_N$ bits of j in the power-of-two profiles and is the identity in 3f. The stored half spectrum uses roots $\zeta_{M_\varphi}^{1+(M_\varphi/H_N)\,\mathrm{ord}(j)}$.

The dual coefficient-vector norm bounds inverse-transform coefficients by $r_{\text{hi}}\sqrt{q}$ in power-of-two degrees and $(2/\sqrt{3})r_{\text{hi}}\sqrt{q}$ in degree 768 (Appendices C.6 and E use the coarser factor 2). Rounding and OddSum add at most one to the first bound; plain rounding adds at most 1/2 to the second. Hence $|f_i|, |g_i| \le 127$ for 1f/5f and at most 141 for 3f. Each candidate is evaluated once in directed

Q_{64} intervals, with signed 128-bit endpoints below 2^{112} and 256-bit products. Coefficient rounding, OddSum residual comparisons and directions, and annular comparisons must be determined by their enclosures; otherwise the candidate is rejected. OddSum decisions affect only even sums. This cutoff forms part of $\mathcal{K}_\pi$, whereas [4, Appendix B] refines on the same tape.

The evaluator branches only on public data: comparisons and guards update masked flags, all 32 phase-table entries are scanned, and the OddSum search runs for every power-of-two candidate. A shared interval graph, rounding points, constant enclosures, and tie rules ensure identical finite and reference classifications under the arithmetic contracts. An arithmetic-guard violation terminates generation as an operational failure.

The generator returns the first candidate satisfying the annular test, the completion policy, modulo-q invertibility, and Eq. (1), within 2^{16} seeds. Exhaustion or operational failure returns $\perp$.

The full-(F, G) comparison solver uses 32-fractional-bit nearest-plane reduction with leading-limb windows and 13/11-bit steps at degrees 512/1024; it requires $|F_i|, |G_i| \le 127$, so $\|fG - gF - q\|_\infty \le 2N\,127^2 + q < p_{\mathrm{NTT}}$ makes its modular equation check exact. Here $p_{\mathrm{NTT}} = 2147473409$ (Section 5.1).

The degree-768 tower is $768 \to 256 \to 128 \to \cdots \to 2 \to 1$. The full comparison policy lifts a canonical bottom Bézout solution exactly and reduces in Q_{256}, checking the equation after each step. Its variable-time variant allows 128 iterations per level and keeps only updates that decrease the coefficient-square norm.

The hardened full policy runs a fixed public number of reduction rounds at each tower level, using NTT/CRT arithmetic:

Degree	768	256	128	64	32	16	8	4	2
Rounds	2	2	2	2	3	4	7	12	19

In nonfinal rounds, the first zero quotient or non-decreasing norm freezes the selected state; the remaining scheduled rounds still execute with masked selection. A candidate still active in the final scheduled round is rejected if that round's quotient is nonzero; this round computes no further update. The schedule and this rejection rule therefore belong to the profile: the variable-time and hardened policies define different reference key laws $\mathcal{K}_\pi$, and Assumption 1 applies to each separately.

Completing the square gives $|a_i|^2 \le (4/3)\|(a, 0)\|^2_{\mathrm{can}}$ in degree 768; the power-of-two factor is one. Annularity and Eq. (1) imply signed-16-bit coefficient bounds. Conjugate symmetry gives $|\Phi_j(u)| \le \frac{\sqrt{|\Phi_j(F)|^2 + |\Phi_j(G)|^2}}{\sqrt{\nu_j}} \le \alpha^2\sqrt{Nk_N/2}$; summing $|\phi(u)|^2 \le \alpha^2(|\phi(F)|^2 + |\phi(G)|^2)/q$ over the embeddings and applying Eq. (1) gives $\|u\|^2_{\mathrm{can}} \le \alpha^4 k_N$, so $\|u\|_\infty \le \alpha^2\sqrt{k_N} < 25$ in power-of-two degrees and $\|u\|_\infty \le (2/\sqrt{3})\alpha^2\sqrt{k_N} < 19$ in degree 768, and $\|u_{\mathrm{int}}\|_\infty = p_u\|\hat{u}\|_\infty < 2^{26}$. Together with $\|\hat{B}\| \le \beta(p'_u) < \tilde{s} < 140$, these range and positivity bounds support directed preparation, the carrier bounds of Section 4.2, and the ledger of Table 9 for every admitted key.

The completion threshold is determined by the canonical covering radius $r_{\mathrm{cov}}(R)$. Its square is $N/4$ for the orthonormal coefficient lattice and $N/6$ for the product of unit-minimum A_2 lattices in degree 768. Exact closest-vector size reduction leaves a projection residual u with $\|u\|_{\mathrm{can}} \le r_{\mathrm{cov}}(R)$, giving

$$\|(F,G)\|_{\mathrm{can}}^2 = \frac{1}{N}\sum_{\phi}\left(q^2/\nu_\phi + |\phi(u)|^2\nu_\phi\right) \le q\alpha^2(1 + r_{\mathrm{cov}}(R)^2).$$

Equation (1) therefore admits every such completion. The full degree-768 solver terminates according to coefficient-square decrease; coefficientwise rounding provides a residual-square bound of $3N/8$ in the canonical metric. Each returned completion consequently undergoes the canonical norm test as an additional admission predicate, which forms part of the policy-specific key law.

For the full policies, the same common-tape induction applies under their arithmetic and completion contracts. Arithmetic certificates, seeded replay and equation tests support the implementation premise.

F-only completion. Pornin's policy retains only F during recursion [26]. The reference fixes the bottom solution, reduction schedules, quotient precision, roots, rounding, coefficient limits and rejection predicates, and evaluates this finite-precision graph in unbounded carriers. Each reduction produces an integer quotient k, rejects by a prescribed rule, or terminates with an operational failure.

The desktop 1f/5f implementation uses signed 128-bit Q_{80} values, 256-bit products, and 20-bit reduction steps. All limbs of f are scanned when converting to the normalized Fourier representation. For integer Q_{80} components a, b, the stored inverse components are the truncations towards zero of $a2^{112}/(a^2+b^2)$ and $-b2^{112}/(a^2+b^2)$, representing the reciprocal scaled by 2^{-48}. The product with F therefore retains 32 effective fractional bits for rounding k. Products and k round to nearest with ties towards positive infinity; inverse-transform halvings round upwards. Zero numerical denominators, values outside the declared word ranges, quotient coefficients of magnitude at least 2^{28}, and removal of a nonredundant limb reject the candidate. The quotient bound keeps the signed 64-bit sums of the degree-8 integer subtraction and the inputs of the NTT-based subtraction in range. The Cortex-M4 implementation evaluates the same Q_{80} operations on pairs of 64-bit words and produces the same keys. The Q_{80} work occurs below the top recursion level, at degrees at most $N/2$. At degree N, the solver reconstructs G modulo 2147473409 and performs a joint nearest-plane reduction using a Q_{32} quotient [26]. Two temporary vectors of 512 signed 128-bit entries occupy 16 KiB in total and are cleared on every solver exit.

Lemma (soundness of F-only completion). Suppose the integer lifts and updates are exact. Recover G modulo $p = 2147473409$, rejecting if f is not invertible modulo p, and require $|F_i|, |G_i| \le 127$ before narrowing their representations. Together with the unchanged annular, modulo-q unit, and completion-norm checks, every returned key satisfies the admission conditions of Section 4.1. This conclusion does not require k to be the exact real nearest-plane quotient.

Proof. At a power-of-two level $n \ge 2$, write $a^\star(X) = a(-X)$ and let ι_n embed $R_{n/2}$ into R_n by substitution $X \mapsto X^2$, where $R_n = \mathbb{Z}[X]/(X^n + 1)$. Define

the relative norms by $\iota_n(f') = ff^\star$ and $\iota_n(g') = gg^\star$. A lower-level solution $f'G' - g'F' = q$ lifts to

$$F = g^\star \iota_n(F'), \qquad G = f^\star \iota_n(G'), \qquad fG - gF = \iota_n(f'G' - g'F') = q.$$

Thus, starting from the exact bottom Bézout solution, an integer polynomial G exists even when it is not stored. For every $k \in R_n$, the update $F \leftarrow F - kf$ has implicit counterpart $G \leftarrow G - kg$ and preserves the equation. At the top level, compute the centered lift $\widetilde{G}$ of $(q + gF)/f \bmod p$. Its residual $D = f\widetilde{G} - gF - q$ vanishes modulo p. The coefficient checks and the candidate bound $|f_i|, |g_i| \le 127$ give

$$\|D\|_\infty \le 2N\,127^2 + q = \begin{cases} 16528385, & N = 512, \\ 33044481, & N = 1024, \end{cases} \quad < p.$$

Consequently $D = 0$ over the integers. Since $f \neq 0$ and R_N embeds in the cyclotomic field, this lift equals the implicit integer quotient. The remaining admission checks give the same norm and geometry hypotheses used by the preparation and signing proofs. □

Mixed-degree F-only completion. The same invariant extends through the tower $768 \to 256 \to \cdots \to 2 \to 1$. For each extension, let C_f, C_g be the products of the nonidentity conjugates, so that $fC_f = \iota(f')$ and $gC_g = \iota(g')$. The lift $F = C_g \iota(F')$ has implicit counterpart $G = C_f \iota(G')$, and $fG - gF = q$ follows as above. The top extension is cubic; the quadratic conjugation at degree two sends X to $1 - X$. Intermediate reduction computes F/f in Q_{256} and retains only updates that decrease the coefficient-square norm $\|F\|_2^2$, using the public round counts above. In particular, the degree-four budget remains 12 rounds; a nonzero quotient in the final active round rejects the candidate.

At degree 768, F must fit signed 64-bit coefficients. Recover G by a centered lift modulo $2147346433 \cdot 2147309569$, rejecting nonunit f, then apply the two-round joint reduction. The final exact integer equation check is essential: the CRT modulus alone does not prove uniqueness for these intermediate coefficient bounds. Joint updates preserve any reconstruction residual, so a wrapped quotient cannot pass that check. Final coefficient and completion-norm checks give the same admission hypotheses as the full solver.

Coefficient and carrier checks precede narrowing. For $|x| \le L = 2^{w-1} - 1$, sums have magnitude at most $2L$ and complex-product components at most $2L^2 < 2^{2w-1}$, requiring signed $(w+1)$- and $2w$-bit temporaries. Larger completion bounds require a larger CRT modulus or an independent equation check. Quotient cutoffs belong to the policy; unexpected capacity and resource failures remain terminal.

Completion-policy coupling. Fix a policy H whose finite backend implements its integer graph exactly until terminal failure. On a common candidate tape, the candidate map gives identical classifications and pairs; induction through completion and the 2^{16} candidate trials gives Eq. (9). The reference uses the same rounding, schedule and rejection rules, including the 12-round degree-four cutoff. Assumption 1 applies directly to the resulting accepted-key law; returned-key validity alone does not transfer hardness from another policy.

In the reference generator, with independent ideal candidate tapes and a fixed oracle environment, let $a(E)$ be the one-trial probability of returning a key in E, and let r be the probability of ordinary rejection. If the first success occurs at trial t, then

$$\Pr[T_{\text{first}} = t,\ K \in E] = r^{t-1}a(E), \qquad 1 \le t \le 2^{16}.$$

Writing $a > 0$ for the total one-trial acceptance probability, summing over t gives the conditional key law $a(E)/a$. Replacing each rejection factor by the probability of a specified rejected-attempt transcript gives the corresponding joint law. Thus, conditional on reference success in this fixed environment, discarded-attempt transcripts and the trial count are independent of the winning key. Conditioning away implementation failures need not preserve this independence; the finite transfer uses the unconditioned coupling. The accepted attempt still requires the prescribed fixed arithmetic schedules and permitted observations.

Thus Theorems 1–3 and Corollary 1 apply to each policy under its own accepted-key hardness assumption. Preparation and signing bounds are uniform over admitted keys, so changing completion acceptance does not add a signing-error term.

C.3 Width-matched records and acceptance rounding

Both signers use 64 fractional bits. Cholesky uses no common acceptance scale; Crown's scalar scale is given below. An exact common scale in $(0, 1]$ cancels from the accepted law but reduces acceptance, while finite threshold rounding is charged separately. For $j \ge 0$ and $v \in \{0, 1\}$, put $\psi_{\text{prop}}(j, v) = v + (2v - 1)j$.

Fix a real scalar center c and put $k = \lfloor c \rfloor$. Draw $J \leftarrow P_0$ and an independent fair bit V, and propose $z = k + \psi_{\text{prop}}(J, V)$. Set $E_0 = \frac{(z-c)^2}{2r_{\text{in}}^2} - A_0 J^2/2^{64}$ and accept with probability e^{-E_0}. Since $|z - c| \ge J$ and $A_0/2^{64} \le 1/(2r_{\text{in}}^2)$, $E_0 \ge 0$ and the acceptance probability is valid. The accepted mass is $e^{-(z-c)^2/(2r_{\text{in}}^2)}/(2Z_0)$.

For a real pair center (c_0, c_1), draw independent $J_i \leftarrow P_i$ and fair bits V_i, for $i = 0, 1$. Propose $z_1 = \lfloor c_1 \rfloor + \psi_{\text{prop}}(J_1, V_1)$, set $\mu_0 = c_0 - (z_1 - c_1)/2$, and propose $z_0 = \lfloor \mu_0 \rfloor + \psi_{\text{prop}}(J_0, V_0)$. Put $E = \frac{(z_0-\mu_0)^2}{2r_{\text{in}}^2} + \frac{3(z_1-c_1)^2}{8r_{\text{in}}^2} - (A_0 J_0^2 + A_1 J_1^2)/2^{64}$ and accept jointly with probability e^{-E}. The proposal mapping is bijective and $E \ge 0$. For deviations $x_i = z_i - c_i$, the probability that one trial accepts and returns this pair is $\frac{1}{4Z_0Z_1} \exp\left(-\frac{x_0^2 + x_0x_1 + x_1^2}{2r_{\text{in}}^2}\right)$. Conditioning on acceptance therefore gives the same canonical pair Gaussian as Lemma 1.

Scalar exponent and evaluator (1f, 5f). Put $S = 2^{64}$, $D = A_0 = \lfloor S/(2r_{\text{in}}^2) \rfloor$, and let $f \in \{0, \ldots, S-1\}$ be the Q_{64} code of the center fraction, so $c = k + f/S$. The displacement code is $\delta = f$ for $V = 0$ and $\delta = S - f$ for $V = 1$; the endpoint $f = 0$, $V = 1$ represents $\delta = S$ exactly, and zero fractions are not moved. Then $(z - c)^2 = J^2 + 2J\delta/S + \delta^2/S^2$, and the implemented exponent is the integer

$$E = \left\lfloor \frac{D}{S} \left(2J\delta + \left\lfloor \frac{\delta^2}{S} \right\rfloor \right) \right\rfloor,$$

so that E/S differs from E_0 by less than $(1+D/S)/S$ plus the public width error $256|D/S - 1/(2r_{\text{in}}^2)|$. Since $0 \le J \le 15$ and $0 \le \delta \le S$, $0 \le E \le 31D < 13\ell_{64}$ with $\ell_{64} = \mathsf{Round}(2^{64} \ln 2)$. Range reduction computes $q_{\text{exp}} = \lfloor E/\ell_{64} \rfloor \le 12$ and $r_{\text{exp}} = E - q_{\text{exp}}\ell_{64}$ using a reciprocal estimate with one correction. The degree-12 Q_{63} polynomial of fixed-point Falcon [6] is evaluated by initializing $a = C_0$ and, for $i = 1, \ldots, 12$, setting $a \leftarrow C_i - \lfloor r_{\text{exp}} a / 2^{64} \rfloor$. The evaluator produces the 56-bit threshold directly as $\lfloor a / 2^{q_{\text{exp}}+7} \rfloor$. Interval Taylor enclosures on 1024 subintervals of the reduced range, plus every Horner floor, bound the relative error by $4.285 \cdot 10^{-16}$; every accumulator lies in $[0, 2^{63}]$. The threshold shift is at most 19 bits.

Pair exponent and evaluator (3f). Set $S = 2^{64}$, $D_0 = \mathsf{Round}(S/(2r_{\text{in}}^2))$ and $D_1 = \mathsf{Round}(3S/(8r_{\text{in}}^2))$. The displacement codes $\Delta_0 = 2S(z_0 - \mu_0)$ and $\Delta_1 = 2S(z_1 - c_1)$ preserve the conditional center's extra bit. The exponent is

$$E_{\text{pair}} = \mathsf{Round}\left(\frac{D_0\Delta_0^2 + D_1\Delta_1^2}{4S^2}\right) - (A_0 J_0^2 + A_1 J_1^2).$$

Proposal masses are $\lfloor 2^{96} P_i(j) \rfloor$ for $j \ge 1$, with the remainder assigned to zero. Range reduction uses $\ell_{64} = \mathsf{Round}(2^{64} \ln 2)$, followed by a degree-20 Q_{64} Taylor polynomial and downward rounding to the 2^{96} probability scale. Directed intervals bound coefficient, evaluation and table errors.

Lemma C.1 (threshold rounding). Let the integer $b_{\text{acc}} \ge 1$ be the acceptance precision in bits, g_x a probability law on proposal states x, and $p_x \in [0,1]$ their evaluator acceptance probabilities. Put $\alpha_{\text{acc}} = \sum_x g_x p_x$, $q_x = 2^{-b_{\text{acc}}} \lfloor 2^{b_{\text{acc}}} p_x \rfloor$, and $\epsilon = 2^{-b_{\text{acc}}} < \alpha_{\text{acc}}$. For the normalized accepted laws $P_{\text{eval}}(x) \propto g_x p_x$ and $P_{b_{\text{acc}}}(x) \propto g_x q_x$, and any $a \ge 2$,

$$G_{b_{\text{acc}}} = \sum_{p_x > 0} g_x \min\{p_x, \epsilon, a\epsilon^2/(2p_x)\},$$

$$\log_2 R_a(P_{b_{\text{acc}}} \| P_{\text{eval}}) \le \frac{1}{a-1} \log_2 \left(1 + \frac{(a-1)G_{b_{\text{acc}}}}{\alpha_{\text{acc}}(1 - \epsilon/\alpha_{\text{acc}})^a}\right).$$

Proof. For $p_x > 0$, set $d_x = (p_x - q_x)/p_x$ and $m = \mathbb{E}_{P_{\text{eval}}} d_x \le \epsilon/\alpha_{\text{acc}}$. All expectations below use P_{eval}. The likelihood ratio is $(1 - d_x)/(1 - m)$. For $0 \le d \le 1$, $h_a(d) = (1-d)^a - 1 + ad$ lies between zero and $\min\{a(a-1)d^2/2, (a-1)d\}$. Hence

$$R_a(P_{b_{\text{acc}}} \| P_{\text{eval}})^{a-1} = 1 + \frac{\mathbb{E} h_a(d_x) - h_a(m)}{(1-m)^a} \le 1 + \frac{(a-1)G_{b_{\text{acc}}}}{\alpha_{\text{acc}}(1 - \epsilon/\alpha_{\text{acc}})^a}.$$

The direction of comparison permits zeroed outcomes; data processing covers the returned scalar or pair. □

Normalized evaluator error. Let Q be the exact accepted law, and suppose the evaluator changes its unnormalized masses by a factor w satisfying $0 < m \le w \le M$. Its normalized law is $P_w = Qw/\mathbb{E}_Q w$. Applying the log-partition argument of Eq. (C.3) to $\log w$ gives

$$\log_2 R_a(P_w \| Q) \le \frac{a}{8 \ln 2} \left(\ln \frac{M}{m}\right)^2.$$

Indeed, the second derivative of $\ln \mathbb{E}_Q[w^t]$ is a variance bounded by $\frac{1}{4}(\ln(M/m))^2$; subtracting its affine interpolation at 0 and 1 and integrating gives the stated bound. A common multiplicative bias cancels on normalization, so the interval width in log scale, rather than its distance from one, controls this loss. Directed Taylor enclosures of the exponential polynomial, every Horner rounding, exponent and range-reduction errors, and the final scale multiplication supply m, M. Scalar and crown Q_{64} evaluation use signed enclosures; the degree-20 evaluators may use the symmetric interval $[1-\delta, 1+\delta]$. If finite masses or threshold floors give the pointwise bound $P \le CP_w$, then $\log_2 R_a(P\|Q) \le \log_2 R_a(P_w\|Q) + a\log_2 C/(a-1)$. This charges proposal rounding and omitted support without assuming a two-sided likelihood bound for zeroed outcomes. Lemma C.1 separately charges the complete acceptance word.

The scalar count reference in Eq. (10) uses $p_* = r_{\mathrm{in}}\sqrt{2\pi}/(2Z_0)$, and the pair reference uses $p_* = \pi r_{\mathrm{in}}^2/(\sqrt{3}Z_0Z_1)$. Poisson summation and directed evaluator errors bound acceptance uniformly over centers. Evaluating Eq. (10) at the interval endpoints gives the count divergences.

Scalar evaluator of the Crown signer. Crown shares the proposals and pair exponent above. Its scalar acceptance carries the public factor $\kappa_{64} = \texttt{0xf8c3bbb32382276c}/2^{64}$, obtained by rounding Falcon-512's $\sigma_{\min}/r_{\mathrm{in}}$; pairs use scale one. The scalar exponent is the one-coordinate form of E_{pair}. A degree-nine Q_{63} polynomial follows range reduction by ℓ_{64}, with κ_{64} applied before threshold rounding. The scalar count reference is $p_{*,64} = \kappa_{64} r_{\mathrm{in}}\sqrt{2\pi}(1+b_{64})/(2Z_0)$ with $b_{64} = 18806/10^{18}$. The pair reference is unchanged; finite evaluation, scaling and threshold errors are charged in their respective stages.

C.4 Cholesky sampler: construction and finite stages

Proof of Lemma 2. We adapt [32, Theorem 1 and Appendix A] to the canonical metric, with separate smoothing errors and the effective preparation precisions, working in standard-deviation units. The ring-specific bounds are given below; the Gaussian comparison applies in any real inner-product space.

Setting. Let $V = R \otimes \mathbb{R}$ with the canonical inner product, and give V^2 its product inner product. The lattices R, R^2 and Λ_h are full rank. The commutative real algebra V contains $\mathbb{K}$ and the entries of the exact Cholesky factor A. Its complexification decomposes into the N embedding eigenlines; a matrix $H \in V^{2\times 2}$ therefore has operator norm $\|H\| = \max_k \sigma_{\max}(\Phi_k(H))$ and real Frobenius norm $\|H\|_F^2 = \mathrm{tr}(H^*H) = \sum_{k<N} \|\Phi_k(H)\|_F^2$. Here $*$ combines the ring adjoint with transpose. For a positive-definite self-adjoint operator Ψ and center c, put

$$\varrho_{\Psi,c}(x) = \exp\bigl(-\tfrac{1}{2}\langle x-c, \Psi^{-1}(x-c)\rangle_{\mathrm{can}}\bigr),$$

so that $D_{\Lambda,\sigma,c}$ is $\varrho_{\sigma^2 I,c}$ normalized on Λ, and $\varrho_{\Psi,c}(\Lambda) = \sum_{x\in\Lambda} \varrho_{\Psi,c}(x)$. Write $\Psi \ge \eta_\varepsilon(\Lambda)$ if $\sum_{w\in\Lambda^\vee\setminus\{0\}} e^{-2\pi^2\langle w,\Psi w\rangle_{\mathrm{can}}} \le \varepsilon$; for $\Psi = \sigma^2 I$ this is $\sigma \ge \eta_\varepsilon(\Lambda)$ in the

sense of Section 3.1, and $\Psi \succeq \sigma^2 I$ with $\sigma \geq \eta_\varepsilon(\Lambda)$ implies $\Psi \geq \eta_\varepsilon(\Lambda)$. Poisson summation gives

$$\varrho_{\Psi,c}(\Lambda) = \frac{\det(2\pi\Psi)^{1/2}}{\operatorname{covol}(\Lambda)} \sum_{w \in \Lambda^\vee} e^{-2\pi^2 \langle w, \Psi w\rangle_{\mathrm{can}}} e^{2\pi \mathrm{i} \langle w, c\rangle_{\mathrm{can}}}, \tag{C.1}$$

and therefore, whenever $\Psi \geq \eta_\varepsilon(\Lambda)$, $\varrho_{\Psi,c}(\Lambda) \in [\frac{1-\varepsilon}{1+\varepsilon}, 1]\, \varrho_\Psi(\Lambda)$ for every center c [24, Lemma 2.4]. Put $\kappa(\varepsilon) = \frac{1+\varepsilon}{1-\varepsilon}$, $r = r_{\mathrm{in}}$, and $\sigma_{\mathrm{sig}} = r\tilde{s}$.

Step 0: the exact laws. Lemma 1 and Appendix C.3 give exact online ring Gaussians. The perturbation is $D^{\mathrm{can}}_{R^2, L_{\mathrm{p}} r}$, with independent coefficients in power-of-two rings. In degree 768 it factors over canonical pairs (a, b) with mass proportional to $e^{-(a^2+ab+b^2)/(2\sigma^2)}$, where $\sigma = L_{\mathrm{p}} r$. The bijection $e = b \bmod 2$, $\mathsf{k} = (b-e)/2$, $\mathsf{t} = a + \mathsf{k}$ gives $a^2 + ab + b^2 = (\mathsf{t} + e/2)^2 + 3(\mathsf{k} + e/2)^2$. Thus parity has weight $\Theta_\sigma(e/2)\Theta_{\sigma/\sqrt{3}}(e/2)$, and conditioned on e, the independent draws are $\mathsf{t} \leftarrow D_{\mathbb{Z},\sigma,-e/2}$ and $\mathsf{k} \leftarrow D_{\mathbb{Z},\sigma/\sqrt{3},-e/2}$, as in Section 4.2.

Step 1: operator bounds. Write $\lambda_R = \lambda_{\max}(G_R)$, equal to 1 in the power-of-two rings and 3/2 in degree 768. For $h \in V$ with coefficients bounded by e, its embedding values are bounded by Ne and Parseval gives $\sum_k |\phi_k(h)|^2 \leq \lambda_R N^2 e^2$. The errors $\delta_u = u - \hat{u}$ and $E = A - \hat{A}$ have coefficient bounds $1/(2p'_u)$ and $1/(2p'_A)$ (Section 4.1). Set $d_u = N/(2p'_u)$ and $d_A = N/(2p'_A)$ in degree 768; in the power-of-two rings use the sharper $d_u = \csc(\pi/(2N))/(2p'_u)$ and $d_A = \csc(\pi/(2N))/(2p'_A)$ of [32, Lemmas 2 and 3].

At each embedding, the Gram matrix of $\hat{B}$ has determinant q^2 and trace $\gamma(1 + |\phi_k(\delta_u)|^2) + q^2/\gamma$, with $\gamma = |\phi_k(f)|^2 + |\phi_k(g)|^2 \in [q/\alpha^2, \alpha^2 q]$. Maximizing this convex expression over γ and applying the quadratic formula, as in [32, Appendix A.1], gives $\|\hat{B}\| \leq \beta$ and $\sigma_{\min}(\hat{B}) \geq q/\beta$, where

$$\beta = \beta(p'_u) = \frac{\sqrt{q}}{2}\Big(\sqrt{(\alpha + \alpha^{-1})^2 + \alpha^2 d_u^2} + \sqrt{(\alpha - \alpha^{-1})^2 + \alpha^2 d_u^2}\Big).$$

Since $\tilde{s} > \beta$, $\Sigma_p \succeq (\tilde{s}^2 - \beta^2) I \succ 0$. The entry bounds on the triangular matrix E and the identity $AA^* = \Sigma_p$ give

$$\|E\| \leq E_* := \frac{1+\sqrt{5}}{2} d_A, \qquad \|E\|_F \leq F_* := \frac{\sqrt{3\lambda_R}\, N}{2p'_A},$$
$$\|A\| \leq A_* := \sqrt{\tilde{s}^2 - q^2/\beta^2}.$$

The identity $\Sigma_e = AE^* + EA^* - EE^*$ therefore yields

$$\|\Sigma_e\| \leq (2A_* + E_*)E_* =: \bar{\tau}\tilde{s}^2, \qquad \|\Sigma_e\|_F \leq (2A_* + E_*)F_*.$$

In degree 768, $F_* = 3N/(2\sqrt{2}\, p'_A)$. For the power-of-two rings these are [32, Lemma 3] with p'_u, p'_A replacing p_u, p_A. The definition of Σ_e and the lower singular-value bound imply

$$1 + \|\hat{B}^{-1}\hat{A}\|^2 \leq \frac{\beta^2 \tilde{s}^2 (1 + \bar{\tau})}{q^2}. \tag{C.2}$$

Step 2: smoothing. Lemma 1 bounds the coefficient-basis Gram–Schmidt lengths by one, giving $\eta_\varepsilon(R) \le \rho_h(N,\varepsilon)$, $\eta_\varepsilon(R^2) \le \rho_u(N,\varepsilon)$ and $\eta_\varepsilon(\Lambda_h) \le \alpha\sqrt{q}\,\rho_u(N,\varepsilon)$. The three smoothing conditions are

$$\begin{aligned}
&\text{(C1)} \quad r \ge \rho_h(N,\varepsilon_p),\\
&\text{(C2)} \quad \frac{L_{\mathrm{p}} r q}{\beta\tilde{s}\sqrt{1+\bar{\tau}}} \ge \rho_u(N,\varepsilon_v),\\
&\text{(C3)} \quad r\tilde{s}\sqrt{\frac{(1-\sqrt{\varepsilon_\Lambda/2})(1-\bar{\tau})}{1+(2a-1)\bar{\tau}}} \ge \alpha\sqrt{q}\,\rho_u(N,\varepsilon_\Lambda),
\end{aligned}$$

and the certificates check them with directed arithmetic for the stated $\varepsilon_p, \varepsilon_v, \varepsilon_\Lambda$ and the orders a of Table 9. They imply $r \ge \eta_{\varepsilon_p}(R)$, $L_{\mathrm{p}} r q/(\beta\tilde{s}\sqrt{1+\bar{\tau}}) \ge \eta_{\varepsilon_v}(R^2)$, and, for every $t \in [0, 2a]$, $(1-\sqrt{\varepsilon_\Lambda/2})V_t \ge \eta_{\varepsilon_\Lambda}(\Lambda_h)$ with V_t defined in Step 4, because $V_t \succeq r^2\tilde{s}^2\frac{1-\bar{\tau}}{1+(2a-1)\bar{\tau}} I$ there.

Step 3: output decomposition. Fix $z \in \Lambda_h$, write $(x_1, x_2)^T = B^{-1}z$, and put $\widetilde{P}_{B,c}(z) = P_{B,c}(\bar{c} - z)$. For each perturbation $v \in R^2$, the sampler of Figure 7 uses $\hat{c} = \hat{B}^{-1}(\bar{c} - \hat{A}v/L_{\mathrm{p}})$ and the online centers $d_2 = \hat{c}_2$ and $d_1 = \hat{c}_1 - \hat{u}x_2$. Since $(x_1 + \hat{u}x_2, x_2)^T = \hat{B}^{-1}z$, its exact output law is

$$\widetilde{P}_{B,c}(z) = \sum_{v\in R^2} \frac{\varrho_{L_{\mathrm{p}}^2 r^2 I}(v)\,\varrho_{r^2\hat{B}\hat{B}^*}(z - \bar{c} + \hat{A}v/L_{\mathrm{p}})}{\varrho_{L_{\mathrm{p}}^2 r^2 I}(R^2)\,\varrho_{r^2 I, d_2}(R)\,\varrho_{r^2 I, d_1}(R)}.$$

Put $G = (\hat{B}\hat{B}^*)^{-1}$,

$$C = (I + \hat{A}^* G \hat{A})^{-1}, \qquad m = -C\hat{A}^* G(z - \bar{c}).$$

The exact-law identity of [32, Appendix A.3] holds in the canonical metric: completing the square in v/L_{p} gives

$$\begin{aligned}
&\sum_{v\in R^2} \varrho_{L_{\mathrm{p}}^2 r^2 I}(v)\varrho_{r^2\hat{B}\hat{B}^*}(z - \bar{c} + \hat{A}v/L_{\mathrm{p}})\\
&\qquad = \varrho_{r^2(\tilde{s}^2 I - \Sigma_e)}(z - \bar{c})\varrho_{L_{\mathrm{p}}^2 r^2 C, L_{\mathrm{p}} m}(R^2).
\end{aligned}$$

By (C1), each of the two center-dependent online normalizers lies between $\kappa(\varepsilon_p)^{-1}\varrho_{r^2 I}(R)$ and $\varrho_{r^2 I}(R)$, uniformly over the perturbation and the previous sample. Equation (C.2) and (C2) place the last Gaussian mass above smoothing, giving a factor in $[\kappa(\varepsilon_v)^{-1}, 1]$ relative to its centered mass. Consequently,

$$\widetilde{P}_{B,c}(z) = Q_e(z)\frac{a_z}{\mathbb{E}_{Q_e}[a_z]}, \qquad a_z \in [\kappa(\varepsilon_v)^{-1}, \kappa(\varepsilon_p)^2],$$

where Q_e is $\varrho_{r^2(\tilde{s}^2 I - \Sigma_e),\bar{c}}$ normalized on Λ_h. Thus $\log a_z$ has range length at most $\ell = 2\log\kappa(\varepsilon_p) + \log\kappa(\varepsilon_v)$. For $F(t) = \log\mathbb{E}_{Q_e}[a_z^t]$, $F''(t)$ is the variance of $\log a_z$ under the law proportional to $Q_e a_z^t$ and is at most $\ell^2/4$. The function

$H(t) = F(t) - tF(1)$ vanishes at 0 and 1. Integrating its curvature bound gives the extension of [32, Lemma 4] to separate smoothing errors:

$$\log R_b(\widetilde{P}_{B,c}\|Q_e) = \frac{F(b) - bF(1)}{b-1} \le \frac{b\ell^2}{8}, \qquad b > 1. \tag{C.3}$$

Step 4: comparison with the target. Let $Q_0 = D_{\Lambda_h, r\tilde{s}, \bar{c}}$. The Hölder step in [32, proof of Theorem 1], together with (C.3), gives

$$\log R_a(\widetilde{P}_{B,c}\|Q_0) \le \frac{a\ell^2}{4} + \log R_{2a}(Q_e\|Q_0).$$

Let $\theta_1, \dots, \theta_{2N}$ be the eigenvalues of $\tilde{s}^{-2}\Sigma_e$. For $t \in [0, 2a]$, put

$$V_t = r^2(\tilde{s}^2 I - \Sigma_e)\big(I + (t-1)\tilde{s}^{-2}\Sigma_e\big)^{-1}.$$

The parameter matrices of Q_0 and Q_e are V_0 and V_1, and $V_t^{-1} = (1-t)V_0^{-1} + tV_1^{-1}$. Thus $\varrho_{V_1,\bar{c}}^{2a}\varrho_{V_0,\bar{c}}^{1-2a} = \varrho_{V_{2a},\bar{c}}$. The eigenvalues of V_t are $r^2\tilde{s}^2(1-\theta_i)/(1+(t-1)\theta_i)$, where $|\theta_i| \le \bar{\tau}$; hence $V_t \succeq r^2\tilde{s}^2(1-\bar{\tau})/(1+(2a-1)\bar{\tau})I$. Applying (C.1) to V_0, V_1, V_{2a} cancels the common covolume and splits $\log R_{2a}(Q_e\|Q_0)$ into a determinant term and a lattice-mass term. The spectral calculation in [32, proof of Theorem 1] bounds the first by

$$\frac{a\|\Sigma_e\|_F^2}{2\tilde{s}^4(1+\bar{\tau})(1-(2a-1)\bar{\tau})}.$$

For the second, Step 2 gives $(1 - \sqrt{\varepsilon_\Lambda/2})V_t \ge \eta_{\varepsilon_\Lambda}(\Lambda_h)$ for all $t \in [0, 2a]$, the smoothing margin required by the Poisson-series derivative estimate. Let $\varphi(t)$ be the logarithm of the dual sum in (C.1) at $\Psi = V_t$ and center $\bar{c}$, equivalently

$$\varphi(t) = \log \frac{\operatorname{covol}(\Lambda_h)\, \varrho_{V_t,\bar{c}}(\Lambda_h)}{\det(2\pi V_t)^{1/2}}.$$

The Poisson-series derivative argument in [32, proof of Theorem 1] gives $|\varphi''(t)| \le 5\delta^2$, where $\delta = \bar{\tau}/(1-(2a-1)\bar{\tau})$. Subtracting the affine interpolant at 0 and 1 bounds the lattice-mass term by

$$\frac{\varphi(2a) - 2a\varphi(1) + (2a-1)\varphi(0)}{2a-1} \le 5a\delta^2.$$

These two arguments use only the eigenvalue bounds and lattice smoothing, and so apply to the canonical lattices of all three profiles. Adding the bounds and applying the bijection $s = \bar{c} - z$ proves Lemma 2. □

Finite stages. The perturbation tables are compared through their exact finite masses: each generator encloses the infinite normalizer with a geometric tail bound and evaluates the one-sided Rényi moment at its selected order, charging both truncation and mass rounding. At order 65536 the bounds are $1.774226 \cdot 10^{-27}$ and $4.430906 \cdot 10^{-29}$ bits per coordinate in 1f and 5f, and

$1.018481 \cdot 10^{-27}$ bits per 3f pair, including parity. For center rounding, the uniform log-normalizer Hessian gives at most $a2^{-128}/r_{\mathrm{in}}^2$ bits per scalar and $4a2^{-128}/r_{\mathrm{in}}^2$ bits per canonical pair. The evaluator stage uses the normalized-weight bound of Appendix C.3, including the finite proposal tables; Lemma C.1 charges the complete 56-bit threshold. The count stage evaluates Eq. (10) at the certified acceptance endpoints relative to the public reference p_*. Table 9 lists the independently selected orders and lifetime bounds for the Q_{64} profiles of Table 3.

Table 9. Rényi orders a_i and upward-rounded lifetime bounds D_i on $\log_2 R_{a_i}$ at $C_S = 2^{65}$, in composition order.

Stage	1f		3f		5f	
	a_i	D_i	a_i	D_i	a_i	D_i
Construction	412	$3.111 \cdot 10^{-1}$	282	$9.084 \cdot 10^{-1}$	297	$8.617 \cdot 10^{-1}$
Perturbation tables	651496368	$6.838 \cdot 10^{-5}$	1390005085	$3.016 \cdot 10^{-5}$	650531058	$6.052 \cdot 10^{-6}$
Center rounding	1408540695	$9.044 \cdot 10^{-8}$	1148779525	$2.213 \cdot 10^{-7}$	1407217145	$1.807 \cdot 10^{-7}$
Proposals and evaluator	167663	$7.771 \cdot 10^{-4}$	21364673	$3.929 \cdot 10^{-5}$	167506	$1.553 \cdot 10^{-3}$
56-bit acceptance	2742111	$4.646 \cdot 10^{-5}$	2796789	$9.089 \cdot 10^{-5}$	2739535	$9.282 \cdot 10^{-5}$
Observable counts	9606	$1.327 \cdot 10^{-2}$	2480462	$1.025 \cdot 10^{-4}$	9596	$2.650 \cdot 10^{-2}$

3f centers. For each exact center c of Section 4.2, two-digit division computes $\lfloor 2^{64}c \rfloor$ exactly, including negative and integral centers. Each coordinate error is below 2^{-64}. The conditional half-center stays exact at Q_{65}. The stored Q_{64} constants equal direct nearest-even rounding; the limb and Horner identities of Appendix C.5 preserve the degree-20 evaluator's thresholds and the transcript of 32-byte records with 56-bit acceptance words.

In 3f, Table 9's six-stage composition gives a loss at most 1.813429 bits at $b_{\mathrm{ref}} = 256$ (Table 3); the pair acceptance probability exceeds 0.6076.

All bounds are uniform over admitted keys, syndromes and preceding states of the stopped view at $C_S = 2Q_S$, under the arithmetic and observation premises of Sections 4.1 and 4.3.

For the one-trial output law, the certificates evaluate the stage bounds at order two where appropriate and use Rényi monotonicity for the remaining output stages. Pinsker's inequality and the triangle inequality for total variation then bound the distance from the ideal sampler by $5.3 \cdot 10^{-12}$, $9.0 \cdot 10^{-12}$, and $8.5 \cdot 10^{-12}$ in 1f, 3f, and 5f.

Scalar and pair acceptance exceed 0.767 and 0.60 uniformly over centers, giving lifetime 1024-proposal-cap probabilities below 2^{-2076} in 1f/5f and 2^{-1279} in 3f. The total-variation bounds above give rejection below 0.42000000001; hence $2^{64}(0.42000000001)^{272} < 2^{-276.418544}$ bounds outer-cap exhaustion. The global stop is below 2^{-1024} at $Q_S = 2^{64}$. Together with the nonce bound of Appendix C.1, these terms give $\varepsilon_{\mathrm{joint}} < 2^{-276}$, separately from the losses in Tables 3 and 9.

C.5 Crown sampler: construction and finite stages

Proof of Lemma 3. Work in the real canonical space $R \otimes \mathbb{R}$, and put $r = r_{\mathrm{in}}$, $\sigma = \sigma_{\mathrm{sig}}$, and $v_R = \mathrm{covol}(R)$. Let A_i be the positive operator whose quadratic form is $\|x\widetilde{b}_i\|^2_{\mathrm{can}}$, where $\widetilde{b}_1 = b_1$ and $\widetilde{b}_2 = b_2 - ub_1$ (Section 2.1). Its eigenvalues at embedding j are ν_j for $i = 1$ and q^2/ν_j for $i = 2$. The target coordinate covariance is $\Sigma_i = \sigma^2 A_i^{-1}$. Eq. (12) makes $T_i = \Sigma_i - r^2 I$ positive definite. The displayed perturbations in $\mathsf{PreSmp}^{\mathrm{cr}}$ have covariance T_i in this real space, including for degree 768.

Condition on the previously sampled coordinates, and let d be the unperturbed Gram–Schmidt center for stage i. For the continuous Gaussian density ϕ_T of covariance T, the stage's output mass at $x \in R$ is

$$p_i(x \mid d) = \int \phi_{T_i}(y) \frac{\exp(-\|x - d + y\|^2_{\mathrm{can}}/(2r^2))}{\sum_{z \in R} \exp(-\|z - d + y\|^2_{\mathrm{can}}/(2r^2))}\, dy.$$

Poisson summation and $r \geq \eta_{\varepsilon_p}(R)$ bound every denominator between $(1-\varepsilon_p)a_r$ and $(1+\varepsilon_p)a_r$, where $a_r = (2\pi r^2)^{N/2}/v_R$. Gaussian convolution of the numerator is $(2\pi r^2)^{N/2}\phi_{\Sigma_i}(x - d)$. Hence

$$\frac{v_R \phi_{\Sigma_i}(x-d)}{1+\varepsilon_p} \leq p_i(x \mid d) \leq \frac{v_R \phi_{\Sigma_i}(x-d)}{1-\varepsilon_p}.$$

Multiply these conditional bounds in sampling order x_2, x_1. Field Gram–Schmidt orthogonality gives

$$\sum_{i=1}^{2} \langle x_i - d_i^0, A_i(x_i - d_i^0)\rangle_{\mathrm{can}} = \|\bar{c} - x_1 b_1 - x_2 b_2\|^2_{\mathrm{can}},$$

where d_i^0 are the unperturbed centers, including the dependence of d_1^0 on x_2. The density normalizers depend only on B, σ, so the joint mass is a common constant times the target Gaussian weight and a factor in $[(1+\varepsilon_p)^{-2}, (1-\varepsilon_p)^{-2}]$. Normalizing the joint mass divides by an average of factors in the same interval, yielding the claimed ratio. The affine map from (x_1, x_2) to s is bijective, and the argument applies in the canonical metric of both ring families. □

The proof improves the exponent six of [29, Theorem 5.10] and gives $P^{\mathrm{hyb}}_{B,c} = Qw/\mathbb{E}_Q w$, where $Q = D_{\mathcal{L}_h(c),\sigma_{\mathrm{sig}}}$ and $(1+\varepsilon_p)^{-2} \leq w \leq (1-\varepsilon_p)^{-2}$. Hence $\log w$ has range $\ell_{\mathrm{h}} = 2\ln((1+\varepsilon_p)/(1-\varepsilon_p))$. Equation (C.3) gives $\log_2 R_a \leq a\ell_{\mathrm{h}}^2/(8\ln 2)$, uniformly over keys, syndromes and preceding states, proving Eq. (13). Also $\delta_p = ((1+\varepsilon_p)/(1-\varepsilon_p))^2 - 1 < 2^{-35}$ bounds one-trial total variation and therefore the change in acceptance probability.

Crown masses and evaluator. The crown masses are

$$n_J = \left\lceil 8625 e^{-J^2/64} \right\rceil \quad (1 \leq J < 80), \qquad n_0 = 2^{16} - \sum_{J=1}^{79} n_J,$$

so that $n_J \geq 8625 e^{-J^2/64}$. Choose J with probability $n_J/2^{16}$, draw a uniform $U \in \{0, \ldots, 2^{55} - 1\}$ and a fair sign, and put $M_{\mathrm{cr}} = J2^{55} + U$ and

$X = \pm M_{\mathrm{cr}} 2^{-53}$. The exact acceptance probability is $(8625/n_J)e^{-M_{\mathrm{cr}}^2/2^{116}}$, excluding negative zero, and the accepted grid law is proportional to $e^{-X^2/1024}$. Scaling each accepted draw by $\omega\tau_{i,j}$ leaves this rejection law independent of the key. The integer evaluator uses $\lambda_J = \mathsf{Round}\left(2^{64}\left(J^2/64 + \ln(n_J/8625)\right)\right)$ and $E_{\mathrm{cr}} = \lambda_J + 16JU + \mathsf{Round}(U^2/2^{52})$. For $J \geq 25$, $n_J = 1$ and $\lambda_J = \lambda_{25} + (J^2 - 625)2^{58}$; scanning the first 25 offsets and selecting this tail formula with masks gives fixed memory addresses. Crown perturbations start on the Q_{53} grid; the 1f and 5f Q_{64} profiles include one Q_{53} rounding before scaling, whereas 3f converts the grid directly.

Let P_{grid} be the accepted grid law with exact acceptance probabilities, and let τ_{grid} be the mass of the full grid Gaussian omitted outside $|x| < 320$. For $Y \sim \mathcal{N}(0, 512)$, let Q_{rnd} be the law of Y rounded to the 2^{-53} grid. Conditional on $X = x$, draw an auxiliary lift $\widetilde{X}$ from Y conditioned on the rounding cell of x. Then $|\widetilde{X} - X| \leq 2^{-54}$, and the likelihood ratio is constant on each cell. Gaussian cell integration gives $P_{\mathrm{grid}}(x)/Q_{\mathrm{rnd}}(x) \leq e^{2^{-118}}/(1-\tau_{\mathrm{grid}})$. The lift preserves the likelihood comparison, and its displacement contributes to the center error.

The guard $\|x_2\|_2 \leq 2^{14}$ is checked before forming the center for x_1; Appendix E.2 states the integer and failure handling common to both signers.

Transfer. The stages, ordered from the ideal signer toward the implementation as in Section 4.3, are: the construction stage of Eq. (13); L_3, crown evaluator error and the replacement of grid outputs by continuous perturbations; L_2, the replacement of the computed centers by exact ones; L_1, the discrete evaluator law against exact Gaussians at the computed centers; L_{prec}, acceptance rounding; and L_{count}, the observable counts. Each stage uses its own order: a_1, a_2, a_3 for the output stages and $a_{\mathrm{h}}, a_{\mathrm{prec}}, a_{\mathrm{count}}$ for construction, acceptance and counts. Put $\theta_i = (a_i - 1)/a_i$. Then the recurrence of Section 4.3 gives

$$\begin{aligned} \Gamma_{\mathrm{fin}} &= \theta_{\mathrm{count}} L_{\mathrm{count}} + \theta_{\mathrm{count}}\theta_{\mathrm{prec}} L_{\mathrm{prec}} \\ &\qquad + \theta_{\mathrm{count}}\theta_{\mathrm{prec}}\theta_1(L_1 + \theta_2 L_2 + \theta_2\theta_3 L_3), \\ \Theta_{\mathrm{fin}} &= \theta_{\mathrm{count}}\theta_{\mathrm{prec}}\theta_1\theta_2\theta_3, \\ \Gamma &= \Gamma_{\mathrm{fin}} + \Theta_{\mathrm{fin}}\theta_{\mathrm{h}} D_{\mathrm{hyb}}, \qquad \Theta = \Theta_{\mathrm{fin}}\theta_{\mathrm{h}}. \end{aligned} \tag{C.4}$$

and Eq. (11) holds with these Γ, Θ. Table 10 evaluates the loss $(1-\Theta)b_{\mathrm{ref}} + \Gamma$ at $C_S = 2Q_S$, and also the finite-stage loss $(1-\Theta_{\mathrm{fin}})b_{\mathrm{ref}} + \Gamma_{\mathrm{fin}}$ relative to the exact Hybrid-Sampler. Section 4.3 compares these losses with the Cholesky signer's.

The bounds below hold uniformly over admitted keys, syndromes and preceding states, with $\|x_2\|_2 \leq 2^{14}$ and $C_S = 2Q_S$.

Each outer trial makes $n_d = 2N$ scalar calls, or $n_d = N$ pair calls in 3f, and $n_c = 2N$ real crown calls. Let d_d, d_c bound the per-call output log-divergences at orders a_1 and a_3 for discrete and crown draws, respectively; d_c includes crown evaluator error and the grid lift. Let $d_{\mathrm{prec},d}, d_{\mathrm{prec},c}$ bound the acceptance-rounding terms at order a_{prec}, and d_{count} the discrete count term at order a_{count}. For $i = 1, 2$, let e_i bound the canonical-norm error in center d_i, and write d_{ctr} for

Table 10. Transfer losses of the Crown signer in bits for a reference-event bound $2^{-b_{\mathrm{ref}}}$, at $C_S = 2Q_S$, rounded upward. *Finite stages* compares the implementation with the exact Hybrid-Sampler; *Construction* gives Lemma 3 alone; *Complete transfer* compares the implementation with the ideal signer, as in Table 3. The loss columns quantify the multiplicative factor in Eq. (11); additive failures are bounded separately by $\varepsilon_{\mathrm{joint}}$.

Profile	Q_S	b_{ref}	Finite stages	Construction	Complete transfer
1f	2^{64}	128	0.475354	0.424309	0.898876
3f	2^{64}	256	0.599493	1.199714	1.797804
5f	2^{64}	256	1.267474	1.199714	2.464213

the bound on the combined center-comparison log-divergence in one trial. Then

$$d_{\mathrm{ctr}} \le a_2 \kappa_{\mathrm{cov}} (e_1^2 + e_2^2)/(2r_{\mathrm{in}}^2 \ln 2),$$
$$L_1 = C_S n_d d_d, \qquad L_2 = C_S d_{\mathrm{ctr}}, \qquad L_3 = C_S n_c d_c,$$
$$L_{\mathrm{prec}} = C_S (n_d d_{\mathrm{prec},d} + n_c d_{\mathrm{prec},c}), \qquad L_{\mathrm{count}} = C_S n_d d_{\mathrm{count}}.$$

The center comparison uses the exponential-family log-partition identity and a covariance bound $\kappa_{\mathrm{cov}} r_{\mathrm{in}}^2$ in the canonical metric that holds uniformly over all centers, including the center-dependent normalizers; the certificates prove $\kappa_{\mathrm{cov}} < 1 + 2.1 \cdot 10^{-13}$ for the scalar sampler and $\kappa_{\mathrm{cov}} < 1 + 7.2 \cdot 10^{-8}$ for the degree-768 pair sampler. The error bounds cover preparation, transforms, projection, scaling, and the crown lift under the guard $\|x_2\|_2 \le 2^{14}$ in coefficient norm. A guard violation returns a failure flag.

Table 11. Upward-rounded center-error and lifetime divergence bounds of the Crown signer for Table 10's budgets; Eq. (C.4) composes them. Calculations use unrounded enclosures; displayed entries are rounded upward independently.

Profile	e_1	e_2	L_1	L_2	L_3	L_{prec}	L_{count}
1f	$1.936 \cdot 10^{-12}$	$1.503 \cdot 10^{-14}$	$3.507 \cdot 10^{-2}$	$8.578 \cdot 10^{-2}$	$3.508 \cdot 10^{-2}$	$1.602 \cdot 10^{-3}$	$8.060 \cdot 10^{-2}$
3f	$4.755 \cdot 10^{-12}$	$8.367 \cdot 10^{-14}$	$3.908 \cdot 10^{-5}$	$2.979 \cdot 10^{-1}$	$7.306 \cdot 10^{-6}$	$2.454 \cdot 10^{-3}$	$1.022 \cdot 10^{-4}$
5f	$5.267 \cdot 10^{-12}$	$4.815 \cdot 10^{-14}$	$7.004 \cdot 10^{-2}$	$3.296 \cdot 10^{-1}$	$7.009 \cdot 10^{-2}$	$3.204 \cdot 10^{-3}$	$1.610 \cdot 10^{-1}$

The orders, in composition order $(a_{\mathrm{h}}, a_3, a_2, a_1, a_{\mathrm{prec}}, a_{\mathrm{count}})$, are:

1f $(603, 3638, 1487, 3634, 977633, 1579)$
3f $(426, 41100920, 856, 21747348, 1214231, 2487623)$
5f $(426, 3635, 772, 3629, 976351, 1577)$

Transform errors. For every stored root $\widehat{\zeta}$, directed intervals enclose $|\widehat{\zeta} - \zeta|$ over the complete public root table. In units of 2^{-64}, the maximum errors are below 0.642430, 0.698572 and 0.668734 at degrees 512, 768 and 1024, respectively. The length-two and length-four FFT stages use only 1 and $\pm$i; additions, subtractions and these rotations are exact, so their error recurrence is

$E' \le \sqrt{2}E$, with no new rounding term. In the power-of-two rings, the forward input coefficients are integers, so their initial multiplication by stored roots is also exact on the fixed-point grid; only root error is charged there. Later products, inverse normalization, projection, grid scaling and preparation retain their full rounding allowances. Annularity, completion and conjugate symmetry give $|\gamma_j| \le \alpha^2\sqrt{Nk_N/2}$ (Appendix C.2), which bounds error amplification at the projection. Substitution into the transform recurrence yields Table 11's center errors without changing the operation graph or guard.

3f center transfer. The bound of Appendix E.1 includes both preparation roundings. Propagating it through the Q_{64} transforms, direct grid scaling and degree-20 evaluator, with the conditional center at Q_{65}, gives the 3f center errors in Table 11. The online and Crown acceptance words remain 56 and 72 bits, respectively.

The optimized integer operations preserve this arithmetic for every input in the certified ranges. Put $S = 2^{64}$. For the magnitude $m = l + Sh$ of a pair displacement, with $0 \le l < S$, $0 \le h \le 65$, and target coefficient $0 < d < 2^{63}$, the four-limb product reconstructs $m^2 d = (l^2 + 2Slh + S^2h^2)d$ exactly; every carry calculation fits in unsigned 128 bits and the sum of two products is below 2^{205}. A single guard, sticky, and parity test rounds the sum at bit 130. For the Horner product, write $a = a_0 + Sa_1$, where $0 \le a \le S$ and $0 \le x < S$. Then $\lfloor xa/S \rfloor = \lfloor xa_0/S \rfloor + xa_1$, with remainder $xa_0 \bmod S$; applying the same rounding rule gives exactly the specified Q_{64} product. Thus the word operations and their four-lane evaluation preserve the thresholds, including halfway cases. The proposal records, lane order and rewind positions are unchanged, so they preserve the sampled law and its count transcript. This refinement assumes admitted keys, correct compilation of two's-complement integer operations with arithmetic right shifts, and the execution-observation model of Section 4.1.

At order 65536, the per-call threshold-rounding terms are below $3.024154 \cdot 10^{-29}$ bits for scalars, $7.516279 \cdot 10^{-29}$ for pairs, and $3.397845 \cdot 10^{-26}$ for crowns. At each selected order, Lemma C.1 is evaluated again: crown integration divides each block into 256 intervals of 2^{47} positions, using directed evaluation and analytic tail bounds.

Let $\varepsilon_{\mathrm{supp}}$ bound the guard-violation probability in one reference trial and $\varepsilon_{\mathrm{local}}$ the total probability of exhausting a scalar, pair, or crown proposal cap. Including all failure outcomes gives

$$\varepsilon_{\mathrm{joint}} \le \varepsilon_{\mathrm{local}} + Q_S(0.4200000002)^{272} + \varepsilon_{\mathrm{nonce}} + 2^{\Gamma_{\mathrm{fin}}}(C_S\varepsilon_{\mathrm{supp}})^{\Theta_{\mathrm{fin}}},$$

where $\varepsilon_{\mathrm{supp}}$ refers to the exact Hybrid-Sampler, so the guard tail transfers through the finite stages only.

Applying the per-trial total-variation calculation below to Table 11, including the construction, cap and support terms, gives distance below $2 \cdot 10^{-10}$ from the ideal signer. Hence the acceptance probability exceeds 0.5799999998 for every profile. Thus $2^{64}(0.4200000002)^{272} < 2^{-276.418544}$ bounds lifetime outer-cap exhaustion.

Guard tail. Put $W = 2^{14}$ and let $d_2^0 = \Phi^{-1}((\Phi_j(c)\chi_j)_j) = cf/q$ be the second center before perturbation. Annularity gives $\|d_2^0\|_{\mathrm{can}} \le M_N = \alpha(q -$

$1)\sqrt{N/q}$, multiplied by $\sqrt{3/2}$ in degree 768, for every admitted key and syndrome. Each of the N real normal draws behind y_2 has magnitude above 1024 with probability at most $2e^{-1024}$. Otherwise $\tau_{2,j} < 2$ and $\omega = \sqrt{N/1024}$ give $\|d_2\|_2 \le D_N = j_N\big(M_N + 2\sqrt{2}\cdot 1024\sqrt{N/1024}\big)$, where $j_N = 1$ in power-of-two degrees and $j_N = \sqrt{2}$ in degree 768 converts the canonical norm to the coefficient norm. A guard violation then requires a coefficient of $x_2 - d_2$ of magnitude above $m_N = (W - D_N)/\sqrt{N} > 0$. Let v_N be the diagonal entry of G_R^{-1}, namely 1 in power-of-two degrees and 4/3 in degree 768. Completing the square in the moment-generating function of $D^{\mathrm{can}}_{R,r_{\mathrm{in}},d_2}$ and bounding the shifted normalizer by Poisson summation give

$$\varepsilon_{\mathrm{supp}} \le 2Ne^{-1024} + 2N\kappa(\varepsilon_p)\exp\Big(-\frac{m_N^2}{2r_{\mathrm{in}}^2 v_N}\Big).$$

Since $D_N < 4933$, 9879, and 7260 at degrees 512, 768, and 1024, $\varepsilon_{\mathrm{supp}} < 2^{-1466}$ in every profile. The per-call cap terms are 0.255^{1024}, 0.4^{1024}, and 0.08^{256}. The local-cap and transferred-support contributions are below 2^{-856} for every profile. With the 272-trial outer cap above and $\varepsilon_{\mathrm{nonce}}$ from Appendix C.1, $\varepsilon_{\mathrm{joint}} < 2^{-276}$.

Dividing a lifetime log-divergence L by C_S gives its per-trial bound. Rényi monotonicity and Pinsker's inequality then bound the total variation in that transition by $\sqrt{(\ln 2)L/(2C_S)}$. Summing over the finite stages and adding the per-trial cap and support contributions gives a distance below $1.52\cdot 10^{-10}$ from the exact Hybrid-Sampler in every row. Adding the construction distance $\delta_p < 2^{-35}$ of Lemma 3 keeps the distance from the ideal signer below $2\cdot 10^{-10}$.

C.6 Exact response reconstruction

For Falcon, let $R_0 = 28$ and $R_{j+1} = 3R_j + 1$. A worst-case recursion over the prepared tree bounds the combined response residual by $1024(R_{\log_2 N - 1}+1)+2^{20}$. Adding the target bound $q-1$ gives the product bounds in Section 5.1.

The Cholesky signer computes the spectra of $s_0 = c - x_1 g - x_2 G$ and $s_1 = x_1 f + x_2 F$ modulo the single auxiliary prime $p_0 = 2147346433$, reusing the spectrum of x_2, then applies inverse transforms and centered lifts. In degree 768, the transforms evaluate at the two factors of $X^{768} - X^{384} + 1$ modulo p_0 and interpolate before the lift.

One residue suffices because the bound applies to the reduced response polynomials for every coordinate pair the sampler can return, including trials later rejected by the norm or codecs. Write $(s_0, s_1) = p - \hat{B}e$, where $p = \hat{A}v/L_{\mathrm{p}}$ is the perturbation and e the two-stage online displacement, including fractional-center rounding. The finite perturbation and proposal supports bound v and e deterministically, and Section 4.1 bounds $\hat{A}$ and $\hat{B}$ for every admitted key. Directed evaluation gives response coefficients below 102,979, 407,631, and 166,328 in 1f, 3f, and 5f, far below $p_0/2$; the canonical squared norm fits in 64 bits.

Crown signer. As for Cholesky, centered lifting modulo p_0 reconstructs the combined products exactly, including rejected trials. The shared transforms reduce the degree-768 products modulo $X^{768} - X^{384} + 1$. Set $\gamma = \Phi^{-1}((\gamma_j)_j) =$

$-u$ and $\widetilde{b}_2 = b_2 + \gamma b_1$. Then $(s_0, s_1) = (\Phi^{-1}(y_1) - \eta_1)b_1 + (\Phi^{-1}(y_2) - \eta_2)\widetilde{b}_2$, where y_i are the crown perturbations of $\mathsf{PreSmp}^{\mathrm{cr}}$ and η_i collect the center errors and the discrete proposal displacements. Both basis vectors have embedding lengths at most $\alpha\sqrt{q} < 137$. Crown support and scaling give complex perturbation coordinates below 1000 in magnitude; the finite proposal supports and center-error bounds give $|\eta_{i,k}| < 51$ for each coefficient. Thus $|\Phi_j(\eta_i)| < 51N$. Inverse embedding bounds coefficients by the largest embedding magnitude in power-of-two degrees and by twice that magnitude in degree 768. With $c_N = 2$ for $N = 768$ and $c_N = 1$ otherwise, the two reconstruction products have coefficients below

$$2c_N \cdot 137(1000 + 51N) + (q - 1).$$

This gives 7,440,976, 22,024,352, and 14,595,664 at degrees 512, 768, and 1024, all below $2^{25} < p_0/2$. The bound uses prepared-factor errors at most 2^{-60}, componentwise $|\chi|, |\psi| \le 1$, $|\gamma| \le 2048$, and the guard $\|x_2\|_2 \le 2^{14}$.

D Runtime of Lazy Bernoulli Sampling

Lazy Bernoulli comparisons save random bytes, but their early stopping can make runtime depend on the accepted sample. Section 6 of [17, full version] claims that its sampler's total runtime is independent of its output, the perfect isochrony of [17, full version, Definition 5]. The lazy test in [17, full version, Algorithms 1 and 2] violates this claim in the following cost model.

Fix (σ, μ) and use independent proposals with fresh uniform comparison bytes. Let g_z be the proposal law, p_z the actual acceptance probability given z, and $\alpha = \sum_z g_z p_z > 0$. The ideal lazy test compares bytes against the binary expansion of p_z until the first unequal byte. Falcon rounds p_z to a multiple of 2^{-64} after approximation and compares at most eight bytes. Given proposal z, either test accepts after one byte with probability $\lfloor 256 p_z \rfloor / 256$.

Assign the same fixed positive cost to each proposal outside the comparison and a fixed positive cost to each compared byte. This model is consistent with the hypothesis of [17, full version, Theorem 9] that elementary operations are isochronous. Let Z be the returned value and R the total runtime. If some proposal can accept after one byte, the minimum runtime r_* occurs exactly when the first proposal does so. Since $\Pr[Z = z] = g_z p_z / \alpha$ and $\Pr[R = r_*, Z = z] = g_z \lfloor 256 p_z \rfloor / 256$, every z with $g_z p_z > 0$ satisfies

$$\Pr[R = r_* \mid Z = z] = \alpha \, \frac{\lfloor 256 p_z \rfloor}{256 p_z}.$$

This probability need not be constant in z: it vanishes when $0 < p_z < 1/256$, while for $p_z \ge 1/256$ it lies between $\alpha(1 - 1/(256 p_z))$ and α.

For a concrete witness, take $\sigma_{\min} = 5/4$, $\sigma = 3/2$, $\sigma_{\max} = 3641/2000$, and $\mu = 1/4$. The ideal half-Gaussian base sampler and exact exponentials give

$$p_0 = \tfrac{5}{6} e^{-1/72} \approx 0.821839, \qquad p_8 = \tfrac{5}{6} \exp\Big(\frac{49}{2\sigma_{\max}^2} - \frac{961}{72} \Big) \approx 0.002161.$$

For this ideal witness, the Gaussian mass function Θ of Appendix A gives

$$\alpha = \frac{\sigma_{\min}\Theta_\sigma(\mu)}{2\sigma\sum_{j\geq 0} e^{-j^2/(2\sigma^2_{\max})}} \approx 0.563204554.$$

Both proposals have positive mass, but $\lfloor 256p_0 \rfloor = 210$ and $\lfloor 256p_8 \rfloor = 0$. Thus $\Pr[R = r_* \mid Z = 0] \approx 0.562158$, whereas $\Pr[R = r_* \mid Z = 8] = 0$, disproving independence. The smoothing premise of Theorem 9 holds at $n = 512$ with

$$\varepsilon = \Big(\frac{\exp(2\pi^2\sigma^2_{\min})}{1024} - 1\Big)^{-1} \approx 4.12635 \cdot 10^{-11}.$$

Exact rational bounds on the exponentials establish $p_0 > 1/256$ and $0 < p_8 < 1/256$, even under acceptance-probability errors of at most $2^{-40}p_z + 2^{-64}$. At the same σ and μ, the fixed-point Falcon routine of Section 5.1 uses its degree-dependent $\sigma_{\min}$ and gives leading threshold bytes 215 and 218 for $z = 0$ at degrees 512 and 1024, and zero for $z = 8$ at both degrees. In $2 \cdot 10^7$ Falcon-512 sampler calls, the accepting comparison read one byte for 99.96% of outputs $z = 0$ and for none of the observed outputs $z = 8$.

The data-processing step in [17, full version, Appendix C] needs the runtime law conditional on the proposal count and acceptance; its threshold-independent marginal does not suffice. Likewise, the signature/runtime independence used before Eq. (26) in [17, full version, Appendix D] needs a joint argument.

The counterexample does not disprove the statistical isochrony bound with respect to (σ, μ) in Theorem 9 or establish an attack on Falcon. Assessing cumulative leakage in complete signatures requires a separate analysis. Complete-word comparisons with a fixed instruction schedule (Section 4.2) remove this comparison-length correlation.

E Precision Choices for Antrag

E.1 Key generation and preparation

A Q_{96} midpoint-radius pair (c, r) represents $[(c-r)2^{-96}, (c+r)2^{-96}]$ (Table 12). Division by ν computes the exact floor of the center quotient using one certified Newton reciprocal per embedding, shared by both components; its radius uses $\nu \geq 2^{12}$. Square roots use guarded integer arithmetic.

Cholesky signer ($\mathsf{PreKey}^{\mathrm{ch}}$). Let $\ell_{00} = \sqrt{\sigma_{00}}$ be the first diagonal entry of the Cholesky factor of Σ_p at an embedding (Figure 6, step 7). For the Cholesky quotient, both operands are scaled by 16; $\ell_{00} \geq 10$ then gives the lower bound $16\ell_{00} \geq 2^7$ on the scaled pivot. In degree 768, the inverse transform multiplies by an enclosure of $2/\sqrt{3}$. Cholesky square roots divide the radicand by 128^2, apply the guarded Q_{96} square root, and multiply by 128; every radicand exceeds 100 and lies below $\tilde{s}^2 < 140^2$, inside the square-root contract with margin.

Integer majorants bound the final radii of u below 2^{40} and those of A below 2^{50}, 2^{52}, and 2^{50} units of 2^{-96} in 1f, 3f, and 5f, and all centers fit in signed

Table 12. Arithmetic used in key generation and preparation. Each stage computes at a single working precision; Appendix C.2 specifies the cutoff and completion schedules.

Stage	Arithmetic	Output
Candidate generation	Directed Q_{64} intervals, one pass; an undetermined decision rejects the candidate	Rounded (f, g) and annular decision
Completion, 1f/5f	F-only integer recursion; Q_{80} Fourier quotient with scaled inverse, then final joint reduction in Q_{32}	Integer (F, G)
Completion, 3f	Exact integer lifts and Q_{256} reduction; mixed-degree F-only recursion with a fixed round cutoff	Integer (F, G)
Key admission	Exact integer and modular tests	Valid basis and public key
Key preparation (Cholesky signer)	Q_{96} midpoint-radius, one pass; one rounding of $\hat{u}$ and $\hat{A}$, then exact integer maps	Cached $\hat{u}$, $\hat{A}$, maps, and spectra
Key preparation (Crown signer)	Q_{96} midpoint-radius, one pass; 1f/5f round once to the formats of Table 15; 3f rounds to Q_{72}, then narrows once to Q_{64}	Cached projections, widths, and spectra

128-bit words. These bounds give the radius allowances and effective precisions of Section 4.1.

The integer maps of Section 4.2 are formed from the key polynomials and the integer encodings $u_{\mathrm{int}} = p_u\hat{u}$ and $A_{\mathrm{int}} = p_A\hat{A}$. Desktop preparation computes them exactly modulo three 31-bit primes congruent to 1 modulo 18432, namely p_0, p_1, and 2147235841, and reconstructs each coefficient by a centered CRT. Let M be the product of these primes. For every admitted key, $\|\hat{B}\| < \tilde{s} < 140$ at every embedding bounds $\hat{b}_2$, and $AA^* \preceq \tilde{s}^2 I$ bounds every Cholesky entry by $\tilde{s}$. Since inverse embedding bounds coefficients by the largest embedding magnitude, and by twice that magnitude in degree 768, this gives the coefficient bounds $\|m_{11}\|_\infty, \|m_{12}\|_\infty < 2^{25}$ and $\|A_{\mathrm{int}}\|_\infty < 2^{53}$, the latter over all entries; Appendix C.2 gives $\|u_{\mathrm{int}}\|_\infty < 2^{26}$. Each map coefficient is a sum of at most $6N$ products of operand coefficients, so its magnitude is below 2^{91}, below $M/2 > 2^{91.99}$. Fixed-schedule range checks verify the CRT condition from the operand magnitudes. The compact preparations use the same first two primes and third prime 2147297281 at degrees 512/1024 or 2147304961 at degree 768. Their products also satisfy $M/2 > 2^{91.99}$, so the desktop and compact preparations reconstruct the same integer maps.

Crown signer ($\mathsf{PreKey}^{\mathrm{cr}}$). Preparation is evaluated once in this arithmetic. With $\nu \geq 2^{12}$, a public lower bound 2^{-8} on residual radicands reduces the radius calculations to multiplications and shifts. The represented values satisfy $|c|2^{-96} < 2^{26}$ and $r2^{-96} < 2^{-32}$, so the integers c and r fit in signed 128-bit and unsigned 64-bit words, respectively. Conjugate symmetry gives $|\gamma_j| \leq$

$\alpha^2\sqrt{Nk_N/2}$, and the residual widths satisfy $\tau_{i,j} \le 1.32\alpha^2 < 2$ and $\omega \le 1$. Every final ball has diameter below 2^{-w}, with $w = 66, 62, 63$ for 1f, 3f, and 5f; γ has the widest enclosure. In 1f/5f the coefficients round to Q_{64} and widths to Q_{62}. Rounding a ball's center to s fractional bits gives error at most $2^{-s-1}+2^{-w-1}$; in the 1f and 5f Q_{64} profiles, the Q_{62} widths therefore have error at most $2^{-63}+2^{-67}$ (1f) or $2^{-63}+2^{-64}$ (5f).

In 3f, coefficients and widths round first to Q_{72} and then to Q_{64}, giving error at most $2^{-65}+2^{-73}+2^{-63}$ per component. Appendix C.5 propagates this error through the Q_{64} transforms and sampler.

These preparation and carrier bounds hold for every admitted key, so their guards introduce no additional rejection under $\mathcal{K}_\pi$. Crown's separate online x_2 guard has the probabilistic tail charged in Appendix C.5.

E.2 Signing arithmetic

Table 13 gives fractional precisions and coarse magnitude bounds; signed representations also need a sign bit. Appendix C.6 gives tighter response bounds.

Table 13. Fixed-point formats and integer range bounds for signing with the Cholesky signer.

Quantity	1f	3f	5f
Factor scales (p_u, p_A)	$(2^{16}, 2^{43})$	$(2^{16}, 2^{44})$	$(2^{16}, 2^{44})$
Center fraction	Q_{64}	Q_{64}	Q_{64}
Exponential polynomial	degree 12, Q_{63}	degree 20, Q_{64}	degree 12, Q_{63}
High numerator H_2 (bits)	25.4	26.9	26.0
High numerator H_1 (bits)	49.7	52.7	51.6
Low numerators Λ_j (bits)	57.9	60.7	59.9
Response coefficient (bits)	17	19	18

Cholesky signer ($\mathsf{PreSmp}^{\mathrm{ch}}$). Table 13 reflects the following precision constraints; divergence bounds are lifetime bounds at $C_S = 2^{65}$ (Table 9).

- *Factors and carriers.* The scales $p_u = 2^{16}$ and $L_{\mathrm{p}} = 2$ follow [32]. In 1f/5f, factors use 43/44 fractional bits, one more than [32, Table 2], so that the effective precision p'_A includes the Q_{96} radius allowance. In 3f, $p_A = 2^{44}$ gives $\|\Lambda_j\|_\infty < 2^{60.7}$, within one bit of $p_0 p_1/2 > 2^{60.99}$; another factor bit would exceed this certified range. The two primes serve every ring, and all reconstructed numerators fit signed 64-bit words; H_2 needs only p_0.
- *Centers and evaluators.* Center fractions use Q_{64}, with the 3f conditional half-center at Q_{65}; the center-rounding bound is below $2.3\cdot10^{-7}$. The shared 1f/5f evaluator uses a two-floor Q_{64} exponent and degree-12 Q_{63} polynomial without a common scale, giving a proposal/evaluator stage bound at most $1.553 \cdot 10^{-3}$ bits (Table 9). In 3f, the joint pair acceptance test cancels both proposal weights; its degree-20 Q_{64} evaluator gives loss below $4.0 \cdot 10^{-5}$.

– *Perturbation tables.* The 96-bit masses use magnitude cutoffs 28/29 in 1f/5f and parity-conditioned tables in 3f. Their lifetime contribution is below $6.9 \cdot 10^{-5}$. All profiles support $Q_S = 2^{64}$ (Table 3).

The perturbation draws v are integers, and the perturbation $p = \hat{A}v/L_{\mathrm{p}}$ is never formed, since the centers use the integer maps of Section 4.2.

Relation to [32]. Table 14 compares the choices. At degree 1024, [32] obtains a 1.5-bit bound with a larger inner width, perfect integer sampling and exact Cholesky rounding; Table 3's 1.777-bit bound includes preparation and finite-sampling errors.

Table 14. Choices of [32, Section 4] and of the Cholesky signer. Slash-separated entries are for degrees 512 and 1024, respectively. Widths are standard deviations; [32] states $r = 3.305/3.34$ in the convention $e^{-\pi x^2/r^2}$.

Quantity	[32]	Cholesky signer
Inner width; $\tilde{s} = \sigma_{\mathrm{sig}}/r$	1.3185/1.3325; 127.49/138	$r_{\mathrm{in}} = 1.315$ at all degrees; 127.97/136.87
Factor scale p_A	$2^{42}/2^{43}$	$2^{43}/2^{44}$; 2^{44} at degree 768
Factor preparation	`long double`; analysis assumes coefficient error at most $1/(2p_A)$	Q_{96} midpoint-radius; analysis uses p'_A (Section 4.1)
Auxiliary primes	2147473409, 2147389441 ($\equiv 1 \bmod 2048$)	p_0, p_1 ($\equiv 1 \bmod 18432$)
Carrier-bound premises	$\|f\|_\infty, \|g\|_\infty, \|F\|_\infty, \|G\|_\infty \leq 127$; $\|u_{\mathrm{int}}\|_\infty \leq p_u/2$ (size-reduced u)	Annularity and Eq. (1)
Sampling-table masses	88-bit	96-bit
Online magnitude support	$0 \ldots 14$	Scalar (1f, 5f): $0 \ldots 15$; pair (3f): $0 \ldots 15$ and $0 \ldots 17$
Zero fractional code	Mapped to 1 (fraction 2^{-64})	Kept at zero
Acceptance word	Not specified	56-bit complete word
Shared	$\alpha = 1.15/1.23$, $p_u = 2^{16}$, $L_{\mathrm{p}} = 2$, perturbation cutoffs 28/29, high/low center split, exponential polynomial of [6]	

Crown signer ($\mathsf{PreSmp}^{\mathrm{cr}}$). The Crown signer's formats differ in its Fourier center state, its widths and scale, and its exponential polynomials (Table 15).

Integers and failures. In both signers, the online discrete samples, the reconstructed responses, and the squared norms are exact integers. Norm or codec rejection starts a fresh trial; outer-cap exhaustion and operational faults return failure without releasing a signature. The Crown signer also returns failure on a violation of the guard $\|x_2\|_2 \leq 2^{14}$ (Appendix C.5). Corollary 1 accounts for these outcomes.

Table 15. Fixed-point formats and coarse integer range bounds for the Crown signer. $Q_{m.f}$ denotes a signed value with f fractional bits and absolute value below 2^m; rows marked "bits" give magnitude bounds for choosing integer types. The last row is the certified bound used by the compact signer of Table 6.

Quantity	1f	3f	5f
Transform and center state	$Q_{39.64}$	$Q_{42.64}$	$Q_{43.64}$
Widths τ, scale ω	$Q_{1.62}$	$Q_{1.64}$	$Q_{1.62}$
Crown grid $\to$ scaled output	$Q_{53} \to Q_{64}$	$Q_{53} \to Q_{64}$	$Q_{53} \to Q_{64}$
Exponential polynomial degree	9	20	9
Exponential polynomial arithmetic	Q_{63}	Q_{64}	Q_{63}
Transform product (bits)	168	171	172
Raw response coefficient (bits)	39	43	41
Raw squared norm (bits)	87	97	93
Compact working state (Table 6)	$Q_{28.64}$	$Q_{29.64}$	$Q_{30.64}$

F Benchmark Methods and Additional Results

F.1 Measurement protocol

For Table 5, each signer configuration and processor uses three process repetitions, with implementation order reversed in the middle repetition. Each configuration measures 3,000 randomized key generations, 2,700 core preparations, and 8,100 signatures and 810,000 verifications per ordinary or MRM mode. Signing and verification use three keys, batches of 300, and 32-byte explicit messages. The recoverable payload is 809, 1,220, or 1,631 bytes at degrees 512, 768, or 1024, respectively, equal across schemes. Signing includes tape expansion, retries, hashing, reconstruction, encoding and library cleanup; OS entropy acquisition is excluded, and every timed signature is verified outside the timer. Verification includes public-key decoding and hashing without a key cache. Core preparation starts from a validated integer basis in preallocated storage and includes geometry, numerical checks and all required key and map caches. Wire decoding, key admission and public derivation, protocol hashes, system allocation and final owner cleanup are excluded. Shared public tables are initialized before timing. Key generation includes entropy, rejection, completion, admission and serialization. Rates are total counts divided by total elapsed time, retaining every batch; Antrag's shared keygen and verification rows pool both signers. The Falcon-768 rows were remeasured on both hosts with the same harness, counts and repetitions after its key-generation changes (Section 5.1: the shared completion and the 16-bit key law); the benchmark keys therefore differ from the earlier runs, while the preparation, signing and verification code is unchanged and those rows moved within run-to-run variation. Desktop completion-policy comparisons use 1,000 randomized key generations per configuration, with the same build flags and timing boundaries. The Cortex-M4 comparison uses identical seeds.

Across the 27 signing batches per configuration and mode, sample standard deviations relative to mean time range from 0.42–8.06% for Antrag and 0.15–0.74% for Falcon on M5, and 0.95–10.11% and 0.42–6.06%, respectively, on i7. The largest range across three process means, divided by the overall mean, is 16.5% for signing and 33.5% for core preparation on M5, versus 11.0% and 2.5% on i7. These variations limit conclusions from small timing differences.

Cortex-M4 measurements use Thumb-2 soft-float code, two flash wait states, prefetch and caches. The DWT counter times 100 four-byte messages under one prepared key with deterministic seeds and retries. Signing uses no heap or mutable library state. SRAM is workspace plus observed stack high-water mark, excluding harness buffers. Code flash includes public constants, runtime, startup and harness, but excludes the prepared key and whole functions or tables unused by the signing run: 3.9–9.7 KB for Antrag, 14.1 KB for FN-DSA and none for Falcon-768. FN-DSA's degree-indexed roots additionally count at half size for degree 512 (12.3 KB), so this is a signing-only footprint rather than the complete image size. The Falcon-768 Cortex-M4 code reproduces the SDK's keys, prepared keys and signatures byte for byte on the board. Its prepared key stores one LDL branch and derives the mirrored one while signing. The Falcon-768 row of Table 6 is measured like the Antrag rows (GCC 14.2.1, LTO, 84 MHz; 100 lazy signatures of one key, 30 keys, five preparations); its code flash follows the rule above. Before the measurement, Unicorn with the Cortex-M4 cycle model of the compact Antrag toolchain (zero wait states) on a clang 14 `-O3` image without link-time optimization estimated 265, 26.6 and 19.9 Mcc, 1%, 14% and 7% below the board.

The Cholesky key stores $22N$ 32-bit NTT residues ($72 + 88N$ bytes); Crown stores packed geometry and four NTT arrays. Both add the $136+14N/8$-byte v1 prefix. The measured FN-DSA build, c-fn-dsa-fx32.64 at commit `10e9c03` [27], uses Cortex-M4 assembly and an expanded key of $(3\log_2 N + 25)N + 105$ bytes in flash. Both schemes use the same XKCP permutation; FN-DSA's internal restart count is unavailable.

The lazy option reads acceptance coins most significant byte first until a byte differs. It preserves acceptance probabilities but changes byte consumption, seeded signatures and timing. The ChaCha20 variant of Section 5.4 generates each lane with RFC 8439 ChaCha20 and block counter zero. The first 32 SHAKE256 output bytes of its v1 tape input form the key; the next eight bytes followed by four zero bytes form the 12-byte nonce. It preserves proposal and retry rules, changes seeded signatures, and changes SRAM by at most 96 bytes.

The compact Crown signer's Fourier working state uses 12 bytes per real component (Q_{64}). Conditional on the Crown certificates and the x_2 norm guard, a rational certificate bounds every working component by 258451169, 509299801, and $830131036 < 2^{30}$ at degrees 512, 768, and 1024 (Table 15). The target cache is bounded by 2^{23} and uses 11 bytes per component.

F.2 Reimplementation of [32]

The reimplementation follows Algorithm 3 of [32], with 88-bit perturbation and proposal tables, their center conversion, and exact NTT/CRT centers; factors are prepared in Q_{96} interval arithmetic. We use 64-bit acceptance thresholds and, where indicated, variable-length encoding, neither specified in [32]. Table 16 uses AVX2 SHAKE256 and complete-word acceptance. The ChaCha20 comparison in Section 5.3 instead uses their parameters, norm bounds and primes, a modular norm for the untransmitted component, and Falcon's lazy acceptance. Each configuration uses one key per degree, identical signing-seed schedules and 127-byte messages, with four rounds of eight batches of 512 signatures. CPU timings include tape expansion, retries, hashing, response reconstruction, encoding and cleanup; preparation, verification and replay checks are outside the timer. At our Gaussian widths, the squared-norm bounds of [32] correspond to tail factors of about 1.049 and 1.073 at degrees 512 and 1024; we retain 1.042 for the security estimates of Table 2.

Table 16. Cumulative changes to the reimplementation of [32] on i7-13700 (GCC 15.2, AVX2 SHAKE256): CPU time and trials per ordinary prepared signature. Each "+" row builds on the preceding row; differences depend on their order and are not independent operation costs.

Configuration	512		1024	
	μs	Trials/sig.	μs	Trials/sig.
Cholesky signer	97.948	1.046	187.610	1.020
Reimplementation, matched	99.459	1.042	190.269	1.022
+ norm-only filter, variable codec	97.284	1.028	184.897	1.005
+ [32] squared-norm bound	95.895	1.013	184.187	1.000
+ [32] Gaussian parameters, factor precision, and primes	96.026	1.012	184.452	1.001
+ modular norm, untransmitted component	94.998	1.012	182.749	1.001

G A Deterministic Certificate Mode

This appendix records a deterministic variant of the PKG certificate, a proof skeleton for its ideal form in the classical random-oracle model, and the exponents the resulting bound supports. The user signature stays randomized, and nothing here enters the bounds of Sections 3 and 4.

G.1 Counter-recovery signing

The variant replaces the random nonce of SignMRM by the trial index. Let $J = 272$ be the per-signature attempt cap of Section 4.3, so $J < q$. The nonce has

one coefficient and the recoverable field M_0 has $N - \lambda_{\mathrm{MR}} - 1$; the oracles $H^{\mathrm{d}}_{0,\pi,h}$ and $H^{\mathrm{d}}_{1,\pi,h}$ are domain-separated from those of Section 2.4. For $j = 0, \ldots, J-1$ in turn, the signer forms

$$c_{0,j} = H^{\mathrm{d}}_{0,\pi,h}(j, M_0, M_1), \qquad c_j = \big(c_{0,j},\, (H^{\mathrm{d}}_{1,\pi,h}(c_{0,j}) + (j, M_0)) \bmod q\big),$$

samples a response for c_j from a private tape derived from the secret key, the message, and j, and returns the first response in $\mathcal{A}_\pi$; exhaustion returns failure. No random source is used. Verification recovers (j, M_0), rejects $j \geq J$, and otherwise follows VerifyMRM. Because j is a recovered coordinate, every trial has its own public target, the accepted index is public, and a message always yields the same signature. With the embedding of Section 3.4, the mode increases recovery capacity by $\gamma_{\mathrm{MR}} - 1$ key coefficients. For the profile pairs of Table 4, complete signatures take 1,916, 2,929, and 3,844 bytes in the convention of Table 4.

G.2 Proof skeleton

In the ideal signer, trial (M, j) with $M = (M_0, M_1)$ draws its response from $D^{\mathrm{can}}_{\mathcal{L}_h(c_j),\sigma_{\mathrm{sig}}}$ using an independent private record that is fixed across repeated queries. Oracles take the message directly, as in Theorem 2.

Proposition G.1 (ideal deterministic certificate). Assume Eq. (3). Consider an adversary running in time t that makes signing queries on at most $Q_S \geq 1$ distinct messages and at most Q_0, Q_1 queries to $H^{\mathrm{d}}_{0,\pi,h}, H^{\mathrm{d}}_{1,\pi,h}$. Put $A_D = Q_0 + JQ_S + 1$. For every order $a > 1$ and $\theta = (a-1)/a$,

$$\mathrm{Adv}^{\text{UF-CMA}}_{\text{A-MRM}^{\mathrm{d}}} \leq e(Q_S+1)\left\{r_u(a)^{A_D}\ \mathrm{Adv}^{A_D\text{-A-R-ISIS}}_{\pi}\right\}^{\theta} + \frac{A_D(Q_1+1) + \binom{A_D}{2}}{q^{\lambda_{\mathrm{MR}}}},$$

with the A-R-ISIS advantage evaluated at expected local time $t + \widetilde{O}(A_D + Q_1)$.

Proof skeleton. Call a queried pair (M, j) a cell; there are at most A_D cells, counting final verification.

1. *Distinct targets.* Equal targets of one message have equal prefixes, hence equal masks and equal indices modulo q; since $J < q$, the indices coincide.
2. *Simulated cells.* For a fresh cell, draw $s \leftarrow D^{\mathrm{can}}_{R^2,\sigma_{\mathrm{sig}}}$, set $c = s_0 + s_1 h \bmod q$, answer $H^{\mathrm{d}}_{0,\pi,h}$ with c_0, program $H^{\mathrm{d}}_{1,\pi,h}(c_0) = c_1 - (j, M_0) \bmod q$, and store s with its filter decision. Stop if c_0 already carries an $H^{\mathrm{d}}_{1,\pi,h}$ value. A signing query walks $j = 0, 1, \ldots$, creates missing cells, and returns the first accepted stored response or failure; no cell is reprogrammed. As in Appendix A.3, a simulated cell differs from the ideal signer's only in its syndrome law, with ratio $U(R_q)/Q_{\mathrm{syn}}$. This is charged at every cell, including those opened by hash queries, because any of them may later be signed.
3. *Partition.* Following [5], mark each message independently with probability $1/(Q_S+1)$. Cells of marked messages receive A-R-ISIS challenge syndromes instead, and a signing query on a marked message aborts. Marks are independent of the ideal transcript, so with probability at least $1/(e(Q_S+1))$ the forged message is marked and no signed message is.

4. *Extraction.* On that event and without a prefix collision, the pair (j^*, M_0^*) recovered from the forgery names a cell of the marked message, and the forgery is a preimage in $\mathcal{V}_\pi$ of that cell's challenge.

Let B_{pref} denote the displayed prefix term and E the event of a successful, collision-free forgery whose message is marked while every signed message is unmarked. Sequential Rényi composition and Hölder's inequality (Section 3.1) give

$$\frac{\epsilon_{\mathrm{det}} - B_{\mathrm{pref}}}{e(Q_S+1)} \le \Pr_{\mathrm{ref}}[E] \le \left(r_u(a)^{A_D} \Pr_{\mathrm{red}}[E] \right)^{\theta},$$

where ϵ_{det} is the ideal forgery probability. Both experiments stop on a marked signing request or a prefix collision; the probabilities are not conditioned on avoiding these stops. A union bound over fresh uniform prefixes, including a fresh final H_0^{d} guess, gives B_{pref}. On E, extraction solves an A-R-ISIS challenge, proving the stated bound. □

The accepted index and exhaustion belong to the simulated view, and no acceptance lower bound is used. The skeleton omits the steps that Corollary 1 supplies for randomized signing (private-tape replacement, digest composition, and the fixed-point transfer), one PKG key issuing both certificate modes, and availability when messages are chosen after inspecting public targets.

G.3 Numerical reading

When the A-R-ISIS advantage is at most $2^{-\lambda}$, the leading term is at most $2^{-b_{\mathrm{det}}}$ with $b_{\mathrm{det}} = \max_a \theta(\lambda - A_D \log_2 r_u(a)) - \log_2(e(Q_S+1))$, over the integer orders $2 \le a \le 512$ of Section 3.5. Table 17 uses the selected classical costs of Table 2 and $Q_S = 2^{32}$ issued certificates.

Table 17. Leading-term exponents of Proposition G.1 at $Q_S = 2^{32}$; b_{cal} is the randomized classical exponent of Table 2 at $Q_S = 2^{64}$ and $C_S = 2^{65}$. The loss is $b_{\mathrm{cal}} - b_{\mathrm{det}}$ at $Q_0 = 2^{64}$. A dash marks a vacuous bound.

Profile	b_{cal}	b_{det}, $Q_0 = 2^{64}$	Loss	b_{det}, $Q_0 = 2^{72}$	b_{det}, $Q_0 = 2^{80}$
1f	121.172	87.97	33.21	79.35	–
3f	192.909	160.07	32.84	138.54	–
5f	261.566	228.83	32.74	203.67	–

The deterministic reduction has two sources of loss. The partition factor costs $\log_2(e(Q_S+1)) < 33.45$ bits. The Rényi exponent A_D counts hash queries, whereas randomized signing charges signing trials only: its cost is below one bit for $A_D \le 2^{65.5}$, $2^{62.8}$, and $2^{62.4}$ in 1f, 3f, and 5f, and at this Q_S the bound is vacuous by $A_D = 2^{80}$. Proposition G.1 remains valid at larger query budgets, but its leading term is informative only when $b_{\mathrm{det}} > 0$; a vacuous bound does not imply an attack. For PKG-generated keys one may hope to charge signing cells

only. That needs a bound on the conditional min-entropy of the certified message given the adversary's view before issuance (earlier keys and certificates, and all oracle answers), and it must charge the probability that a cell of a later-certified message was queried in advance, about $Q_0 Q_S$ times the guessing probability. The entropy of $\mathcal{K}_\pi$ alone does not suffice; neither the required entropy bound nor the tighter reduction is established here.